Macroeconomics under Partially Identified Expectations

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Abstract

Can a macroeconomic counterfactual be identified if we know the distribution of expectations mechanisms but not which households hold them? This paper shows that it need not be. Under a scalar responsiveness channel and a compatible resource condition, the unknown assignment enters equilibrium through one statistic, responsiveness among unconstrained households. Rearrangement gives sharp bounds on this statistic, while endogenous household composition makes its admissible values an equilibrium object. In a benchmark economy, the identified set for the aggregate spans 25.93 percent of its midpoint. Without a restriction linking observed groups to the economic state, average responsiveness by income or liquidity group does not narrow the set, even with arbitrarily large samples. A classification that refines the state partition can identify the statistic from group shares and group averages using repeated cross-sections. In the benchmark, proportional income incidence makes the observable spending contrast zero under every admissible assignment.

Keywords: partial identification, heterogeneous agents, expectations, assignment, general equilibrium.

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*Center for the Economics of Human Development, Department of Economics, University of Chicago. Email: cchavezpadilla@uchicago.edu. Codex and Fable 5 were used for proofreading, for writing and checking the code behind the numerical exhibits and the estimation, and for other tasks of execution. The research question, the model, the theorems and their proofs, the empirical design, the reading of the results and all remaining errors are the author's own.

1 Introduction

This paper starts with a simple question: Would two economies that look identical in standard survey marginals respond identically to the same policy? Not necessarily. Consider, for example, two economies that report the same share of hand-to-mouth households and the same distribution of expectations: in both, half the population updates like a rational forecaster and half discounts news about the aggregate by the same amount. What the survey sees are the same marginals in both cases. However, it does not show how these characteristics are distributed jointly. So, the attentive forecasters may be savers in one economy and households living off current income in the other, and that difference alone could change the response to a monetary shock. Balance sheets determine who can respond to a change in income (Kaplan et al., 2014; Auclert, 2019), while expectations determine how households respond when their information is incomplete (Coibion and Gorodnichenko, 2015; Gabaix, 2020). What matters for the aggregate, then, is also how these two characteristics line up across households.[1] The two marginals leave that joint allocation open. We call this allocation the assignment.

Knowing how common each expectations mechanism is need not be enough for policy analysis when we do not know which households hold each one. Even if the prevalence and behaviour of every mechanism are known exactly, the macroeconomic counterfactual may still fail to be point identified. What remains unknown is how these mechanisms are assigned across households with different economic responses. Throughout the paper, we summarize this heterogeneity with responsiveness, the loading that determines how strongly expectations affect spending. Responsiveness can follow an arbitrary distribution in most of the analysis. Finite-support distributions and the binary rational-or-alternative case then appear as special cases.

The first result identifies when a problem over the full joint distribution can be reduced to a single statistic. Constrained households consume current resources, so responsiveness does not affect their choices. Among unconstrained households, expectations heterogeneity enters the forward-looking decision only through a scalar loading, which we call responsiveness. This definition does not impose linearity in any particular survey measure. Resources and transition-saving are otherwise independent of responsiveness. Hence, for equilibrium, an assignment is summarized by the constrained share and the responsiveness of unconstrained households. Any two assignments with the same values of these objects yield the same equilibrium.

If resources or transitions depend on responsiveness through another channel, this one-statistic reduction can fail. Conditional on the responsiveness distribution and the constrained share, the assignment statistic has sharp rearrangement bounds. Its attainable values form a closed interval whose endpoints are quantile integrals generated by the two extreme couplings.[2] The bounds keep a closed form when only the mean and the support of responsiveness are known, and a two-point distribution at the ends of the support attains them. That distribution therefore generates the

[1] This relationship has been documented by Broer et al. (2026) jointly for expectations and wealth.

[2] Atoms are allowed and continuity is unnecessary. The bounding arguments follow the classical results in (Fréchet, 1951; Hoeffding, 1940; Cambanis et al., 1976; Manski, 2003).

most assignment ambiguity among all distributions with the same mean and support.

General equilibrium carries its own identification problem, which must be solved jointly with the admissible values of the assignment statistic, and that creates a fixed-point problem. When household composition responds to the assignment, the constrained share at which the rearrangement bounds are evaluated is itself an equilibrium outcome. Computing the bounds once at a frozen composition is not a conservative shortcut, because it can admit assignments that no equilibrium supports and exclude assignments that are feasible. This paper develops the conditions under which the equilibrium-consistent feasible set is an interval and the aggregate is bounded by extremal feasible assignments. We show that in the reference economy used for the quantitative illustration, the proportional income incidence fixes composition across assignments, so the frozen and equilibrium-consistent calculations coincide exactly.

In the specified heterogeneous-agent economy, an accounting identity shows how the assignment of responsiveness shapes the aggregate response. For unconstrained households, saving is income minus consumption; combined with goods-market clearing and stationarity, this makes the effect of responsiveness on consumption offset its effect on saving inside the equilibrium comparative static. Under the model's forward-looking loading, responsiveness located among unconstrained households raises the absolute aggregate response, whereas responsiveness located among constrained households lowers it. Policy therefore transmits least effectively when the households most responsive to aggregate news are also the households least able to act on that response.

How much is left undetermined need not be small. In the reference economy the identified set for the aggregate spans 25.93 percent of its midpoint. Holding mean responsiveness and every other primitive fixed, spreading the two-point distribution into a uniform distribution over the same support narrows the span to 17.28 percent, while giving every household the same responsiveness eliminates assignment uncertainty altogether. Among these three marginals, the two-point rational-or-alternative case produces the widest set. The mean-and-support bound extends the comparison, since no responsiveness distribution with that mean and support produces a wider one. A natural default is to impute independence between economic state and responsiveness. This places the aggregate near the middle of the identified set even though the true response can lie almost fifteen percent away. At proportional income incidence, the independence imputation also makes the constrained share cancel from the aggregate response. The imputed model therefore does more than misstate a magnitude: it removes a transmission channel that is present under other admissible assignments.

The same distinction determines which data are informative. Average responsiveness by income or liquidity group does not identify the assignment, even with an arbitrarily large sample. Such averages restrict the joint distribution of responsiveness and the observed group, whereas the assignment statistic depends on the joint distribution of responsiveness and the economic state. Without a restriction linking observed groups to economic states, the relationship between group membership and state can vary while leaving the observed group averages unchanged. Those averages therefore do not restrict the assignment statistic. By contrast, when the classification

refines the state partition, group shares and group averages point identify the statistic. A repeated cross-section is then sufficient because identification requires only group averages.

The empirical analysis asks how much an actual household survey recovers the theoretical statistic. We use the European Central Bank's Consumer Expectations Survey because it records expectations and spending for the same household in the same month and allows us to construct a predetermined liquidity proxy for the economic state and cross it with income quintiles. Under external macroeconomic signals, the data reveal no detectable cross-cell heterogeneity in either expectations or spending. Inverting the test on the projected spending contrast produces an upper bound on the observable contrast. That contrast, however, is not the covariance required by the theory. Mapping it into the model requires one exclusion restriction and four mappings: 1) from reported spending to consumption, 2) from observed signals to the news entering the loading, 3) from reported liquidity to the structural state, and 4) from the estimation sample to the model population. The calculation conditions on them. Whether the observable bound restricts the aggregate then depends on how the model maps responsiveness into spending responses, and the reference economy shows how this last step can fail. With proportional income incidence, and a counterfactual in which the joint law of state and responsiveness moves only with composition, the observable spending contrast is zero under every admissible assignment even though the aggregate response still varies with the assignment. In this benchmark, the survey contrast does not vary with the assignment dimension that drives the aggregate response. Here the observable contrast has insufficient rank to recover the relevant assignment variation, regardless of statistical precision. Away from proportional income incidence, the contrast becomes sensitive to the assignment again, and depending on the maintained bridge scale the data can reject the maintained combination, partially restrict the assignment, or leave the identified set unchanged.

The closest macroeconomic literature establishes that the pairing between behavioural sensitivity and economic exposure can affect aggregate outcomes. Guerreiro (2023) summarizes the interaction between cyclical exposure and beliefs using a correlated-disagreement object. Patterson (2023) expresses the aggregate marginal propensity to consume as an average plus a covariance with individual income sensitivity, imputes the pairing from demographics, and finds that it raises the aggregate propensity out of labour income by six percentage points. Luetticke et al. (2026) pair elicited expectations with model-implied responses and find that the covariance amplifies the aggregate consumption decline by about fifty-six percent. Bilbiie et al. (2025) observe consumption, income, and wealth for an entire population and find a covariance close to zero. Macaulay (2026) characterizes the general-equilibrium effect of heterogeneous expectations through a small set of cross-sectional statistics under a specified process for expectation formation. Other work determines the pairing inside a fully specified environment (Broer et al., 2026; Pfäuti and Seyrich, 2026), and the interaction between belief heterogeneity and the wealth distribution appears more broadly in Bigio et al. (2025) and Datta and Sethi (2023).

A related identification literature studies joint distributions known only through their marginals. Franguridi and Liu (2025) characterizes sharp identified sets for parameters defined by moments of

such a distribution and develops inference for them. Liu et al. (2026) estimates a heterogeneous-agent block using the fact that individual decisions reach the aggregate through a finite vector of statistics. Luo et al. (2026) bounds an unknown bilateral coupling in a production network without closing the assignment through equilibrium.

This paper studies the identification problem left when the pairing is unknown. With a rank-one responsiveness channel, the joint distribution enters through a single assignment statistic. At higher rank, several statistics may be needed, although behaviorally redundant directions can reduce the dimension relevant for equilibrium. The paper characterizes the identified set for these objects and allows the state distribution to respond endogenously to the candidate assignment. The result is an equilibrium-consistent identification problem in which low dimension does not imply point identification.

Sections 2 to 4 develop the assignment problem, embed it in general equilibrium, and derive the restrictions from the household-level model. Section 5 studies how survey data restrict the assignment statistic. Section 6 concludes. Appendix E develops inference for the Consumer Expectations Survey exercise. Appendix F reports the dated-news application using the Consumer Finance Monthly survey.

2 The assignment problem

The identification problem is posed here for an arbitrary distribution of expectations responsiveness. The two-type economy that motivates it appears later as a special case.

A unit mass of households is indexed by two characteristics. The first is an economic state s, which we take to be binary, $s \in \{C, S\}$, where C denotes a household whose consumption tracks current resources and S denotes a household on an intertemporal margin. The second is an expectations responsiveness $\rho \in [0, \bar{\rho}]$ with $\bar{\rho} < \infty$, which scales how strongly the household incorporates news about the aggregate into its decision. It is measured in loading units and normalized so that rational expectations correspond to $\rho = 1$, with values below one representing underreaction and values above one overreaction.[3]

The population is described by a joint law $\mathbb{P}(s, \rho)$ with marginals

$$h \;=\; \mathbb{P}(s = C), \qquad F_\rho \;=\; \text{the law of } \rho, \tag{1}$$

and we write Q_ρ for the quantile function of F_ρ. The sharp bounds of Theorem 2 need only $\mathbb{E}[\rho] < \infty$. Boundedness is used where it is invoked, in (6) and in Theorem 3, and it cannot be dropped there: under F_ρ exponential with unit mean, $Q_\rho(u) = -\log(1-u)$ is unbounded and (7) gives $\overline{M}'(h) = \log(1-h)$, which diverges as the constrained share approaches one, and $\underline{M}'(h) = \log h$, which diverges as it approaches zero. We refer to $\mathbb{P}(s, \rho)$ as the *assignment*. The two marginals do not determine it. Any law with those marginals is admissible, and the set of such

[3]The normalization is a choice of units. Since ρ is the object through which expectations reach behaviour, two households whose beliefs differ in a way that leaves this loading unchanged are, for the aggregate, the same household.

laws is the transportation polytope of couplings of F_ρ with a Bernoulli marginal.

Definition 1 (Assignment)**.** An assignment is a joint law $\mathbb{P}$ on $\{C, S\} \times [0, \bar{\rho}]$ whose marginals are h and F_ρ. The identified set for any aggregate is its range over admissible assignments, holding the marginals fixed.

This section grants that h and F_ρ are known exactly, so what it establishes is conditional identification at a known pair of marginals. The aggregate can fail to be identified even then. From Section 3 onward h is an equilibrium outcome and is no longer something to condition on.

Let $y(s, \rho)$ denote the household-level object that aggregates, and suppose for the moment that it does not depend on the aggregate itself. The aggregate is $\mathbb{E}_{\mathbb{P}}[y(s, \rho)]$, and its dependence on the assignment is governed by the cross-difference of y in its two arguments. Write $d(\rho) = y(C, \rho) - y(S, \rho)$ for that cross-difference. Since $\mathbb{E}_{\mathbb{P}}[y] = \mathbb{E}[y(S, \rho)] + \mathbb{E}_{\mathbb{P}}[d(\rho)\mathbf{1}\{s = C\}]$, the coupling reaches the aggregate only through the second term. If y is additively separable, so that $y(s, \rho) = a(s)+b(\rho)$, then d is constant, $\mathbb{E}_{\mathbb{P}}[y] = \mathbb{E}[a] + \mathbb{E}[b]$ depends on the two marginals alone and the assignment is irrelevant. If d is non-constant on a set of positive F_ρ measure, the extreme couplings give different values and the aggregate moves.

Remark 1 (When statistical matching is harmless). With $h \in (0, 1)$, imputing a joint distribution by matching on the two marginals leaves an aggregate unaffected if and only if $d(\rho)$ is constant F_ρ-almost surely. Additive separability is the leading case of that, and it is a property of the response function that can be checked without observing the joint. The necessity half holds almost surely and can fail pointwise, since a response that fails separability only on an F_ρ-null set is invisible to every admissible coupling: with F_ρ uniform and y differing across states only at a single value of ρ, no coupling can see it. And if either marginal is degenerate the coupling is pinned and matching is harmless however non-separable the response is, which is Proposition 1 read the other way. Imputation of one joint is common practice, and the condition says when that practice is safe for the aggregate at hand.

Nonseparability alone is not enough. If either marginal is degenerate the coupling is pinned and the aggregate is point identified however non-separable the response is. With $h \in \{0, 1\}$ there is only one economic state present and the joint is determined. With F_ρ degenerate at a single value every admissible assignment delivers the same aggregate.

Proposition 1 (When the assignment matters)**.** *The interval of Theorem 2 has positive width if and only if $h \in (0, 1)$ and F_ρ is non-degenerate. Holding composition fixed, set identification of the aggregate therefore requires both a non-separable response and two non-degenerate marginals, and neither condition implies the other. Remark 3 shows that once composition responds to the assignment, a degenerate responsiveness marginal can still leave the identified set for the aggregate with positive width.*

Proof. The width is $\overline{M}(h) - \underline{M}(h) = \int_h^1 Q_\rho - \int_0^{1-h} Q_\rho$. At $h = 0$ both integrals equal $\int_0^1 Q_\rho$ and at $h = 1$ both are zero, so the width vanishes. For $h \in (0, 1)$ the difference is zero exactly when Q_ρ is

constant almost everywhere on $[0,1]$, which is degeneracy of F_ρ. Conversely a non-degenerate F_ρ with $h \in (0,1)$ gives a strictly positive difference because the comonotone and antitone couplings place distinct quantile ranges on the unconstrained. □

Outside the degenerate cases the statistic is set identified. Whether the aggregate is set identified too depends on the response as well, since a separable one is invariant to the coupling however wide the interval for the statistic is. How wide the set becomes once the aggregate is an equilibrium object, and no longer an expectation, is the subject of what follows.

The heterogeneous-agent economy below has two features that are standard in that literature. Constrained households consume current income and carry no forward-looking term, so responsiveness does nothing for them. Unconstrained households are forward looking, and expectations heterogeneity reaches their decision through a single scalar loading. Neither feature requires the response to be linear in any particular index of expectations. Since the assignment can only act through that loading, nothing is lost by taking the responsiveness variable to be the loading itself, and we do so throughout. Under these two features the aggregate depends on the assignment only through a single number.

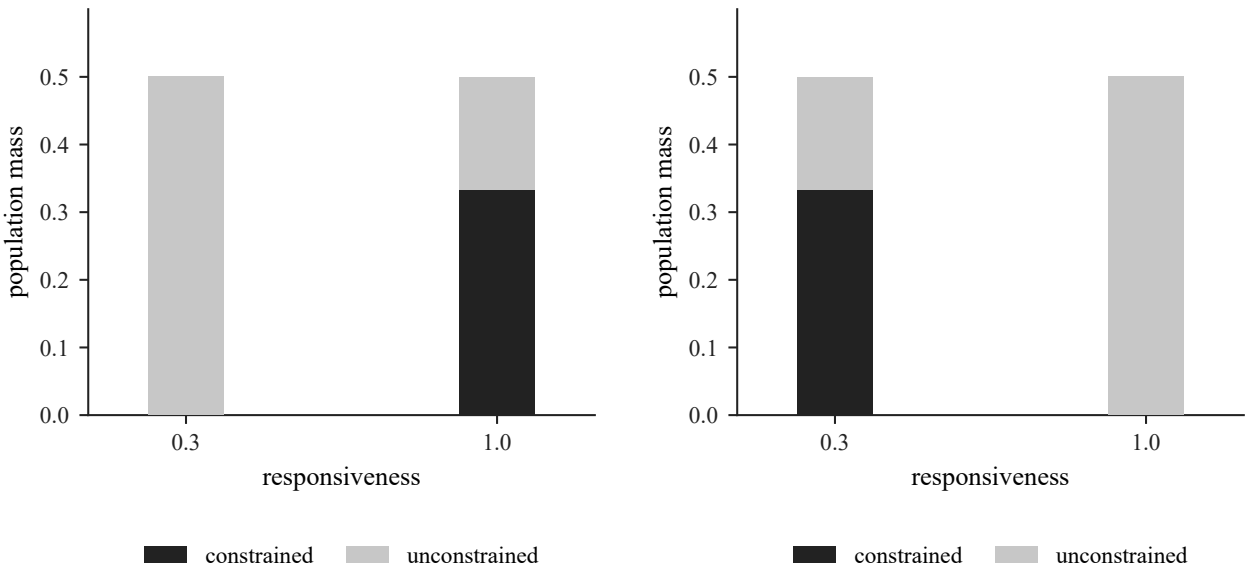
Definition 2 (Assignment statistic)**.**

$$M_S \;=\; \mathbb{E}_{\mathbb{P}}[\rho\,\mathbf{1}\{s = S\}]\,, \tag{2}$$

the expectations responsiveness located among unconstrained households.

M_S is not a normalized average. It carries the mass of the unconstrained as well as their responsiveness, and it is the product $\mathbb{P}(s = S)\,\mathbb{E}[\rho \mid s = S]$. Under independence of the two characteristics it equals $(1-h)\,\mathbb{E}[\rho]$, which is the benchmark against which sorting is measured.

Figure 1 shows two populations that agree on the distribution of responsiveness and on the share of households that are constrained, and differ only in which households are which. No survey reporting either distribution can tell them apart. The aggregate is not the same in the two.



(a) Responsiveness among the constrained

(b) Responsiveness among the unconstrained

Figure 1: Same marginals, different assignments

Notes. Two responsiveness types with mass one half each and a constrained share of 0.3333. The bars have the same heights in both panels because the responsiveness distribution is the same, and the dark area is the same size because the constrained share is. Only its placement differs. In panel (a) every constrained household is of the responsive type and in panel (b) none is. The assignment statistic is 0.317 in (a) and 0.550 in (b), which is the whole of the difference between the two economies. They are the extremal assignments of the economy Section 4 solves, and the aggregate responses they deliver, 0.02623 and 0.03404, are the endpoints of an identified set spanning 25.93 percent of its own midpoint.

Theorem 1 (Sufficient assignment statistic)**.** *Suppose constrained households consume ξY, with no dependence on ρ, and unconstrained households consume*

$$c_S(\rho) \;=\; A(Y,r) \;+\; B(Y,r)\,\rho, \tag{3}$$

that unconstrained households have current resources Y, with no dependence on ρ, so that saving per unconstrained household is $Y - c_S(\rho)$, and that the distributional state evolves through a hazard depending on the average of that saving. Then the joint law $\mathbb{P}(s,\rho)$ enters the equilibrium system only through the pair (h, M_S), for any functions A and B. Two assignments with the same h and the same M_S induce the same equilibrium, whatever else distinguishes them.

Proof. Aggregate consumption is

$$C \;=\; h\,\xi Y \;+\; \int_{s=S} (A + B\,\rho)\,d\mathbb{P} \;=\; h\,\xi Y \;+\; (1-h)\,A \;+\; B\,M_S,$$

where the second equality uses only that the loading enters once and holds for every h, including $h = 1$, where $M_S = 0$ and the last term drops. For $h < 1$ average saving per unconstrained household is $Y - A - B\,M_S/(1-h)$ by the resource condition, again a function of (Y, h, M_S) alone, so both blocks of the equilibrium system depend on $\mathbb{P}$ only through (h, M_S). At $h = 1$ that

average is not defined, and nothing is lost: the state marginal is degenerate, the only admissible assignment places every household in C, and the conclusion holds trivially. Any further feature of the assignment, including higher moments of ρ among the unconstrained and the shape of F_ρ within each state, is invisible to the equilibrium. □

The resource condition is necessary. Restricting consumption alone leaves the saving side free, and a hazard reading a saving object that carries its own dependence on responsiveness can separate assignments that agree on (h, M_S). Let F_ρ put mass $1/3$ on each of $\{0, 1, 2\}$ and take $h = 1/2$. The constrained masses $(1/6, 1/6, 1/6)$ and $(1/4, 0, 1/4)$ are both admissible and both give $M_S = 1/2$, while $\mathbb{E}\big[\rho^2 \mathbf{1}\{s = S\}\big]$ is $5/6$ under the first and $2/3$ under the second. If unconstrained resources were $Y + \rho^2$ instead of Y, a saving-dependent hazard would tell the two apart. The economy of Section 4 satisfies the condition, since (24) defines the saving object as a function of Y, h and M_S.

Theorem 1 restricts nothing about how behaviour depends on an index of expectations a survey might report. Responsiveness names the loading itself, so the affine form is a definition. What the theorem requires is that expectations heterogeneity enter the unconstrained decision through one scalar loading, with A and B otherwise unrestricted and possibly nonlinear in the aggregate. Whenever heterogeneity has that rank-one form, the entire dependence of the equilibrium on the matching between economic states and expectations collapses to one moment of the assignment.

Remark 2 (The relevant dimension is a rank). The scalar-loading condition is a rank-one restriction, and its failure is graded. If unconstrained consumption takes the finite-rank form

$$c_S(\rho) \;=\; A(Y, r) \;+\; \sum_{j=1}^{J} B_j(Y, r)\, T_j(\rho), \tag{4}$$

the same aggregation step shows that the joint law enters only through the constrained share and the vector $(M_1, \ldots, M_J)$ with $M_j = \mathbb{E}[T_j(\rho)\mathbf{1}\{s = S\}]$, under the same resource condition, so that the saving entering the hazard is a function of the aggregate, the constrained share and that vector. What has to be bounded is therefore not the whole joint distribution. Its relevant dimension is the rank with which expectations heterogeneity enters behaviour. At $J = 1$ this is Theorem 1 and the identified set for the statistic is an interval. At $J > 1$ the admissible values form a convex body, the image of the coupling set under J linear functionals, whose dimension is at most J and can be strictly less. Where F_ρ has finite support Proposition 2 gives that dimension exactly, as the rank of the loading-difference matrix, and the body is an interval whenever that rank is one, which happens for any collection of loadings that are affine functions of one another on the support. For a general F_ρ the matrix is not defined and the claim is the inequality alone.

That rank is the dimension of the identified set for the vector. It bounds what the equilibrium needs without always equalling it, because the coefficients in (4) can leave a direction of the vector behaviourally redundant: two loadings entering through the same coefficient leave the equilibrium reading their sum and nothing else, at any rank. Writing $\mathcal{B}$ for the span of $(B_1, \ldots, B_J)$ over the domain the experiment covers, what the equilibrium requires is the dimension of the assignment

variation after projection onto $\mathcal{B}$. It equals the rank when $\mathcal{B}$ annihilates no direction the loading differences span, for which linear independence of the B_j as functions of (Y, r) is sufficient, and is smaller otherwise. The equilibrium analysis has to be redone in the vector only when that projected dimension exceeds one. Section 4.1 does that. The results between here and there are stated for $J = 1$.

How much the reduction achieves can be settled without the equilibrium analysis at $J > 1$. It is a question about identification alone, and it depends on how many types the responsiveness distribution has.

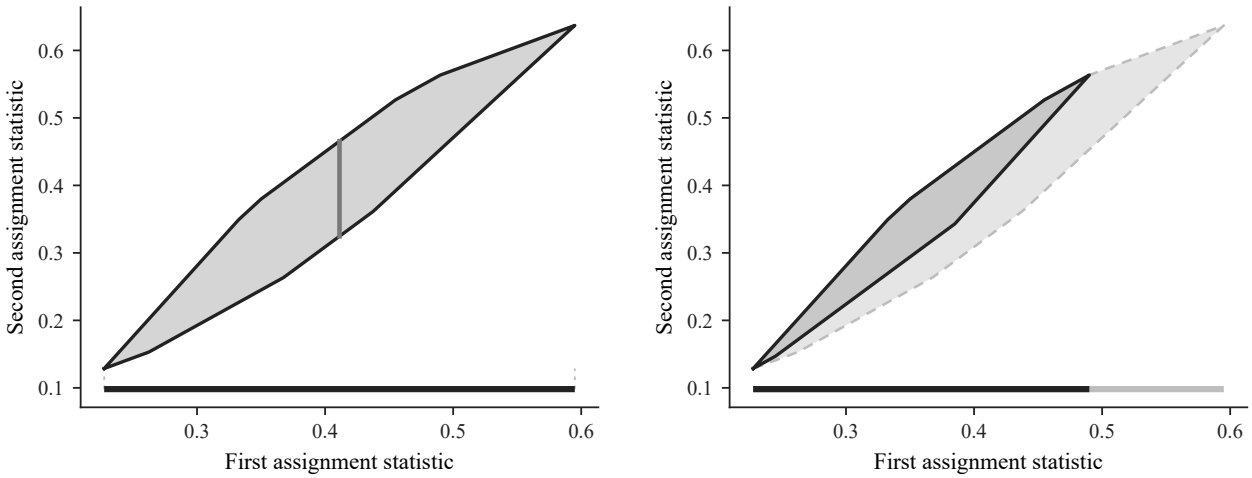
Proposition 2 (Saturation)**.** *Let F_ρ be supported on K points $\rho_1, \ldots, \rho_K$ with masses $f_i > 0$, and let $h \in (0, 1)$. The admissible couplings are $\{(c_1, \ldots, c_K) : 0 \le c_i \le f_i,\ \sum_i c_i = h\}$, where c_i is the constrained mass of type i, and they form a polytope of dimension $K - 1$. Suppose the assignment enters through $(M_1, \ldots, M_J)$ as in (4). Then the identified set for that vector has dimension equal to the rank of the $J \times (K - 1)$ matrix with entries $T_j(\rho_i) - T_j(\rho_K)$, and the couplings consistent with a given value of the vector form a set of dimension at most $K - 1$ less that rank, with equality at every value in the relative interior of the identified set.*

Proof. Taking $c_i = hf_i$ gives a point with $0 < c_i < f_i$ for every i, so the polytope has an interior point in the hyperplane $\sum_i c_i = h$ and its dimension is $K - 1$. Each statistic is affine in c, since $M_j = \sum_i T_j(\rho_i) f_i - \sum_i T_j(\rho_i) c_i$. Eliminating c_K through the adding-up constraint leaves the stated matrix as the linear part of the map. An affine map whose linear part has rank k carries a polytope of dimension $K-1$ onto a set of dimension k, and its fibres have dimension at most $K-1-k$, with equality above the relative interior of the image. The inequality can be strict on the boundary. Under $(c_1, c_2) \mapsto c_1 + c_2$ the square $[0, 1]^2$ has one-dimensional fibres above the interior of the image and single points above each of its two endpoints. □

The rank obeys two bounds. It cannot exceed J, and it cannot exceed $K - 1$, because functions on K points span a space of dimension K and the constant already occupies one direction in it. So no collection of loadings, however large, pushes the rank past $K - 1$, and at that ceiling the vector of statistics determines the coupling and compresses nothing, being another way of writing the joint law. That ceiling is a statement about finite support and has no automatic counterpart when F_ρ has infinite support. The coupling is then infinite dimensional and no finite collection of statistics identifies it generically, though particular values can still pin it: for the baseline loading $T(\rho) = \rho$ with F_ρ continuous, the rearrangement extremizers are almost surely unique.

The bound also disciplines what counts as a second loading. Adding T_2 raises the rank only if it is not an affine function of T_1 on the support, and that condition is exact because rank is discrete. Two loadings close to affine transforms of one another still give rank two, and the identified set still has the dimension that rank supplies: on $\{0, 1, 2\}$ with equal masses, $T_1(\rho) = \rho$ and $T_2(\rho) = 2\rho + \varepsilon\rho^2$ make the matrix of Proposition 2 have determinant 2ε, so the rank is two for every $\varepsilon \neq 0$ however small. Near-affinity reduces width and leaves the dimension unchanged. The second direction is close to redundant, the identified body is thin in it, and a nominal $J = 2$ then adds little to $J = 1$.

Theorem 1 is of a familiar kind. Aggregate responses in heterogeneous-agent economies are governed by a small number of covariances between household sensitivities and balance-sheet exposures, and those covariances are robust to the model generating them (Auclert, 2019). That program then estimates the object. The question here is what is identified when it cannot be estimated, and that calls for its range. That range is an interval under one loading, and under several a body of dimension at most the number of loadings, exactly the rank of the loading-difference matrix where the responsiveness marginal has finite support, and what the sufficient statistics leave undetermined shrinks as restrictions are added (Figure 2).



(a) Two loadings instead of one

(b) With one further restriction

Figure 2: Rank of the loading and what the statistics fix

Notes. 5 responsiveness types and two loadings, the level and its square. Responsiveness takes the values $0, 0.35, 0.7, 1.05, 1.4$ with masses $0.15, 0.25, 0.25, 0.2, 0.15$. With one loading the identified set is the interval drawn as the heavy bar, $[0.2275, 0.5950]$. With two it is the shaded region, and the interval is only its shadow. Knowing the first statistic does not pin the second: at the midpoint of the first, the vertical segment shows the second still ranging over $[0.3246, 0.4655]$, which is 28 percent of its full range. Panel (b) imposes one linear restriction, with the unrestricted region left as a dashed outline. The first statistic contracts to an upper end of 0.4900 and the area of the region falls by 59.2 percent. Both regions are exact, being convex hulls of the images of the polytope's vertices under a linear map.

Theorem 2 (Sharp conditional identification)**.** *Fix F_ρ and $\mathbb{P}(s = S) = 1 - h$ with $h \in (0, 1)$. The set of values of M_S attainable by an admissible assignment is exactly the closed interval*

$$\left[\, \underline{M}(h),\ \overline{M}(h) \,\right], \qquad \underline{M}(h) = \int_0^{1-h} Q_\rho(u)\, du, \qquad \overline{M}(h) = \int_h^1 Q_\rho(u)\, du, \tag{5}$$

every intermediate value is attained, and no assumption on F_ρ beyond a finite mean is required. Atoms are permitted.

Proof. Write $M_S = \mathbb{E}[\rho X]$ with $X = \mathbf{1}\{s = S\}$ a Bernoulli variable with $\mathbb{E}[X] = 1 - h$, the joint being free subject to that marginal. Maximizing and minimizing $\mathbb{E}[\rho X]$ over couplings of F_ρ with a fixed Bernoulli marginal is the classical rearrangement problem, and for the supermodular function $(x, y) \mapsto xy$ the supremum is attained by the comonotone coupling and the infimum by the antitone

one (Hoeffding, 1940; Fréchet, 1951; Cambanis et al., 1976). Since X takes two values, comonotone means placing $X = 1$ on the largest $1 - h$ of the responsiveness distribution, which contributes $\int_h^1 Q_\rho$, and antitone means placing it on the smallest, which contributes $\int_0^{1-h} Q_\rho$.

If F_ρ has an atom straddling the quantile level h, no deterministic cutoff rule has the required marginal. The coupling formulation does not need one, because a joint law on $\{C, S\} \times \mathbb{R}_+$ may split an atom, assigning part of the mass at that value to S and the rest to C. That is the vertex of the transportation polytope which the quantile integral computes, and it is why (5) is written through Q_ρ instead of through a cutoff of F_ρ.

For attainment of intermediate values, let $\mathbb{P}_0$ attain $\underline{M}(h)$ and $\mathbb{P}_1$ attain $\overline{M}(h)$. For $t \in [0, 1]$ the mixture $\mathbb{P}_t = (1 - t)\mathbb{P}_0 + t\,\mathbb{P}_1$ has the same two marginals, because marginals are linear in the joint, and $M_S(\mathbb{P}_t) = (1 - t)\underline{M}(h) + t\,\overline{M}(h)$ is continuous and onto. □

Two consequences are used repeatedly. Since Q_ρ is bounded above by $\bar{\rho}$, both endpoints are Lipschitz in the constrained share,

$$|\underline{M}(h_1) - \underline{M}(h_2)| \;\le\; \bar{\rho}\,|h_1 - h_2|\,, \qquad \left|\overline{M}(h_1) - \overline{M}(h_2)\right| \;\le\; \bar{\rho}\,|h_1 - h_2|\,, \tag{6}$$

hence absolutely continuous and differentiable almost everywhere, with

$$\underline{M}'(h) \;=\; -\,Q_\rho(1 - h), \qquad \overline{M}'(h) \;=\; -\,Q_\rho(h) \tag{7}$$

at every h at which Q_ρ is continuous at the relevant argument, which is all but countably many. The exceptions are the levels at which Q_ρ jumps, which occur exactly where the support of F_ρ has a gap. The two-type benchmark used throughout has a gap between its two atoms, so its lower bound has a kink at one value of the constrained share.[4] Both derivatives are non-positive where they exist, and $\underline{M}(h) \le \overline{M}(h)$. The width of the interval is a function of the two marginals alone, and by Proposition 1 it is positive exactly when $h \in (0, 1)$ and F_ρ is non-degenerate. Homogeneous responsiveness is one way to lose it and a degenerate state marginal is the other: at $h = 0$ both endpoints are the whole quantile integral and at $h = 1$ both are zero, whatever F_ρ is.

Example 1 (The two-type case)**.** Let F_ρ place mass q at $\rho = 1$ and mass $1 - q$ at $\rho = \mu$ with $\mu \in [0, 1)$, which is the rational-or-alternative benchmark. Writing $z = \mathbb{P}(s = C, \rho = 1)$ for the mass of constrained households using the responsive mechanism,

$$M_S \;=\; (q - z) \;+\; \mu\,(1 - h - q + z), \qquad \frac{dM_S}{dz} \;=\; -(1 - \mu) \;<\; 0. \tag{8}$$

The coupling has a single degree of freedom, z, and the interval in Theorem 2 reduces to the Fréchet–Hoeffding interval for a two-by-two table. Sorting responsiveness toward the constrained raises z and lowers M_S.

[4] Atoms of F_ρ produce flat stretches of Q_ρ and cause no difficulty by themselves. Gaps in the support make Q_ρ jump and break pointwise differentiability where atoms do not. This is why the results below are stated with Lipschitz bound functions in place of differentiable ones.

Example 1 connects to the rational-expectations question and serves as the running illustration.

Theorem 2 uses the whole responsiveness distribution. Evidence often delivers less, a mean and a range, and the bounds survive that loss in closed form.

Corollary 1 (Mean and support only)**.** *Suppose all that is known about responsiveness is that* $\rho \in [a, b]$ *with* $0 \le a < b$ *and* $\mathbb{E}[\rho] = m \in [a, b]$*, and let* $\mathbb{P}(s = C) = h \in (0, 1)$*. The values of* M_S *attainable by some responsiveness distribution with that support and mean, coupled in some way with the state marginal, form the closed interval*

$$[\,\max\{(1-h)\,a,\ m - h\,b\},\quad \min\{(1-h)\,b,\ m - h\,a\}\,]\,. \tag{9}$$

For every distribution F_ρ *with that support and mean, the interval of Theorem 2 lies inside (9). Both endpoints are attained by the two-point distribution that places mass* $p = (m-a)/(b-a)$ *at* b *and the rest at* a*.*

Proof. For any such F_ρ the quantile function takes values in $[a, b]$ and integrates to m. Hence $\underline{M}(h) = \int_0^{1-h} Q_\rho \ge (1-h)\,a$, and also $\underline{M}(h) = m - \int_{1-h}^1 Q_\rho \ge m - h\,b$. In the same way $\overline{M}(h) = \int_h^1 Q_\rho \le (1-h)\,b$ and $\overline{M}(h) = m - \int_0^h Q_\rho \le m - h\,a$. For the two-point distribution $Q_\rho = a$ on $[0, 1-p)$ and $Q_\rho = b$ on $[1-p, 1]$. If $h \ge p$ the lower quantile integral is $(1-h)\,a$, and $(m - hb) - (1-h)a = (p-h)(b-a) \le 0$, so it equals the maximum in (9). If $h < p$ it is $(1-p)\,a + (p-h)\,b = m - hb$, and $(1-h)a - (m-hb) = (h-p)(b-a) < 0$, so again it equals the maximum. The upper endpoint follows by the symmetric argument. Every intermediate value is attained by Theorem 2 applied to the two-point distribution. □

The corollary is an elementary consequence of the rearrangement bounds and is not a new mathematical result. What it adds is a statement about the marginal. Among all responsiveness distributions with a given mean and support, the two-point law at the ends of the support generates the most assignment ambiguity, and Example 1 is that law whenever $\mu = a$, $b = 1$ and $q = (m-\mu)/(1-\mu)$. The framework therefore runs on a mean and a range alone, at the cost of reporting the envelope.

The bounds so far take the two marginal distributions as given. Evidence rarely delivers a household-level responsiveness, and what it does deliver are restrictions on the joint distribution. Which restrictions the framework can use decides which data are informative. Suppose households are also observed in groups g with shares π_g, where the grouping is something a survey records, such as an income or liquidity category. The restrictions that data of this kind produce are linear in the joint law: the responsiveness marginal, the state marginal, the group shares, group-average responsiveness $\mathbb{E}[\rho \mid g] = m_g$, and where a second source supplies it, how each group divides across economic states, $\mathbb{P}(s = C \mid g)$.

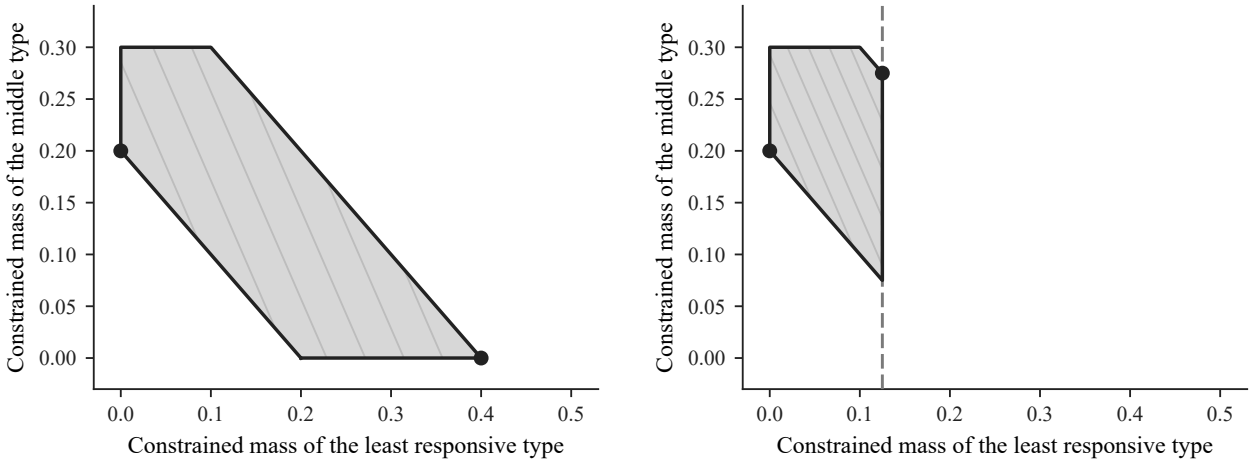
Theorem 3 (Sharp bounds under moment restrictions)**.** *Let* $\mathcal{G}$ *be finite. Let the admissible joints be those laws on* $\{C, S\} \times [0, \bar\rho] \times \mathcal{G}$ *satisfying any finite collection of linear restrictions of the kind listed above, and suppose the restrictions are jointly satisfiable, so that at least one admissible law*

exists, and that $\pi_g > 0$ for every group whose conditional moment is restricted. The admissible laws then form a non-empty weakly compact convex set, and since M_S is a bounded linear functional its image is a compact interval whose endpoints solve two linear optimization problems over probability measures. When F_ρ has finite support these reduce to ordinary finite-dimensional linear programs. Adding restrictions shrinks the admissible set, so the identified sets for M_S are nested, and because the aggregate is a continuous function of the statistic on a regular branch its identified sets are nested by inclusion in the same order.

Proof. Every restriction listed is linear in the joint and the objective $M_S = \mathbb{E}[\rho\,\mathbf{1}\{s = S\}]$ is linear and bounded on the stated support. The admissible set is an intersection of the space of probability measures on a compact product space with finitely many closed half-spaces and hyperplanes, hence weakly compact and convex, and non-empty by hypothesis, so the image of a bounded linear functional is a compact interval and the endpoints are attained. Satisfiability has to be assumed. Restrictions of the listed kinds can conflict: with a single group carrying the whole population, $F_\rho = \delta_0$ forces $\mathbb{E}[\rho \mid g] = 0$, so pairing it with $\mathbb{E}[\rho \mid g] = 1$ admits no law at all and neither endpoint exists. Intermediate values are attained by mixing the two optimizers, which preserves every linear constraint. Nesting is immediate because adding constraints shrinks the admissible set, and it carries to the aggregate as inclusion of images, since a continuous map sends a subset into the image of the superset. Monotonicity is not used, and regularity would not supply it: with $H \equiv h_0$ and $G = 2 + M_S(1 - M_S)$ the branch $Y = 2 + M_S(1 - M_S)$ is unique, continuously differentiable and non-vanishing with $D = 1 > 0$, yet $Y' = 1 - 2M_S$ changes sign on $[0, 1]$. Monotonicity is a hypothesis of Theorem 7, not a consequence of Assumption 1. □

Theorem 2 is the special case with only the two marginals imposed.

The assignment problem is a transportation polytope, whose vertices are the extreme couplings (Figure 3). Which datasets are informative is settled by a negative result.



(a) The two marginals alone

(b) One further restriction

Figure 3: The assignment problem as a transportation polytope

Notes. Three responsiveness types, taking the values $0.3, 0.65, 1$ with masses $0.5, 0.3, 0.2$. A coupling is a point in the polygon that the two marginals cut out of a box, the axes give the constrained mass of the two lower types, and the third is residual. The statistic is linear on that polygon, so its level sets are the parallel lines and its sharp bounds sit at vertices, marked. In panel (a) the bounds are $[0.215, 0.425]$, which are the quantile integrals of Theorem 2. Panel (b) adds one linear restriction of the kind Theorem 3 allows, drawn as the dashed half-plane, and the identified set contracts to $[0.215, 0.329]$, a reduction of 45.8 percent.

Theorem 4 (Group-average responsiveness alone is uninformative)**.** *Add to the two marginals the group shares π_g and the group averages m_g, with no restriction linking the grouping to the economic state. Suppose the three are jointly satisfiable, with $\pi_g > 0$ for every group whose average is imposed. The identified set for M_S is unchanged and remains the interval of Theorem 2.*

Proof. One inclusion is immediate because the constraints are a superset. For the other, take any joint of (ρ, g) meeting F_ρ, the group shares and every group average, of which the truth is one. Define the conditional law of s given ρ, independently of g, by the comonotone coupling that places the top $1 - h$ of the responsiveness distribution into S, splitting the atom at the quantile if needed. This changes no marginal of (ρ, g), so every group-average constraint continues to hold, and it attains $\overline{M}(h)$. The antitone rule attains $\underline{M}(h)$, and mixtures fill the interval. Satisfiability cannot be dropped, for the same reason as in Theorem 3: $F_\rho = \delta_0$ with one group of unit share and $m_g = 1$ admits no law, so the augmented set is empty where Theorem 2 would give a singleton. It is automatic when the group moments are taken from a population, which is the case whenever the theorem is applied. □

The reason is structural. Group averages restrict the joint of responsiveness and group, while M_S depends on the joint of responsiveness and economic state. With nothing tying the grouping to the state, the group dimension can always be rearranged out of the way. A survey reporting average responsiveness by income group, however large and however precisely measured, therefore says nothing about the assignment on its own. What identifies M_S is a classification tied to the economic state.

Theorem 5 (When the grouping reveals the state)**.** *Suppose each group lies wholly inside one economic state, so that the observed partition refines the state partition. Then*

$$M_S \;=\; \sum_{g \subseteq S} \pi_g \, m_g, \tag{10}$$

which is point identified from group shares and group averages alone. No distributional object is required: neither F_ρ, nor the within-group laws, nor any household-level responsiveness.

Proof. Under the hypothesis $\mathbf{1}\{s = S\}$ is a function of g, so $\mathbb{E}[\rho\,\mathbf{1}\{s = S\}] = \sum_{g \subseteq S} \mathbb{E}[\rho\,\mathbf{1}\{g\}] = \sum_{g \subseteq S} \pi_g m_g$. □

Theorem 5 has a practical consequence. A repeated cross-section suffices, because a group average is estimable without following the same household over time.[5] It also gives a sufficient condition for the identification question this paper is about not to arise. The converse fails. By Proposition 1 a degenerate responsiveness marginal point identifies the statistic whatever the grouping looks like, so a classification can cut across the states and still leave nothing to bound.

Definition 3 (Scale-free sorting measures)**.** Assume $\mathbb{E}[\rho] > 0$, which rules out only the case of no responsiveness anywhere, and $\mathbb{P}(s = S) > 0$, equivalently $h < 1$, without which the conditional mean below is not defined. The statistic M_S itself needs neither: it is $\mathbb{E}[\rho\mathbf{1}\{s = S\}]$ and equals zero at $h = 1$. Absolute scale is absorbed into ψ, so normalize by $\bar{m} = \sum_g \pi_g m_g$ and define

$$\Lambda_S \;=\; \frac{M_S}{\mathbb{E}[\rho]} \;=\; \frac{\sum_{g \subseteq S} \pi_g m_g}{\bar{m}}, \qquad R_S \;=\; \frac{\mathbb{E}[\rho \mid s = S]}{\mathbb{E}[\rho]}. \tag{11}$$

Under independence of state and responsiveness $\Lambda_S = \mathbb{P}(s = S)$ and $R_S = 1$. Values of R_S below one place responsiveness disproportionately among constrained households, and values above one among the unconstrained.

3 General equilibrium and endogenous composition

The constrained share plays three roles in this paper. Section 2 holds it fixed and asks what the assignment leaves undetermined given it, so its results are conditional identification at a known marginal. From here on it is an equilibrium outcome, moving with the assignment, and it is not being conditioned on as something observed. Section 5 has it latent, observed only through a classification that misreads households in both directions.

The middle regime raises a question the other two do not. If composition is an equilibrium outcome and someone observes it, that observation says something about the assignment. Where $h_M \neq 0$ on a branch satisfying Assumption 1, h is strictly monotone in the statistic, so $h(M_S) = h^{\text{obs}}$ has at most one solution in $\mathcal{M}$ and an observed constrained share identifies the statistic exactly.

[5] A panel would be needed to recover a household-level responsiveness, which is a within-person object.

Nothing below uses that route, since the empirical section has no observed h. The same response that makes the feasible set a fixed point also makes composition an observable with content about the assignment. Where $h_M = 0$, as at $\xi = 1$, the map is constant and carries none.

The identification problem so far treats the aggregate as an expectation over a fixed response function. Two things change once the aggregate is an equilibrium object. The household response depends on the aggregate that households jointly produce, and the distribution of economic states may itself move with the assignment. Both bear on what remains identified.

Let the aggregate and the distributional state solve

$$Y \;=\; \mathcal{G}(Y, h, M_S), \qquad h \;=\; \mathcal{H}(Y, h, M_S), \tag{12}$$

where G is goods-market clearing and H is the law of motion of the constrained share, evaluated at a stationary distribution. Both depend on the assignment only through M_S, by Theorem 1. Differentiating and applying Cramer's rule,

$$\frac{dY}{dM_S} \;=\; \frac{N}{D}, \qquad N \;=\; \mathcal{G}_M\,(1 - \mathcal{H}_h) + \mathcal{G}_h\,\mathcal{H}_M, \qquad D \;=\; (1 - \mathcal{G}_Y)(1 - \mathcal{H}_h) - \mathcal{G}_h\,\mathcal{H}_Y. \tag{13}$$

Assumption 1 (Regular branch)**.** On a non-empty compact interval I of candidate values of M_S there is a unique, continuously differentiable equilibrium branch $M_S \mapsto (Y(M_S), h(M_S))$ with $D(M_S) > 0$ and $h(M_S) \in (0, 1)$, and $\mathcal{G}$ and $\mathcal{H}$ are continuously differentiable in a neighbourhood of that branch. Economies with two admissible branches are excluded.

The differentiability of the two maps is separate from the differentiability of the branch and does not follow from it. A branch can be unique and continuously differentiable with a positive determinant while a map has partial derivatives at a point without being totally differentiable there, and (13) is then wrong at that point, since the chain rule it applies does not hold. Beyond that, Assumption 1 is stated at this level of generality because the abstract system places no structure on $\mathcal{G}$ and $\mathcal{H}$. It remains an assumption in the heterogeneous-agent economy below. Proposition 6 reduces that economy to a quadratic and narrows the candidates, but it does not deliver existence, uniqueness or a positive determinant, as the degenerate cases recorded there show. That I is a compact interval is part of the assumption too: a disconnected domain would break the feasible set for a reason unrelated to the bounds, and without compactness the extrema below need not be attained.

The abstract condition for the aggregate to be monotone in the assignment is that the numerator of (13) keeps its sign.

Proposition 3 (Monotonicity and extremal assignments)**.** *Under Assumption 1, if N does not change sign on the feasible set, then Y is monotone in M_S on each connected component of that set. If in addition the feasible set is a non-empty compact interval, the sharp bounds on Y are attained at its two endpoints.*

The determinant is positive by Assumption 1, so the sign of the derivative is the sign of N and

no further condition is needed for the conclusion. Writing N as $\mathcal{G}_M(1-\mathcal{H}_h)+\mathcal{G}_h\mathcal{H}_M$ is useful for reading which channel carries the effect, and a positive $1-\mathcal{H}_h$ makes the first term share the sign of the direct channel, but neither is a hypothesis of the proposition. A constant sign of N orders values of the aggregate within a component and says nothing across a gap, because the aggregate can move the other way while the assignment statistic passes through an infeasible interval. Theorem 6 below supplies the interval, and compactness then follows because the feasible set is closed in the compact domain I.

Two features of Proposition 3 need stating.

The condition is on N itself, and its summands are unrestricted. Requiring $\mathcal{G}_M(1-\mathcal{H}_h)$ and $\mathcal{G}_h\mathcal{H}_M$ to agree in sign pointwise is neither necessary nor sufficient. Appendix B reports a numerical experiment over 32,782 economies. It is not sufficient: 33 of them have the summands agreeing at every value of the assignment while the sign of N still turns, and in 32 of those the extremal assignments fail to bound the identified set. In the worst of them N runs from -0.905 to $+0.802$ and the endpoints miss the extremum by 0.122 on an interior range of 0.154. Nor is it necessary, and the violations are not rare: 17,645 economies violate the pointwise requirement, keep the sign of N and bound correctly. Failures of sufficiency are the rare ones.

The relevant object is the feasible set, never an interval computed once. Once h responds to the assignment there is no fixed interval of admissible couplings determined before the equilibrium is solved, because the bounds of Theorem 2 are evaluated at $h(M_S)$.

Definition 4 (Feasible set for the assignment statistic)**.**

$$\mathcal{M} \;=\; \left\{M_S \in I \;:\; \underline{M}(h(M_S)) \;\leq\; M_S \;\leq\; \overline{M}(h(M_S))\right\}. \tag{14}$$

This is a fixed-point object. A value of the assignment statistic is admissible when the equilibrium composition it induces is one at which that value can be achieved by some coupling. Computing the rearrangement bounds once at an initial composition and treating the result as the admissible range is not a conservative approximation. It errs in both directions, admitting assignments no equilibrium supports and excluding assignments that are feasible. In the two-type economy, over 1,548 economies drawn wide, the fixed-point feasible set extends below the frozen-composition lower endpoint in 21.3 percent of cases and above its upper endpoint in 7.5 percent, and the construction moves the feasible set in 28.7 percent of economies.

Write $h_M = dh/dM_S$ for the response of composition to the assignment along the regular branch. One case needs no computation: where h_M vanishes the composition is the same at every admissible assignment, the bounds of Theorem 2 are evaluated at one value throughout, and (14) returns exactly the interval a frozen calculation returns. The two agree there as an identity.

Away from that case the two-type economy settles the comparison, because there the rearrangement bounds are linear in the composition. Proposition 8 in Appendix B settles what a frozen composition delivers when it is itself feasible, meaning it is the composition of some assignment the equilibrium admits. Such an anchor gives an outer set when $h_M > 0$, an inner set

when $h_M < 0$, and the feasible set itself when $h_M = 0$. So freezing is an inner rule or an outer rule according to the sign of the response being ignored, and an analyst who does not know that sign does not know which. An anchor chosen outside the compositions the feasible set generates is covered by neither case, and the same proposition shows that which interval is wider depends on the anchor, not on that sign, so width is not the comparison to make.

The difference is small in the reference economy. That economy sets $\xi = 1$, at which the composition equation loses its aggregate term, so h_M is zero, the frozen and equilibrium-consistent calculations coincide exactly, and the width of 25.93 percent carried into Section 5 owes nothing to Definition 4. Moving ξ away from one switches the feedback on, and Table 6 in Appendix D already does that with every other primitive fixed: the frozen and correct endpoints differ there by at most 0.0026, and the widths those endpoints deliver differ by at most 0.32 percentage points, of which the table prints the equilibrium-consistent one. The aggregate moves the constrained share by little next to the rates at which households cross the constraint, and Appendix B reports how far that response has to go before the difference stops being small.

Two extremal assignments describe the whole range only when $\mathcal{M}$ is an interval. A sufficient condition for that uses h_M and a bound on the support of responsiveness.

Theorem 6 (Connectedness of the feasible set)**.** *Let Assumption 1 hold and let ρ be supported on $[0, \bar{\rho}]$. Write $f_L(M_S) = M_S - \underline{M}(h(M_S))$ and $f_U(M_S) = \overline{M}(h(M_S)) - M_S$, so that $\mathcal{M} = \{f_L \geq 0\} \cap \{f_U \geq 0\}$. Both are Lipschitz, and almost everywhere*

$$f_L'(M_S) \;=\; 1 + Q_\rho(1-h)\, h_M, \qquad f_U'(M_S) \;=\; -1 - Q_\rho(h)\, h_M. \tag{15}$$

Then $\mathcal{M}$ is a compact interval in either of the following cases:

1. *$h_M \geq 0$ throughout I, with no further condition;*
2. *$h_M < 0$ somewhere and $\bar{\rho} \sup |h_M| \leq 1$.*

Proof. Both f_L and f_U are Lipschitz on I, since $\underline{M}$ and $\overline{M}$ are Lipschitz in the constrained share by (6) and $h(\cdot)$ is continuously differentiable by Assumption 1. A Lipschitz function is differentiable almost everywhere, and at every such point the chain rule with (7) gives (15). If $h_M \geq 0$ then, since $Q_\rho \geq 0$, we have $f_L' \geq 1 > 0$ and $f_U' \leq -1 < 0$ wherever the derivatives exist. If $h_M < 0$ then $Q_\rho \leq \bar{\rho}$ gives $f_L' \geq 1 - \bar{\rho}|h_M| \geq 0$ and $f_U' \leq -1 + \bar{\rho}|h_M| \leq 0$ under the stated condition. A Lipschitz function whose derivative is almost everywhere non-negative is non-decreasing, so in either case f_L is non-decreasing and f_U is non-increasing on I. So $\{f_L \geq 0\}$ is an up-set of I and $\{f_U \geq 0\}$ is a down-set. Each is an interval and so is their intersection, and I is a compact interval by Assumption 1, so $\mathcal{M}$ is a compact interval. □

The condition is sufficient. In the numerical validation reported in Appendix B, economies that fail it still produce interval feasible sets. It is checkable from the support of responsiveness and the response of composition, both of which are objects the model delivers. In the economy below the

composition response satisfies $\operatorname{sign}(h_M) = \operatorname{sign}[(1-\xi)Y]$, so the first case applies wherever income incidence is at most one and the aggregate is positive on the branch. A negative rate deviation does not deliver the second of those on its own.

Theorem 7 (Sharp bounds at extremal feasible assignments)**.** *Let Assumption 1 and the conditions of Theorem 6 hold, and let $\mathcal{M} = [\underline{M}^*, \overline{M}^*]$ be non-degenerate. Suppose the aggregate satisfies $dY/dM_S = c(M_S)\,Y$ on $\mathcal{M}$ for a continuous $c > 0$. Then either Y vanishes identically on $\mathcal{M}$, or it vanishes nowhere, never changes sign, $|Y|$ is strictly increasing, and the identified set for the aggregate is*

$$\big[\,\underline{Y},\ \overline{Y}\,\big] \;=\; \Big[\ \min\big\{Y(\underline{M}^*), Y(\overline{M}^*)\big\},\quad \max\big\{Y(\underline{M}^*), Y(\overline{M}^*)\big\}\ \Big], \tag{16}$$

and both endpoints are attained by admissible assignments, meaning joint laws with the responsiveness marginal and the equilibrium state marginal.

Admissibility here is the coupling constraint of Theorem 2 and nothing more. An economy can impose further restrictions on which joint laws it is able to hold, and the heterogeneous-agent economy of Section 4 does. Stationarity of the distributional state removes couplings that leave a responsiveness type entirely constrained (Lemma 1). The interval (16) is then the closure of the stationary identified set, and whether each endpoint belongs to it is settled by Corollary 4.

Proof. The hypothesis is a linear homogeneous differential equation in M_S, so on the interval $\mathcal{M}$ we may write $Y(M_S) = Y(\underline{M}^*)\exp\big(\int_{\underline{M}^*}^{M_S} c(m)\,dm\big)$. The exponential is strictly positive, so the aggregate vanishes identically when $Y(\underline{M}^*) = 0$ and otherwise cannot vanish or change sign and $|Y|$ is strictly increasing. A continuous strictly monotone function on an interval attains its extrema at the endpoints, and by Theorem 2 every value of M_S in $\mathcal{M}$ is attained by an admissible assignment, so both endpoints are attained and the interval is sharp. □

Remark 3 (When the identified set is non-degenerate). Theorem 7 bounds the aggregate but does not by itself say the bounds differ. The dichotomy in its statement carries the argument. The aggregate either vanishes throughout $\mathcal{M}$ or vanishes nowhere on it, and only in the second case is the identified set of interest. A non-degenerate identified set therefore requires $\mathcal{M}$ to have positive length, which depends on the equilibrium and on more than the marginals, and the aggregate not to be identically zero. Non-degeneracy of F_ρ is not on that list. Proposition 1 holds the composition fixed. Once composition responds, a degenerate marginal δ_{ρ_0} leaves the fixed-point condition $M_S = \rho_0\,(1 - h(M_S))$, which can hold on a whole interval: with $F_\rho = \delta_1$ and $h(M_S) = 1 - M_S$ every candidate value is feasible, and the identified set for the aggregate has positive width with a degenerate responsiveness distribution. A sufficient condition for at most one solution is $1 + \rho_0 h_M > 0$ on the branch, which makes the difference strictly increasing. A policy experiment leaving the aggregate at zero produces a singleton identified set however much responsiveness heterogeneity there is and however forward looking demand is, so heterogeneity of mechanisms and an open forward-looking channel are necessary for the assignment to matter at a fixed composition, which is the problem Proposition 1 states, and are not sufficient. The first of the two stops being

necessary once composition responds, as the degenerate marginal above shows: the aggregate varies over the feasible set with no mechanism heterogeneity at all. The second does not. At $\psi = 0$ the aggregate does not move with the statistic at any composition, so the identified set is a point, which Section 4 records as a limit of Theorem 9.

Remark 4 (Without connectedness). If $\mathcal{M}$ is not an interval the conclusion weakens in a specific way. On each connected component the differential equation still holds and $|Y|$ is still monotone, so the sharp extrema over that component sit at its endpoints. Globally the bounds are the minimum and maximum over the endpoint values of all components. What does not follow is that $\inf \mathcal{M}$ and $\sup \mathcal{M}$ bracket Y, because the differential equation is local to a component and carries neither the level nor the sign of Y across a gap.

Theorem 7 rests on one hypothesis, that the aggregate satisfies a linear homogeneous equation in the statistic with a positive coefficient. In the abstract system that is an assumption of the same kind as the sign condition in Proposition 3. The next section derives it from the household-level blocks of a specified economy on a branch where Assumption 1 holds.

The results so far take the assignment as unobserved and bound the aggregate over everything the marginals and the equilibrium allow. In an application one may know more than that. Write T^C for the loading through which expectations reach the consumption decision, which under the scalar-loading condition of Theorem 1 is the responsiveness itself, $T^C = \rho$, so that Definition 2 reads $M_S = \mathbb{E}[T^C \mathbf{1}\{s = S\}]$. The notation is kept because what an application measures is a loading. With it the statistic decomposes as

$$M_S \;=\; \mathbb{E}\big[T^C\,\mathbf{1}\{s = S\}\big] \;=\; \underbrace{\mathbb{E}\big[T^C\big]\,\mathbb{P}(S)}_{M_0} \;+\; \mathrm{Cov}\big(T^C, \mathbf{1}\{s = S\}\big)\,, \tag{17}$$

where the first term is built from the two marginals and the second is the entire content of the matching. The first is identified whenever the marginals are, which with composition fixed makes it a number and with composition endogenous a function of the candidate. The second is not identified at all without the coupling. A bound on the second is the form in which evidence about behavioural heterogeneity enters the macroeconomic conclusion.

One feature of (17) matters for what follows. With the state marginal fixed, M_0 is a constant and a bound on the covariance places M_S in an interval centred on it. Once composition responds, $\mathbb{P}(S) = 1 - h(M_S)$ along the branch, so the centre moves with the candidate and the restriction is no longer a fixed band. Write

$$\varphi(m) \;=\; m \;-\; \mathbb{E}\big[T^C\big]\,(1 - h(m))\,, \qquad \mathcal{B}(\delta) \;=\; \{\, m \in I \;:\; |\varphi(m)| \le \delta \,\}\,, \tag{18}$$

so that $\varphi(m)$ is the covariance evaluated at the candidate m and $\mathcal{B}(\delta)$ is the set the evidence admits. Write D_δ for the diameter of $\mathcal{M} \cap \mathcal{B}(\delta)$. When h does not move, $\varphi(m) = m - M_0$ and $\mathcal{B}(\delta)$ is the interval $[M_0 - \delta, M_0 + \delta]$.

Proposition 4 (A bound on the matching bounds the aggregate)**.** *Suppose* $\left|\mathrm{Cov}\big(T^C, \mathbf{1}\{s = S\}\big)\right| \leq \delta$ *for a known* $\delta \geq 0$*. Then the identified set for the statistic is* $\mathcal{M} \cap \mathcal{B}(\delta)$*, with* $\mathcal{B}$ *from* (18)*, and:*

(i) if Y *is* L*-Lipschitz on* $\mathcal{M}$*, the identified set for the aggregate has diameter at most* $L\, D_\delta$ *and at most* $\operatorname{diam} Y(\mathcal{M})$*, where* $D_\delta \leq 2\delta$ *whenever composition does not respond, with equality when the band* $[M_0 - \delta, M_0 + \delta]$ *lies inside* $\mathcal{M}$*;*

(ii) under the hypotheses of Theorem 7, if $\mathcal{M} \cap \mathcal{B}(\delta) = [a_\delta, b_\delta]$ *is a non-degenerate interval then the identified set is* $[\min\{Y(a_\delta), Y(b_\delta)\},\ \max\{Y(a_\delta), Y(b_\delta)\}]$ *and both endpoints are attained;*

(iii) if in addition φ *has an interior zero* m_0 *of* $\mathcal{M}$ *with* $\varphi'(m_0) = 1 + \mathbb{E}[T^C]\, h_M(m_0) \neq 0$*, and* δ *is small enough that the component of* $\mathcal{B}(\delta)$ *containing* m_0 *lies inside* $\mathcal{M}$*, the width of that set is*

$$W_Y(\delta) \;=\; \frac{2\, c(m_0)\, |Y(m_0)|}{|1 + \mathbb{E}[T^C]\, h_M(m_0)|}\, \delta \;+\; o(\delta), \tag{19}$$

and where composition does not respond $h_M \equiv 0$ *and* $m_0 = M_0$*, so this is* $2c(M_0)|Y(M_0)|\delta + o(\delta)$*, exact at* $2|Y(M_0)|\sinh(c\delta)$ *when* c *is constant.*

Proof. The decomposition (17) evaluated at a candidate m is $\varphi(m)$, so the bound places M_S in $\mathcal{B}(\delta)$, and feasibility places it in $\mathcal{M}$. The identified set for the statistic is the intersection. Part (i) is the definition of a Lipschitz bound applied to that intersection, together with the bound the feasible set already delivers. For part (ii), Theorem 7 gives strict monotonicity of $|Y|$ and constancy of its sign on $\mathcal{M}$, and a continuous strictly monotone function on a sub-interval attains its extrema at that sub-interval's endpoints. Attainment follows because every value of the statistic in $\mathcal{M}$ is attained by an admissible assignment. For part (iii), φ is continuously differentiable near m_0 with $\varphi'(m_0) \neq 0$, so the component of $\{|\varphi| \leq \delta\}$ containing m_0 has diameter $2\delta/|\varphi'(m_0)| + o(\delta)$ by the inverse function theorem. The hypothesis of Theorem 7 is $dY/dM_S = c(M_S)Y$, so a first-order expansion of Y across that component gives (19). With $h_M \equiv 0$ the map φ is the identity shifted, $\varphi'(m_0) = 1$, the component is $[M_0 - \delta, M_0 + \delta]$, and when c is constant the equation integrates to $Y(M_0 \pm \delta) = Y(M_0)e^{\pm c\delta}$ with difference $2Y(M_0)\sinh(c\delta)$. □

The statement separates a local reading from a global one. *Locally* the width grows linearly in δ, provided the local derivative is non-zero, so to first order halving what is not known about the matching halves the macroeconomic uncertainty it generates. *Globally* the curve $\delta \mapsto W_Y(\delta)$ saturates. Once $\mathcal{B}(\delta)$ contains the whole feasible set the intersection in Proposition 4 is all of $\mathcal{M}$, the bound reverts to the marginal one, and further ignorance about the matching adds no width because the marginals were already binding. Part (i) carries both bounds for that reason, and it says the bound is never worse than what the marginals give: information about the matching can only narrow the set.

The constant of proportionality is the local semi-elasticity c of the aggregate with respect to the statistic, which is a statement about the economy. The data do not enter it. Where composition does not respond the elasticity form is $W_Y(\delta)/|Y| = 2c\delta + o(\delta)$, and where it does the same

expression carries the factor $1/|1+\mathbb{E}[T^C]h_M|$. Either way it is scale-free, which makes δ comparable across applications and across economies. The economy of the next section supplies c in primitives and reads its comparative statics.

Remark 5 (Truncation at the boundary). The closed form of part (iii) is exact only while the component of $\mathcal{B}(\delta)$ around m_0 lies inside the feasible set. Near a boundary the relevant endpoints are the endpoints of $\mathcal{M} \cap \mathcal{B}(\delta)$, the set around m_0 is no longer symmetric, and the closed form is replaced by $|Y(b_\delta) - Y(a_\delta)|$. Part (ii) continues to apply unchanged, since it is stated on the intersection instead of on $\mathcal{B}(\delta)$. What is lost is the closed form and the exact linearity. Whether the truncation binds is an equilibrium question.

Corollary 2 (A bound on dispersion)**.** *Suppose instead that* $\mathbb{E}\left\|T^C - \mathbb{E}[T^C]\right\|^2 \leq \delta_T^2$ *for some* $\delta_T \geq 0$*. Then*

$$\left|\mathrm{Cov}\big(T^C, \mathbf{1}\{s = S\}\big)\right| \;\leq\; \delta_T \sqrt{\mathbb{P}(S)\,(1 - \mathbb{P}(S))}, \tag{20}$$

The right-hand side depends on the state marginal, so with composition fixed it is a constant and Proposition 4 applies with δ *replaced by it. With composition endogenous* $\mathbb{P}(S) = 1 - h(m)$ *varies over the candidates and the restriction is*

$$|\varphi(m)| \;\leq\; \delta_T \sqrt{h(m)\,(1 - h(m))}, \tag{21}$$

which is not $\mathcal{B}(\delta)$ *for any scalar* δ*. The proposition then applies either on the set* (21) *in place of* $\mathcal{B}(\delta)$*, or with the scalar* $\delta = \delta_T \sup_m \sqrt{h(m)(1 - h(m))}$*, which is valid because it is a uniform upper bound and conservative because it is attained only where* h *is nearest one half.*

Proof. Write $\mathrm{Cov}(T^C, \mathbf{1}\{s = S\}) = \mathbb{E}\big[(T^C - \mathbb{E}T^C)(\mathbf{1}\{s = S\} - \mathbb{P}(S))\big]$ and apply Cauchy-Schwarz. The second factor has variance $\mathbb{P}(S)(1 - \mathbb{P}(S))$. □

The two bounds refer to different objects. A bound on the covariance is a statement about the matching, which is unobserved. A bound on the dispersion of the behavioural loadings is a statement about heterogeneity, which a household survey can measure. Equation (20) converts the second into the first. The conversion loses most when the population is evenly split between the two states, $h = 1/2$, and loses nothing as either state becomes rare. The conversion is not tight: it is attained only when the loading is perfectly aligned with the indicator, which is the least favourable matching there is. An application that can bound dispersion and not the covariance obtains a conservative version of (19), and the gap between the two measures the value of a direct measurement of the matching.

4 A heterogeneous-agent economy

The economy below is small, so that every derivative in (13) traces to a primitive and one accounting identity signs the comparative static.

A unit mass of households occupies one of two economic states. Constrained households consume current income,

$$c_C \;=\; \xi\, Y, \tag{22}$$

where $\xi > 0$ is the incidence of aggregate income on their resources. They have no forward-looking term, so responsiveness does nothing for them.

Household resources do not exhaust the aggregate. Constrained households receive ξY and unconstrained households receive Y, so the household block absorbs $h\xi Y + (1-h)Y = [1 - h(1-\xi)]\, Y$ and the residual $h(1-\xi)Y$ is absorbed outside it, by a government, an intermediary, a foreign sector or a transfer. This is a reduced-form closure and it lets the incidence of aggregate income across the two states differ from their population shares. At $\xi = 1$ the residual is zero and the block is closed on its own, and average saving among the unconstrained is zero with it, which is the same condition that switches the composition channel off. That case is the *no-composition benchmark*, and it is the economy every illustration of the pure assignment problem below uses. It isolates that problem, since nothing the assignment does runs through the constrained share. It is not the economy the empirical section reads a survey in, and Proposition 5 below gives the reason: the same condition that makes composition unresponsive makes the observable spending contrast unresponsive too.

The alternative closure is not available. Closing the block by giving the unconstrained $(1 - h\xi)\, Y/(1-h)$ makes household resources exhaust the aggregate exactly. Goods-market clearing then leaves average saving among the unconstrained equal to zero, the inflow hazard constant and $h_M = 0$, so the composition channel this section exists to study disappears. Endogenous composition needs a margin through which household saving can be absorbed somewhere other than contemporaneous household consumption, and an allocation with no such margin does not have one. Unconstrained households are on an intertemporal margin,

$$c_S(\rho) \;=\; -\,\sigma r \;+\; \psi\, \rho\, Y, \tag{23}$$

with $\sigma > 0$ the intertemporal substitution term, r the policy experiment, and $\psi > 0$ the strength of the forward-looking channel. This is (3) with $A = -\sigma r$ and $B = \psi Y$. Responsiveness ρ scales the forward-looking channel and nothing else. This gives the assignment its economic content: the mechanism operates only on households who are not constrained, so where it sits in the distribution of economic states decides whether it operates at all.

Households move between states in continuous time. An unconstrained household with responsiveness ρ becomes constrained at an intensity $\lambda_{\text{in}}(\rho)$, and a constrained household leaves at a constant intensity $\lambda_{\text{out}} > 0$. These are rates and not one-period probabilities. The construction in Lemma 1 returns inflow intensities that can exceed one. The model restricts the inflow hazards only through their average among the unconstrained, which falls with average saving there,

$$\lambda_{\text{in}} \;\equiv\; \mathbb{E}[\lambda_{\text{in}}(\rho) \mid s = S] \;=\; \bar{\lambda} \;-\; \theta\, a, \qquad a \;=\; Y + \sigma r - \frac{\psi\, Y\, M_S}{1-h}, \tag{24}$$

where a is saving per unconstrained household, $\theta > 0$ governs how strongly saving protects against becoming constrained, and $\bar{\lambda} > 0$ is the baseline inflow rate. The distributional state is stationary.

The two consumption blocks, (22) and (23), are the ones used in tractable heterogeneous-agent models with type switching, in which unconstrained households face a risk of becoming hand to mouth and therefore carry a precautionary motive. The transition block departs from those models in one respect. Such models normally give every unconstrained household the same inflow hazard. Here (24) restricts only the average and leaves the dependence on responsiveness open. The usual reading closes the identification problem instead of posing it.

Lemma 1 (Which assignments are stationary)**.** *Fix the marginals F_ρ and $h \in (0,1)$, and let the outflow hazard $\lambda_{\text{out}} > 0$ be common.*

(i) If the inflow hazard is also common to the unconstrained, the only stationary joint law is the independent one. The assignment statistic is then point identified at $M_S = (1-h)\,\mathbb{E}[\rho]$.

(ii) If the inflow hazard may vary with responsiveness, every joint law with those marginals that leaves no responsiveness type entirely constrained, meaning $\mathbb{P}(s = S \mid \rho) > 0$ for F_ρ-almost every ρ, is stationary under some profile $\lambda_{\text{in}}(\cdot)$, and the average inflow hazard takes the same value $\lambda_{\text{out}}\, h/(1-h)$ at all of them. At an equilibrium value of h that value is the one (24) assigns, the two conditions being the law of motion written twice, so (24) then holds at every such law.

(iii) A law that places an entire responsiveness type in the constrained state is stationary under no profile.

Proof. For measurable A, stationarity of the mass with $\rho \in A$ requires $\mathbb{E}[\lambda_{\text{in}}(\rho)\,\mathbf{1}\{s = S,\ \rho \in A\}] = \lambda_{\text{out}}\,\mathbb{P}(s = C,\ \rho \in A)$. Under (i) the left side is $\lambda_{\text{in}}\,\mathbb{P}(s = S,\ \rho \in A)$. Taking A to be the whole support gives $\lambda_{\text{in}}(1-h) = \lambda_{\text{out}}\,h$, and substituting $\mathbb{P}(s = S,\ \rho \in A) = F_\rho(A) - \mathbb{P}(s = C,\ \rho \in A)$ gives $\mathbb{P}(s = C,\ \rho \in A) = \lambda_{\text{in}}\,F_\rho(A)/(\lambda_{\text{in}} + \lambda_{\text{out}}) = h\,F_\rho(A)$, which is the independent law. Under (ii), let $\mathbb{P}$ be any target with the stated marginals and write $s(\rho) = \mathbb{P}(s = S \mid \rho) \in (0,1]$, which is where the hypothesis places it. Setting $\lambda_{\text{in}}(\rho) = \lambda_{\text{out}}\,(1 - s(\rho))/s(\rho)$ balances the flows at each ρ, and integrating against F_ρ gives $\mathbb{E}[\lambda_{\text{in}}(\rho) \mid s = S] = \lambda_{\text{out}} \int (1-s)\,dF_\rho/(1-h) = \lambda_{\text{out}}\,h/(1-h)$, which does not depend on the target. That this common value also solves (24) is a separate condition, $\bar{\lambda} - \theta a = \lambda_{\text{out}}\,h/(1-h)$, which is (26) set to h. It holds at an equilibrium constrained share and fails at an arbitrary one, so the last clause of (ii) is a statement about equilibria. For (iii), if $\mathbb{P}(s = S,\ \rho \in A) = 0$ while $\mathbb{P}(s = C,\ \rho \in A) > 0$, the balance condition on A reads $0 = \lambda_{\text{out}}\,\mathbb{P}(s = C,\ \rho \in A) > 0$, which no finite profile satisfies. That cell drains at rate λ_{out} with nothing replenishing it. □

Part (i) explains why the transition block is written as it is. A hazard common to all unconstrained households destroys the assignment instead of leaving it neutral, because non-selective flows remove whatever sorting the initial condition carried, and at a stationary distribution the

coupling is independent. Part (ii) says the coupling is free exactly when flows are allowed to sort, and that the aggregate conditions used below do not restrict that sorting: the average inflow hazard takes the same value whatever the coupling, so (24), goods-market clearing and the law of motion are unchanged. What is left unmodelled is why some households are more exposed to becoming constrained than others at the same level of aggregate saving. That is the assignment problem in this economy, and the paper bounds what follows from leaving it unresolved.

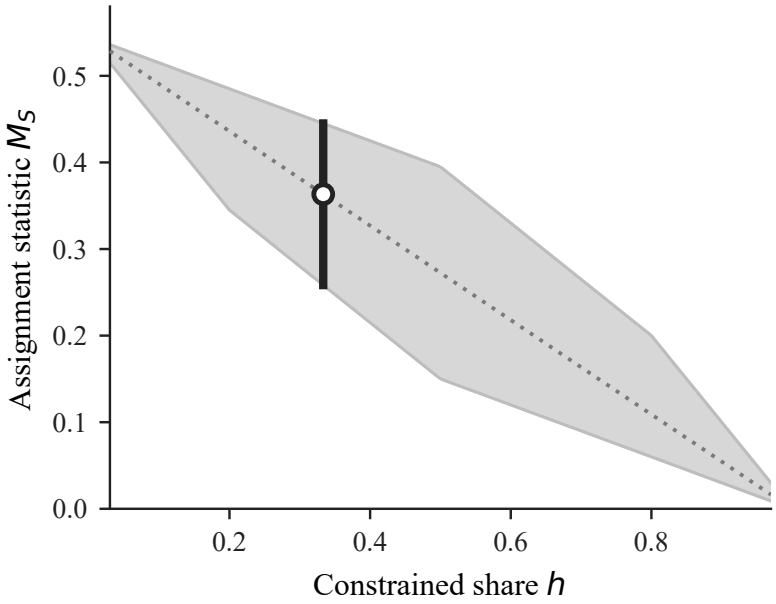


Figure 4: Assignments permitted by stationarity and by a common hazard

Notes. Three responsiveness types, taking the values $0.3, 0.65, 1$ with masses $0.5, 0.3, 0.2$, which are not those of the reference economy. The shaded band is the sharp interval for the statistic at each constrained share, from Theorem 2, and the dotted line inside it is the independent assignment. At the constrained share 0.3333 the heavy segment is the set of assignments that hazards varying with responsiveness can sustain, by Lemma 1(ii). It runs from 0.2583 to 0.4450 and excludes its lower end, where the whole of the most responsive type would be constrained and Lemma 1(iii) admits no hazard profile. Its upper end is attained, since the least responsive type carries more mass than the constrained share. The drawn segment does not resolve the difference. The open marker is the single assignment a common hazard can sustain, 0.3633, by part (i). The segment spans 51.4 percent of that value. There is no time dimension in the figure: both objects are sets of rest points, not paths.

Part (iii) restricts which assignments this economy can hold (Figure 4), and whether it binds at the extremes depends on the marginal. When F_ρ is continuous the comonotone and antitone couplings put a whole interval of responsiveness into the constrained state, so the endpoints of Theorem 2 are approached but not attained here, and the stationary identified set is the sharp interval without its endpoints. When F_ρ has finite support the extremes may be attained and need not be, and Figure 4 shows one of each: its upper end is attained while its lower end is not, since

reaching that one would constrain the whole of the most responsive type. In the two-type case with mass q at $\rho = 1$, a type is entirely constrained only at $z = q$ or $z = h + q - 1$, and both fall outside the Fréchet range for z when $h < q$ and $h + q < 1$. The economy used for the magnitudes below satisfies both, with $h = 0.3333$ and $q = 0.5$, so its endpoints are attained.

Goods-market clearing and the law of motion of the constrained share are

$$\mathcal{G}(Y, h, M_S) \;=\; h\,\xi Y \;+\; (1-h)(-\sigma r) \;+\; \psi\, Y\, M_S, \tag{25}$$

$$\mathcal{H}(Y, h, M_S) \;=\; h \;+\; (1-h)\,\lambda_{\text{in}} \;-\; h\,\lambda_{\text{out}}, \tag{26}$$

where λ_{in} is the average from (24), and the equilibrium is the fixed point (12). By Theorem 1 the assignment enters only through M_S, which makes the two-equation representation possible at all. Solving (25) for the aggregate, wherever $1 - h\xi - \psi M_S \neq 0$,

$$Y \;=\; \frac{-(1-h)\,\sigma r}{1 - h\xi - \psi\, M_S}, \tag{27}$$

which at fixed composition satisfies $\operatorname{sign}(\partial Y/\partial M_S) = -\operatorname{sign}(r)$, so it rises with the assignment statistic for an expansion and falls for a contraction. That observation is not yet the result, because h is not fixed. The result follows once composition moves as well.

The excluded case is the same degeneracy seen elsewhere. At $\xi = 1$ the feasibility identity gives $h = \bar{\lambda}/(\lambda_{\text{out}} + \bar{\lambda})$, so $1 - h\xi - \psi M_S = \left[\lambda_{\text{out}} - (\lambda_{\text{out}} + \bar{\lambda})\psi M_S\right]/(\lambda_{\text{out}} + \bar{\lambda})$, and the denominator vanishes exactly when the linear coefficient of (36) does. Both are the degeneracy recorded after Proposition 6: goods-market clearing then reads $0 = -(1-h)\sigma r$, which every Y satisfies when $r = 0$ and none satisfies otherwise. Assumption 1 excludes it.

The derivatives that (13) needs are immediate from (25) and (26):

$$\mathcal{G}_M \;=\; \psi Y, \qquad \mathcal{G}_h \;=\; \xi Y + \sigma r, \qquad \mathcal{H}_M \;=\; \theta\,\psi\, Y. \tag{28}$$

Responsiveness located among the unconstrained raises their consumption, lowers their saving, and therefore raises the inflow hazard, so $\mathcal{H}_M > 0$ requires $\theta > 0$, $\psi > 0$ and $Y > 0$ together. At $\psi = 0$ the channel is shut and $\mathcal{H}_M = 0$. In an experiment with $Y = 0$ the same is true. Both are limits of the results below.

One identity makes the economy tractable.

Theorem 8 (Feasibility identity)**.** *At any equilibrium of* (12) *with* $h \in (0, 1)$*,*

$$h\,W \;=\; \bar{\lambda}, \qquad W \;=\; \lambda_{\text{out}} + \bar{\lambda} - \theta\, Y\,(1-\xi), \tag{29}$$

and the distribution of responsiveness does not appear. Consequently $W = \bar{\lambda}/h > 0$*. The identity does not use the loading structure of* (3) *at any rank. It holds for an arbitrary unconstrained consumption function.*

Proof. Let $\bar{c}_S$ denote consumption per unconstrained household, whatever function of responsive-

ness and the aggregate generates it. Saving per unconstrained household is $a = Y - \bar{c}_S$, so stationarity of the two-state distribution, $(1-h)\lambda_{\text{in}} = h\,\lambda_{\text{out}}$, reads

$$(1-h)\bar{\lambda} \;-\; (1-h)\,\theta\,Y \;+\; (1-h)\,\theta\,\bar{c}_S \;=\; h\,\lambda_{\text{out}}.$$

Goods-market clearing at a fixed point gives $(1-h)\bar{c}_S = Y(1-h\xi)$. Substituting, the unconstrained consumption term disappears entirely, leaving

$$(1-h)\bar{\lambda} \;-\; (1-h)\theta Y \;+\; \theta Y(1-h\xi) \;=\; h\,\lambda_{\text{out}},$$

and since $-(1-h)+1-h\xi = h(1-\xi)$ this is $(1-h)\bar{\lambda} + \theta Y h(1-\xi) = h\lambda_{\text{out}}$, which rearranges to (29). Positivity follows from $\bar{\lambda} > 0$ and $h > 0$. No feature of unconstrained behaviour survives into the result. □

The two conditions the paper uses are therefore separate. The sufficient statistic of Theorem 1 needs the rank-one loading. The feasibility identity needs none of it: only that constrained households consume current income, that saving is income minus consumption, and that the inflow hazard is linear in average saving.

Within the household block the identity follows from the definition of saving, goods-market clearing and stationarity of the distribution, and that is why the assignment disappears from it, and why the functional form of the unconstrained decision disappears with it. Whatever responsiveness does to an unconstrained household's consumption it does in the opposite direction to that household's saving, and in N the two movements cancel. The cancellation is of terms inside the sum. The summands need not agree: the direct channel $\mathcal{G}_M(1-\mathcal{H}_h)$ and the composition channel $\mathcal{G}_h\mathcal{H}_M$ can carry opposite signs. At the parameter vector of the negative-aggregate example below, where $Y = -1.1341$, they are -3.0158 and $+0.7849$, and they still sum to $\psi Y W$. What accounting delivers is the sign of the total, which the comparative static needs. In the abstract system that sign had to be assumed.

Corollary 3 (The numerator factorizes)**.**

$$N \;=\; \mathcal{G}_M(1-\mathcal{H}_h) + \mathcal{G}_h\mathcal{H}_M \;=\; \psi\,Y\,W \;=\; \frac{\psi\,\bar{\lambda}\,Y}{h}. \tag{30}$$

Proof. From (28), $N = \psi Y(1-\mathcal{H}_h) + (\xi Y + \sigma r)\theta\psi Y = \psi Y\,[(1-\mathcal{H}_h) + \theta\mathcal{G}_h]$. Writing out $1-\mathcal{H}_h$ from (26) and (24), the terms in $\theta\psi Y M_S/(1-h)$ cancel against $\theta\mathcal{G}_h$ and what remains is $\lambda_{\text{out}} + \bar{\lambda} - \theta Y(1-\xi) = W$. The last equality is Theorem 8. □

Theorem 9 (Assignment monotonicity from household feasibility)**.** *Under Assumption 1, with $h \in (0,1)$, $\bar{\lambda} > 0$, $\psi > 0$ and $D > 0$,*

$$\frac{dY}{dM_S} \;=\; \frac{\psi\,\bar{\lambda}\,Y}{h\,D}, \qquad \frac{dh}{dM_S} \;=\; \frac{h\,\theta\,\psi\,(1-\xi)}{D}\,Y, \tag{31}$$

at every point of the branch, including any at which the aggregate vanishes. The first relation is a linear homogeneous differential equation, so either $Y \equiv 0$ on the branch or Y never vanishes on it. In the second case

$$\frac{d \log |Y|}{dM_S} \;=\; \frac{\psi\, \bar{\lambda}}{h\, D} \;>\; 0, \tag{32}$$

so $|Y|$ is strictly increasing in the assignment statistic and Y keeps its sign, and $\operatorname{sign}(h_M) = \operatorname{sign}[(1-\xi)\, Y]$. *In the first case the aggregate does not respond to the assignment at all and the identified set is a single point.*

Proof. The first expression is Cramer's rule (13) with Corollary 3, and it holds wherever the branch is defined. A solution of a linear homogeneous differential equation with a locally bounded coefficient either vanishes identically or never vanishes, which gives the dichotomy. On a non-vanishing branch, dividing by Y gives (32), which is strictly positive under the stated conditions. For the composition response, differentiate $h = \bar{\lambda}/W$ from Theorem 8, using $W_M = -\theta(1-\xi)Y_M$, to obtain $h_M = \bar{\lambda}\,\theta(1-\xi)Y_M/W^2 = (h\,\theta(1-\xi)/W)\, Y_M$, and substitute the first expression together with $\bar{\lambda}/W = h$. □

One object the empirical section reads is already determined here. By (22) a constrained household's spending response is ξ times the aggregate response, identical for every value of its responsiveness, and by (23) an unconstrained household's is $-\sigma\dot{r} + \psi\rho\dot{Y}$. Differentiating the aggregation identity along the equilibrium and separating the average of the individual within-state responses from the movement of composition,

$$\Delta_{\text{true}} \;=\; (\xi - 1)\,\dot{Y} \;+\; \dot{h}\,(\xi Y + \sigma r) \;+\; \psi Y \dot{M}_S, \tag{33}$$

with a dot denoting the derivative along the policy experiment. The last term is the movement of the statistic itself. A counterfactual has to say what happens to the joint law of state and responsiveness while the experiment runs, and $\dot{M}_S$ is fixed by that rule. Holding an assignment fixed while composition responds does not hold the joint law fixed, so even the natural rules give $\dot{M}_S \neq 0$ away from the benchmark. The proposition below covers every rule under which the joint law moves only with composition.

Proposition 5 (The no-composition benchmark has no observable contrast)**.** *Suppose that along the experiment the joint law of state and responsiveness changes only through the constrained share, so that it is held fixed whenever h is. Then at $\xi = 1$, $\Delta_{\text{true}} = 0$ at every admissible assignment, while the aggregate continues to respond to the assignment at every point of a non-vanishing branch.*

Proof. At $\xi = 1$ Theorem 8 reads $h\,W = \bar{\lambda}$ with $W = \lambda_{\text{out}} + \bar{\lambda}$, which involves neither the aggregate nor the assignment, so h takes the same value at every admissible assignment and is constant along the experiment, giving $\dot{h} = h_M = 0$. Under the stated rule the joint law is then constant along the experiment, so $\dot{M}_S = 0$ as well. Every term of (33) then vanishes, the first with $\xi - 1$. The aggregate response is Theorem 9, whose first relation does not vanish on a branch where Y does not vanish identically. □

The hypothesis is a property of the counterfactual, and it is the one the rest of the paper maintains. A policy that reallocates households across states at fixed composition violates it. In the two-type case such a reallocation moves the coupling z of Example 1 with h fixed, and the contrast becomes $-(1-\mu)\psi Y \dot{z}$, which is not zero. Its size is easy to bound in the reference economy. With Y between 0.0262 and 0.0340 on the feasible set, a reallocation large enough to produce the smallest contrast found at $\xi = 1.4$ below would have to move z by 0.59 per percentage point of the policy rate, more than the whole feasible range of z, whose width is 0.333. Within the stated class of rules, at $\xi = 1$ the aggregate remains identified and the observable loses rank. The assignment still moves the aggregate over 25.93 percent of its own midpoint, and this contrast sees none of it. Section 5 takes up what a survey can do about that.

Away from the benchmark the last term of (33) has to be evaluated, and that needs the rule just described. In the two-type case of Example 1, where the coupling is indexed by z, (8) gives $\dot{M}_S = -\mu\dot{h} - (1-\mu)\dot{z}$ in general, so that

$$\Delta_{\text{true}} \;=\; \dot{h}\,(\xi Y + \sigma r - \mu\psi Y) \;+\; \dot{Y}\,(\xi - 1) \;-\; (1-\mu)\,\psi Y \dot{z}. \tag{34}$$

Section 5 works with the rule $\dot{z} = 0$. Since q is fixed, that rule holds $\mathbb{P}(s = C \mid \rho = 1)$ fixed and lets the less responsive type absorb every movement of composition. It is one choice among several that tie the coupling to composition. Holding $\mathbb{P}(s = C \mid \rho = \mu)$ fixed instead gives $\dot{z} = \dot{h}$, and holding the composition of the constrained pool, $\mathbb{P}(\rho = 1 \mid s = C)$, fixed gives $\dot{z} = (z/h)\dot{h}$. Re-solving the equilibrium under each, with the rule imposed on both sides of the derivative, leaves the sign of the contrast unchanged at every incidence in the grid Section 5 uses and changes the scale window reported there by at most 2 percent. The empirical conclusions do not turn on the choice.

Remark 6 (What depends on the specification). Under the general loading (3) the same steps give $N = B\,W$, so

$$\frac{dY}{dM_S} \;=\; \frac{B\,\bar{\lambda}}{h\,D}, \qquad \frac{d\log|Y|}{dM_S} \;=\; \frac{B\,\bar{\lambda}}{Y\,h\,D}, \tag{35}$$

on a non-vanishing branch, so the direction of the assignment effect is the sign of B/Y. The sufficient statistic and the feasibility identity hold for any A and B. The particular choice $B = \psi Y$ makes $B/Y = \psi$ a positive constant, so the direction holds unconditionally.

The logarithmic form does not condition on the sign of the aggregate, whereas a statement about dY/dM_S has to carry the sign of Y with it. Writing the result in logs also makes the multiplicative structure visible: each increment of the assignment statistic scales the aggregate response by a factor instead of shifting it.

The sign of the composition response involves the aggregate as well as the income incidence. Dropping Y from that expression is wrong whenever the policy experiment can move the aggregate in either direction. With the aggregate carried through, the condition of Theorem 6 can be checked: on a branch with $Y > 0$ we have $h_M \geq 0$ exactly when $\xi \leq 1$, so the first case of Theorem 6 applies and the feasible set is an interval with no further condition. The sign of Y is a property of the branch, not of the experiment. A negative rate deviation does not imply it: at $\xi = 1/2$, $\psi = 2$,

$\sigma = 1/4$, $r = -1/2$, $\theta = 1/2$, $\lambda_{\text{out}} = 1/5$, $\bar{\lambda} = 1/2$ and $M_S = 2/5$, the unique feasible root of (36) is $Y = -1.1341$ with $D > 0$ and an interior coupling, and there $h_M = -8.32$. Positivity of the aggregate has to be carried as a condition on the branch. This is the analogue of a restriction that does not close in the two-type economy, and it closes here because the bound $Q_\rho \leq \bar{\rho}$ supplies the missing piece.

Combining Theorem 9 with Theorem 7, the identified set for the aggregate is bounded by the two extremal feasible assignments. Theorem 7 characterises the set of admissible couplings, and Lemma 1 says this economy can hold only part of it. The two statements are reconciled as follows.

Corollary 4 (The stationary identified set)**.** *Let the hypotheses of Theorem 7 hold in the economy of this section, with $\mathcal{M} = [\underline{M}^*, \overline{M}^*]$ non-degenerate and Y not identically zero on it, and suppose that $h_M \geq 0$ on I or that $\bar{\rho} \sup |h_M| < 1$. Call a value of the statistic stationary when some joint law attaining it at the equilibrium composition is stationary under some inflow profile. Then*

(i) Every value in the open interval $(\underline{M}^*, \overline{M}^*)$ is stationary.

(ii) An endpoint at which the corresponding rearrangement bound does not bind, as when it is an endpoint of I lying strictly inside that bound, is stationary. If $\underline{M}^* = \underline{M}(h(\underline{M}^*))$, the endpoint $\underline{M}^*$ is stationary if and only if the antitone coupling at $h(\underline{M}^*)$ leaves every responsiveness type some unconstrained mass, and if $\overline{M}^* = \overline{M}(h(\overline{M}^*))$, the endpoint $\overline{M}^*$ is stationary if and only if the comonotone coupling at $h(\overline{M}^*)$ does.

(iii) The stationary identified set for the aggregate is the interval (16) with each endpoint included exactly when the corresponding value of the statistic is stationary, so its closure is always (16).

Proof. For (i), the stated conditions make f_L strictly increasing and f_U strictly decreasing in the proof of Theorem 6, since their derivatives are bounded away from zero almost everywhere, so an interior value m satisfies $\underline{M}(h(m)) < m < \overline{M}(h(m))$ strictly. At that composition the independent coupling gives $\mathbb{P}(s = S \mid \rho) = 1 - h > 0$ for every ρ and attains $(1-h)\mathbb{E}[\rho]$, which lies weakly inside the rearrangement interval. If m is at least that value, mix the independent coupling with the comonotone one, and otherwise with the antitone one. The weight on the independent coupling needed to reach m is positive because m is strictly inside the interval, so the mixture keeps $\mathbb{P}(s = S \mid \rho) > 0$ for every ρ and is stationary by Lemma 1(ii). For (ii), an endpoint lying strictly inside the rearrangement interval at its composition is reached by the same mixture. At a binding endpoint, a coupling attains $\underline{M}(h)$ only if it places no unconstrained mass above the quantile $Q_\rho(1-h)$ and no constrained mass below it, since otherwise exchanging the two masses lowers $\mathbb{E}[\rho \mathbf{1}\{s = S\}]$, and the split of an atom at that quantile is fixed by the state marginal. The conditional law $\mathbb{P}(s = S \mid \rho)$ is therefore the antitone one F_ρ-almost everywhere, and Lemma 1(ii) and (iii) give the equivalence. The upper endpoint is symmetric. Part (iii) follows because $|Y|$ is strictly monotone in the statistic on $\mathcal{M}$ by Theorem 7. □

Whether the bounds are attained therefore depends on the responsiveness marginal. For continuous F_ρ both extremal couplings put an interval of responsiveness entirely into the constrained state, so neither endpoint is stationary and the bounds are a supremum and an infimum. For finite support satisfying the condition recorded after Lemma 1, both extremes are attained and the bounds are a maximum and a minimum. In between, the endpoints can fail separately, and all three possibilities occur. The set is closed when both extremal couplings leave every responsiveness type some unconstrained mass, half open when exactly one of them does not, and open when neither does. Two atoms with $q = 4/5$, $h = 3/10$ and $\mu = 3/10$ are the middle case. The comonotone coupling, which gives the upper endpoint of the statistic and the lower endpoint $z = 1/10$ of the coupling, leaves the low type entirely constrained, while the antitone coupling leaves both types represented. The reference economy is closed, so the width reported below is attained by stationary assignments at both ends. The direction does not depend on any of that. Responsiveness located among unconstrained households raises M_S and raises $|Y|$, and responsiveness located among constrained households lowers both. A given policy therefore transmits least well when the households who respond most to aggregate news are the ones least able to act on that response, because they are not on the margin where the response operates.

The degenerate cases are limits of the theorem. If $\psi = 0$ the forward-looking channel is closed, dY/dM_S is zero, and the identified set collapses to a point. The statement is in levels because the branch may be the vanishing one, on which the logarithmic derivative is not defined: in a backward-looking economy it does not matter who holds which mechanism, however large the heterogeneity. If F_ρ is degenerate the feasibility condition is the scalar equation $m = \rho_0(1-h(m))$, whose left side minus right side has derivative $1+\rho_0 h_M$. Where that is positive on the branch the equation has at most one solution, so the feasible set has at most one point, which is uniqueness. The condition is immediate wherever composition does not respond, in particular at $\xi = 1$ where $h_M = 0$. Without it a degenerate marginal need not pin the statistic, as Remark 3 shows. And as that remark also records, an experiment that leaves Y at zero produces a singleton identified set whatever else holds.

Proposition 6 (Existence and selection of the branch)**.** *In this economy the equilibrium system reduces to a quadratic in the aggregate. Writing $\zeta = \theta(1-\xi)$, a symbol kept apart from the bridge scale of Section 5, and substituting the feasibility identity into goods-market clearing to eliminate the constrained share,*

$$-\zeta\,(1-\psi M_S)\,Y^2 \;+\; \left[\left(\lambda_{\text{out}}+\bar{\lambda}\right)(1-\psi M_S) - \bar{\lambda}\,\xi - \sigma r\zeta\right] Y \;+\; \sigma r\,\lambda_{\text{out}} \;=\; 0. \tag{36}$$

A real root of (36) is a candidate and not yet an equilibrium: it is an equilibrium only if in addition $W > 0$, $h = \bar{\lambda}/W$ lies in $(0,1)$, and M_S is feasible at that h. When $\xi \neq 1$ there are at most two candidates and none when the discriminant is negative. At $\xi = 1$ the leading coefficient vanishes and (36) is linear, with a unique root when $(\lambda_{\text{out}}+\bar{\lambda})(1-\psi M_S) \neq \bar{\lambda}$. When that coefficient vanishes the equation is an identity if $r = 0$ and inconsistent otherwise.

Proof. The feasibility identity gives $h = \bar{\lambda}/W$ with $W = \lambda_{\text{out}} + \bar{\lambda} - \zeta Y$. Substituting into

$Y\,(1 - h\xi - \psi M_S) = -(1-h)\sigma r$ and clearing the denominator gives (36). At $\xi = 1$ we have $\zeta = 0$. □

Proposition 6 narrows the equilibrium set without delivering Assumption 1, and two things can go wrong. A real root can fail admissibility. At $\xi = 1/2$, $\psi = 1$, $M_S = 1/2$, $\bar{\lambda} = \lambda_{\text{out}} = 1/10$, $\theta = 1/10$, $\sigma = 1$ and $r = -1$ the discriminant is zero and the double root is $Y = 2$, at which $W = 1/10$ and $h = 1$. No household is then unconstrained, so $M_S = \mathbb{E}[\rho\,\mathbf{1}\{s = S\}]$ is zero where the root was solved with $1/2$, and the candidate violates feasibility. It is the conditional mean $\mathbb{E}[\rho \mid S]$ that ceases to exist at $h = 1$, while the statistic itself does not. And the degenerate linear case is reachable. At $\xi = 1$, $\psi = 1$, $M_S = 1/2$ and $\bar{\lambda} = \lambda_{\text{out}} = 1/5$ the coefficient of Y vanishes, so with $r = 0$ every admissible Y is an equilibrium and with $r = -1/10$ none is. The existence and uniqueness clause of Assumption 1 therefore remains a restriction on the economies considered. What the proposition does establish is that away from those cases the equilibrium set is finite and the determinant selects within it: over 5,840 economies drawn widely with ξ away from one and a real discriminant, the number of roots with an interior constrained share and a positive determinant was one in 5,840 cases and two in 0. Both roots are admissible on the interiority criterion alone in most of those economies, so the determinant selects. The quadratic is satisfied at the solved fixed point to 6.8×10^{-14}.

The binary rational-or-alternative case, which is where the question originates, follows in one line.

Corollary 5 (The two-type theorem)**.** *Let F_ρ place mass q at $\rho = 1$ and mass $1 - q$ at $\rho = \mu$ with $\mu \in [0, 1)$ and $q \in (0, 1)$, and let z index the coupling as in Example 1, with z interior to a non-degenerate coupling interval. Both restrictions are what make a derivative in z describe anything: at $q \in \{0, 1\}$ there is one type and no sorting to vary. On a non-vanishing branch with $1 + \mu\, h_M > 0$,*

$$\frac{dY}{dz} \;=\; -\,c(z)\,Y, \qquad c(z) \;=\; \frac{(1-\mu)\,\psi\,\bar{\lambda}}{h\,D\;(1+\mu\,h_M)} \;>\; 0, \tag{37}$$

a linear homogeneous differential equation in the coupling, with $c(z) > 0$ requiring $\psi > 0$ and $\mu < 1$ in addition to $h > 0$, $\bar{\lambda} > 0$ and $D > 0$.

Proof. Once composition responds, h is a function of the coupling, so the derivative of (8) is total,

$$\frac{dM_S}{dz} \;=\; -(1-\mu) \;-\; \mu\,\frac{dh}{dz},$$

and substituting $dh/dz = h_M\, dM_S/dz$ and solving gives $dM_S/dz = -(1-\mu)/(1+\mu h_M)$, which requires $1 + \mu h_M \neq 0$ for the local inverse to exist. The chain rule with dY/dM_S from Theorem 9 then delivers (37). □

The term in h_M is the composition response, and it is absent only when composition does not move. In the reference economy $\xi = 1$ forces $h_M = 0$, so $c(z)$ reduces to $(1-\mu)\psi\bar{\lambda}/(hD)$ and the

numerical results are unaffected. Away from that case the correction is present, and treating the partial derivative of (8) as the total one would misstate the coefficient.

Two remarks remain. First, the feasible set for the coupling,

$$\mathcal{Z} \;=\; \{z \;:\; 0 \le z \le q, \quad z \le h(z), \quad z \ge h(z) + q - 1\}\,, \tag{38}$$

is the image of $\mathcal{M}$ under the map that inverts (8), $z(M_S) = [\,q + \mu(1-q) - \mu\,h(M_S) - M_S\,]\,/(1-\mu)$, which on a regular branch is single-valued and continuous, since Assumption 1 makes h a continuous function of the statistic there. A continuous image of a connected set is connected, so connectedness of $\mathcal{M}$ delivers connectedness of $\mathcal{Z}$. The argument has to run in this direction. Once composition responds, M_S is not an affine function of z, by the same total derivative just recorded, and preimages of intervals under a non-affine map need not be intervals. When composition does not respond, h is constant and (38) reduces to the Fréchet interval (56). In the two-type analysis on its own that had to be assumed. Second, (37) is signed by the cancellation in Theorem 8, which uses only the household budget constraint and stationarity, never the binary structure.

The theory says which way the identified set moves and, under the conditions of Remark 3, when it has positive width. Whether that width is large enough to matter needs numbers. Every quantity that follows is computed from assumed primitives, so the economy is an illustration. The assignments compared below are all stationary states of this economy, since $h < q$ and $h + q < 1$ keep them inside the range Lemma 1(ii) covers, and they differ only in who is exposed to becoming constrained.

Table 1: The reference economy

	Economic meaning	Value	Role in the theory
ξ	Income incidence on constrained households	1.0	Enters the demand channel G_h
ψ	Forward-looking elasticity of savers	0.5	Assignment matters only if positive
μ	Discounting of news by the alternative	0.3	Assignment matters only if below one
σ	Interest elasticity of savers	1.0	Scales the policy experiment
r	Real rate deviation, the experiment	-0.02	Policy experiment, direct demand impulse
q	Prevalence of the rational mechanism	0.5	One marginal of the assignment
θ	Hazard sensitivity to saver saving	0.3	Strength of the wealth channel
λ_{out}	Exit rate from constrained status	0.3	Distributional turnover
$\bar{\lambda}$	Baseline inflow into constrained status	0.15	Fixes W through $h\,W = \bar{\lambda}$
Implied objects			
Feasible assignment set $\mathcal{Z}$		[0.0000, 0.3333]	
Constrained share h on $\mathcal{Z}$		0.3333, constant	
Identified set for Y		[0.02623, 0.03404]	
Width of the identified set		0.00781, or 25.93 percent of the interval midpoint	

Notes. The parameter vector is a reference point for a theoretical illustration and is not estimated, calibrated, or matched to any economy. The response variables are deviations from a steady state and $r < 0$ is a monetary expansion. The shares h and q and the two intensities are levels. The endpoints of $\mathcal{Z}$ are located by bisection on the feasibility restrictions in (56). With $\xi = 1$ the scalar W carries no term in Y, so $h\,W = \bar{\lambda}$ fixes $h = \bar{\lambda}/(\lambda_{\text{out}} + \bar{\lambda})$ at every assignment.

The reference economy has $\xi = 1.0$, which has one consequence. With $\xi = 1$ the scalar W in Theorem 8 carries no term in the aggregate, so the feasibility identity pins

$$h \;=\; \frac{\bar{\lambda}}{\lambda_{\text{out}} + \bar{\lambda}} \;=\; 0.3333, \tag{39}$$

independently of the assignment and of θ. Composition therefore does not move in this particular economy, which means the reference economy cannot exhibit the endogenous-feasibility mechanism. That mechanism is illustrated separately in Appendix B using an economy with $\xi \neq 1$.

The generalization to arbitrary responsiveness distributions makes a comparison available that the two-type model cannot pose. Hold the primitives fixed, hold the mean of responsiveness fixed, and vary only how dispersed responsiveness is. The two-type benchmark of the reference economy has mass 0.5 at $\rho = 1$ and mass 0.5 at $\rho = 0.3$, so mean responsiveness is 0.65. Two comparisons keep that mean.

The first replaces the two atoms by a uniform law on the interval they bound, $\rho \sim U[0.3, 1]$. Its mean is $(0.3 + 1)/2 = 0.65$ by construction, so no economic parameter is added and nothing is recalibrated. The only change is that responsiveness is continuous instead of concentrated at the two endpoints of its support. The second collapses the distribution entirely, $\rho \equiv 0.65$, which is the case of a common responsiveness parameter and is the limit in which assignment stops mattering.

Table 2: Identified sets under three responsiveness marginals with the same mean, closed where the endpoints are limits

Responsiveness marginal	Mean	Support	$\underline{Y}$	$\overline{Y}$	Width over midpoint
Two-point	0.65	$\{0.3, 1\}$	0.02623	0.03404	25.93 %
Uniform	0.65	$[0.3, 1]$	0.02727	0.03243	17.28 %
Common	0.65	$\{0.65\}$	0.02963	0.02963	0.00 %

Notes. All three rows use the reference economy of Table 1 without change, and all three hold mean responsiveness at 0.65. The first two range over $[0.3, 1]$, the two-point law placing its mass at the two ends of it and the uniform spreading mass over the whole of it. Bounds are the quantile integrals of Theorem 2 evaluated at the pinned h of (39), mapped through the equilibrium. The first row reproduces the two-type benchmark exactly.

Holding mean responsiveness fixed, the macroeconomic uncertainty generated by an unknown assignment depends on the dispersion of responsiveness that can be sorted across economic states. The two-point distribution places all its mass at the extremes of the support and gives the widest set among these benchmarks (Table 2), at 25.93 percent of the midpoint of the identified interval. Spreading the same average responsiveness continuously over the interval those atoms bound narrows the set to 17.28 percent. Common responsiveness removes assignment uncertainty altogether. The three sets are nested, which is easier to see on a common scale than from the endpoints (Figure 5).

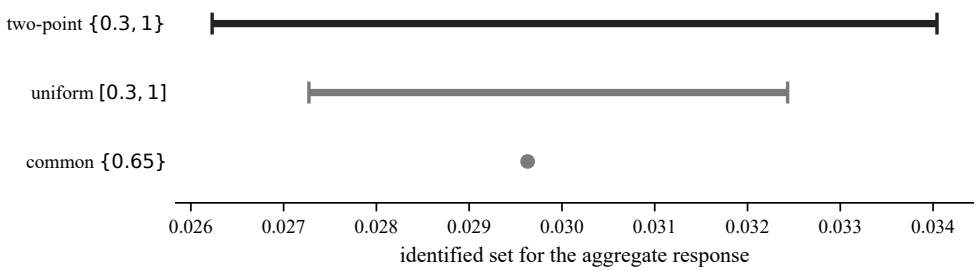


Figure 5: The identified set and the dispersion of responsiveness

Notes. Each segment is the identified set for the aggregate, or its closure where the endpoints are sharp bounds that no stationary assignment attains, under one responsiveness marginal, drawn on a common horizontal scale. Mean responsiveness is the same in all three and the first two share a support. The bottom row is a point.

The ordering extends beyond these three. By Theorem 1 the equilibrium sees the marginal only through the statistic, so on the regular branch $h(M_S)$ and $Y(M_S)$ are the same functions whatever F_ρ is, and the marginal enters the feasible set (14) only through the rearrangement bounds. Corollary 1 places those bounds, at every composition, inside the bounds of the two-point law with the same mean and support. The feasible set of any responsiveness distribution on $[0.3, 1]$ with mean 0.65 is therefore contained in that of the two-point law, and so is its identified set for the aggregate, whether or not composition responds. In the reference economy the corollary gives the statistic the interval $[0.317, 0.550]$, which is the pair of extremal assignments of the two-type benchmark, so the 25.93 percent of the first row is the largest width any marginal with that mean

and support can produce. The dichotomy is the least favourable case among all of them, since all the responsiveness sits at the two ends of the support and can be moved furthest between states.

The uniform case holds the primitives and the mean of the two-type benchmark and adds no parameter, so 17.28 percent illustrates the mechanism. Its endpoints are the sharp bounds of Theorem 2 and are never attained as stationary states, since a continuous marginal puts an interval of responsiveness entirely into the constrained state at either extreme, which Lemma 1(iii) rules out. The row reports the closure of the stationary identified set.

The width states ambiguity alone, so the error a user of such a model makes is a separate quantity. Read the reference experiment as a rate change of 2 percent. Across the identified set the output response runs from 1.31 to 1.70 times that change.

Someone who does not observe the assignment has a ready default, which is to impute the joint distribution by matching the two marginals. That is the independent assignment, and it delivers 1.48. The truth can then sit anywhere from -11.5 to +14.9 percent away from it. The identified set does not contain zero, so no admissible assignment reverses the direction of the experiment. The ambiguity is about magnitude.

The imputation does more than misstate a magnitude. At $\xi = 1$ it sets $M_S = (1-h)\mathbb{E}[\rho]$, so the denominator of (27) becomes $(1-h)\,(1-\psi\mathbb{E}[\rho])$ and the constrained share cancels against the numerator. Moving $\bar{\lambda}$ so that the share ranges over $[0.062, 0.800]$ leaves the aggregate response unchanged to 6.9×10^{-18}. Under an imputed assignment it does not matter whether six percent or eighty percent of households are hand to mouth. In this economy every effect the constrained share has on the aggregate runs through the assignment, which is the object that is not identified, so imputing it deletes the channel instead of approximating it.

Assuming the expectations heterogeneity away is a different error and a larger one. With every household rational the response is 2.00, which is 17.5 percent above the upper endpoint of the identified set and therefore outside it. That comparison changes the responsiveness marginal and leaves the assignment, and it says that heterogeneity moves the answer whatever the sorting turns out to be.

Appendix D moves the two parameters that Theorem 9 and Corollary 5 single out, re-solving the equilibrium at each value so that h and D are free to respond. The width falls to zero as the alternative mechanism approaches the responsive one and grows with the strength of the forward-looking channel. The appendix adds the income incidence ξ, the one parameter that decides whether composition responds at all.

Proposition 7 (The coefficient in primitives)**.** *In the economy of this section, on a branch satisfying Assumption 1, the coefficient in Proposition 4 is*

$$c \;=\; \frac{\psi\,\bar{\lambda}}{h\,D}. \tag{40}$$

If in addition the conditions of Proposition 4(iii) hold at the unsorted point, so that it is interior to a non-degenerate feasible interval with $1+\mathbb{E}[T^C]h_M \neq 0$*, the band is small enough that the relevant*

component does not meet the boundary of that interval, and $Y \neq 0$ there, then

$$\frac{W_Y(\delta)}{|Y|} \;=\; \frac{2\,\psi\,\bar{\lambda}}{h\,D\;|1+\mathbb{E}[T^C]h_M|}\,\delta \;+\; o(\delta), \qquad h_M = 0 \quad \textit{when} \quad \xi = 1. \tag{41}$$

None of those conditions follows from Assumption 1. A degenerate F_ρ satisfies it while feasibility pins a single assignment, so every band gives zero width and no positive coefficient can describe it.

Proof. The comparative static is $dY/dM_S = N/D$ from (13), with $N = \psi Y W$ by the factorization in Appendix A and $W = \bar{\lambda}/h$ by Theorem 8. So $dY/dM_S = cY$ with $c = \psi\bar{\lambda}/(hD)$, which is the hypothesis of Theorem 7 with that coefficient and gives (40). Under the further conditions stated, Proposition 4(iii) applies and gives (41). □

Equation (40) says how much a given amount of ignorance about the matching widens the identified set. As algebra, c is proportional to $\psi\bar{\lambda}$ and inversely proportional to hD. With h and D held fixed, the same ignorance widens the set more when demand is more forward looking, when households move through the constrained state faster, and when the constrained share is smaller, the statistic then being a larger share of a smaller group. Holding the numerator bounded away from zero, c diverges as D approaches zero from above, which marks the neighbourhood Assumption 1 excludes and where general equilibrium feedback amplifies everything.

The practical reading runs the other way. A researcher who can bound the heterogeneity of the behavioural loadings obtains δ through Corollary 2 and then a band on the aggregate through (40). The corollary needs the state marginal: a known constant $\mathbb{P}(S)$ where composition is fixed, and otherwise its path along the branch or a uniform bound on it. If that band is too wide to be useful, (40) also says how much narrower it would become under a given improvement in the measurement of δ, which an assumption about the matching cannot supply.

4.1 The vector case

Remark 2 states the results above for $J = 1$. Finitely many moments still suffice above it, which is Proposition 2 and holds at any rank. What goes is the line. The statistic no longer sits on one, so the ordering behind Proposition 3 and Theorem 7 has no statement to generalise, neither the monotonicity of the aggregate nor the attainment of the bounds at two endpoints. Something else survives.

Theorem 10 (Vector equilibrium inversion)**.** *Let F_ρ be supported on K points with masses f_i, and let the admissible couplings be $\{c : 0 \le c_i \le f_i\}$ with $h = \sum_i c_i$, as in Proposition 2. Suppose that at a fixed aggregate y the equilibrium conditions can be written as expectations under the coupling,*

$$\mathbb{E}_c[g_k(s, \rho;\, y)] \;=\; b_k(y), \qquad k = 1, \ldots, m. \tag{42}$$

Write $P(y)$ for the set of admissible couplings satisfying (42) and $\mathcal{Y} = \{y : P(y) \neq \emptyset\}$. Then $P(y)$ is a polytope for every y, the set $\mathcal{Y}$ is the sharp identified set for the aggregate, and membership in

it is decided by a linear program in K variables with m equality constraints.

Proof. An expectation is linear in the law it is taken under, so

$$\mathbb{E}_c[g_k(\cdot\,; y)] \;=\; \sum_i \Big\{\, c_i\, g_k(C, \rho_i; y) \;+\; (f_i - c_i)\, g_k(S, \rho_i; y) \,\Big\}$$

is affine in c, and each condition in (42) is one hyperplane. The admissible couplings are a box. A box intersected with finitely many hyperplanes is a polytope, and deciding whether one is non-empty is a linear program. Sharpness holds by construction, since y is attainable exactly when some admissible coupling supports it as an equilibrium and the coupling set is never relaxed. □

Market clearing takes this form, being an average of household consumption, and so does the composition condition, being a balance of flows, provided the second is left as a balance and not solved for h, which introduces a ratio and destroys linearity. Those two are not the whole of it. Stationarity of the distributional state is a restriction for every responsiveness set and not only in aggregate, and by Lemma 1 it costs this economy one more: no responsiveness type may be entirely constrained. It is cheap here only because the inflow hazard may vary with responsiveness and the lemma pins nothing but its average. A specification that fixed the hazard by type instead would buy one balance for each type and could determine the coupling outright, leaving nothing to bound.

The polytope of the theorem is the closed set the equalities and the box define, and in the economy of this section it must additionally satisfy a strict support condition that stationarity implies. That condition is $c_i < f_i$, and being strict it does not cut a polytope, so the stationary couplings form a relatively open polyhedral set. Whether one exists at a given aggregate is still settled by a linear programme, through a common slack,

$$\max\ \epsilon \qquad \text{subject to} \qquad (42), \quad 0 \le c_i, \quad c_i \le f_i - \epsilon, \quad \epsilon \ge 0,$$

a stationary coupling existing exactly when the optimum is strictly positive. What strictness costs is attainment. The closed programme can return a bound no stationary coupling reaches, so the extremes of the stationary set may be a supremum and an infimum only, which is the vector form of the case met above under a continuous marginal, where the stationary identified set is the sharp interval without its endpoints.

Which restrictions belong in $P(y)$ is part of specifying the economy. Nothing in the proof refers to the rank or to the geometry of the moments, which is what makes it useful once that geometry fails, and the set it characterises is the equilibrium-compatible one, before any restriction that selects among equilibria.

What higher rank costs is therefore the scalar ordering behind Proposition 3 and Theorem 7, and not the sufficiency of finitely many moments or the sharpness of the characterisation.

5 Empirical restrictions on the assignment

5.1 What the framework predicts

The theory leaves one object unobserved and bounds everything through it. The statistic is

$$M_S \;=\; \mathbb{E}\big[T^C\,\mathbf{1}\{s=S\}\big] \;=\; \mathbb{E}\big[T^C\big]\,\mathbb{P}(S) \;+\; \mathrm{Cov}\big(T^C,\mathbf{1}\{s=S\}\big)\,,$$

whose first term is built from the marginals and whose second is the entire content of the matching. The marginals alone leave the statistic in an interval, and Proposition 4 turns a bound on the covariance into a bound on the aggregate. Where composition responds to the statistic the first term moves with the candidate, and the bound is the set (18) instead of a symmetric band. The economy of Section 4 is the benchmark in which it does not, and the numbers below move away from it, for a reason Proposition 5 gives. The question this section asks is therefore not whether households differ, which is a question about the joint distribution the theory does not need. It is how much of M_S a household survey can recover, and what the macroeconomic band left over is once it has.

The European Central Bank's Consumer Expectations Survey supplies the input. Two heterogeneity maps and an inverted test produce one observable number. Section 5.4 then asks what data would identify the statistic outright instead of bounding it, and reports an exercise on dated financial news whose classification is of that kind.

5.2 From an observable contrast to the statistic

The survey identifies an observable response contrast. The assignment statistic itself stays unmeasured. Five maintained conditions carry that contrast to the model. One is an exclusion and the other four are mappings.

The *exclusion* comes first because the others presume it. Country-by-round effects remove whatever the signal does to every cell alike, so what identifies the map is the difference across cells within a country and a round. Any channel that moves cells differently therefore enters the estimate, and a scale does not remove it: a scale multiplies the structural part and leaves the rest where it is. What is assumed is that, conditional on the included effects and on a predetermined cell, the differential association between the signal and the spending revision runs only through the news the loading responds to. Section 5.4 carries the same kind of restriction for its own design under the name exposure exogeneity.

The four mappings follow: an outcome and dynamics bridge from the spending question to consumption, a signal bridge from the observed macro series to the news entering the loading, a classification bridge from reported liquidity status to the structural state, and a representativeness bridge from the estimation sample to the model population. Each is stated below, and the calculations condition on them.

The *outcome* bridge runs from a spending question to consumption. Of the four spending

questions the two open-ended ones are used here, because a percentage change is closer to something the model can meet than an ordinal category is. It is not the identity, and the gap is more than one of units. Expenditure is not consumption, so the spending concept and the normalisation per household have to be reconciled. More than that, the outcome in (47) is a difference between consecutive rounds of a twelve-month growth rate whose windows overlap, so the coefficient is a revision in an annual rate over a window that overlaps the previous one. That rate is a realised one in the retrospective question and an expected one in the forward-looking question, and both are used. The bridge is therefore an operator,

$$\Delta_{\text{obs}} \;=\; \mathcal{L}_C(\text{structural consumption response})\,, \tag{43}$$

with $\mathcal{L}_C$ carrying the twelve-month horizon, the overlap of the windows, expenditure against consumption, and any difference in timing and normalisation. It collapses to a scalar under a proportionality restriction on the response. The alternative question is ordinal, and its scale carries no structural meaning under any scoring, so it does not discipline the band. It does corroborate the pattern: with the categories placed in economic order and scored three ways, one of them compressing the extremes and one expanding them, no configuration rejects across either ordinal outcome or either signal family, so the absence of detectable heterogeneity is not an artefact of the scoring.

The *signal* bridge, $\mathcal{L}_Z$, runs from one standard deviation of an observed macro series to one unit of the news the loading responds to, and is an operator for the same reason: an observed series is not an identified shock and its comovement with the aggregate is not a structural response. Where it is represented by a scale, fix the orientation once. Let κ be measured in units of the structural loading per unit of the observed response, so that the bound multiplies by κ. If one instead writes the model's news as κ_Z times the observed signal, then κ is the reciprocal of κ_Z and a larger κ_Z tightens the bound. Only the product of the outcome and signal scales enters, so they are carried as one composite.

The *classification* bridge runs from an observable partition to the structural state. The theory writes $s = S$ for the unconstrained household, so write $\mathbf{1}_C$ for the indicator of the constrained one and note that the two indicators sum to one, so $\operatorname{Cov}(T^C, \mathbf{1}\{s = S\}) = -\operatorname{Cov}(T^C, \mathbf{1}_C)$ and the two magnitudes coincide. Consistently with (17), $\operatorname{Cov}(T^C, \mathbf{1}_C) = M_0 - M_S$. The bridge is stated for $\mathbf{1}_C$, which the observable partition proxies. The letter D is reserved for the equilibrium determinant of (13).

Assumption 2 (Non-differential classification)**.** Let $\widehat{C}$ be the classification, with $\mathbb{P}(\widehat{C} = 1 \mid \mathbf{1}_C = 1) = 1 - e_1$ and $\mathbb{P}(\widehat{C} = 1 \mid \mathbf{1}_C = 0) = e_0$, and let

$$\mathbb{E}\Big[\widehat{C} \mid y, T^C, \mathbf{1}_C\Big] \;=\; \mathbb{E}\Big[\widehat{C} \mid \mathbf{1}_C\Big] \;=\; e_0 + (1 - e_0 - e_1)\,\mathbf{1}_C. \tag{44}$$

The measured response y appears in the conditioning set alongside the loading, and both are needed. Conditioning on the loading alone delivers the identity for a contrast in loading units

and not for one in the units the survey measures, since the outcome bridge does not make the response a function of the loading and the state. A classification that reads the response within a true state satisfies the weaker version exactly and still breaks the correction: with a constant loading, rates of a quarter each and a response independent of the state, the weaker version holds by construction while the correction returns a contrast of one half where the truth is zero. The two rates alone deliver neither version. They fix $\mathbb{E}[\widehat{C} \mid \mathbf{1}_C]$ and leave the classification free to read the loading or the response within a value of the true state. Independence of $\widehat{C}$ from (y, T^C) given the state is sufficient and stronger than needed. The response channel is the one that binds here, since the outcome bridge does not make the response a function of the loading and the state, so a household that misreports its liquidity for reasons connected to how much its spending moved breaks the correction even when the loading is uninformative. Taking the loading channel as the tractable case and writing $e_0(\rho) = e_0 + \tau\,(\rho - \mathbb{E}[\rho])$ for the differential part with e_1 fixed, the contrast recovered below acquires the additive term $\tau\psi\dot{Y}(1-h)\,\mathrm{Var}(\rho)/\,[h(1-e_0-e_1)]$, which is present even when the assignment is independent. At $\tau = 0.1$ it is 0.0261. The contrast it sits against never falls below 0.7016 over the swept assignments in the economy the exercise uses, and the artefact averages 3 percent of that contrast, and at the benchmark of Proposition 5 it is the whole of the recovered contrast, because the true one is zero. Assume also

$$e_0 + e_1 < 1, \tag{45}$$

which costs no generality because $e_0 + e_1 > 1$ means the proxy is negatively related to the state and one relabels it, writing $\Delta_T = \mathbb{E}[T^C \mid \widehat{C} = 1] - \mathbb{E}[T^C]$ for the contrast in loading units between the classified group and the average gives

$$\left|\mathrm{Cov}\big(T^C, \mathbf{1}_C\big)\right| \;=\; \frac{\gamma\,|\Delta_T|}{1 - e_0 - e_1}, \qquad \gamma = \mathbb{P}(\widehat{C} = 1), \tag{46}$$

with γ the share classified as constrained. The survey does not observe Δ_T. It observes a spending contrast, and (50) below is this same correction written in those units. The latent constrained share $\mathbb{P}(\mathbf{1}_C = 1)$ cancels. A state that is structurally unobserved has no identified prevalence, so what the sample measures is the share *classified* as constrained. Condition (45) carries the content. As $e_0 + e_1$ approaches one the bound diverges, because at that point $\widehat{C}$ is independent of $\mathbf{1}_C$, the classification carries no information about the structural state, and it cannot bound the matching at all however sharp the observable contrast happens to be.

The *representativeness* bridge runs from the estimation sample to the population the model describes. The covariance in (17) is an expectation over that population, while every ingredient of the observable contrast, the response map and the cell shares alike, is computed on the sample of consecutive-round revisions. The bridge is the equality of the two, that Δ_{obs} evaluated on the estimation sample equals its counterpart on the population the model describes. Preserving the cell shares under selection is not sufficient for that, since selection can move the responses inside a cell and so move the contrast itself. The survey supplies a respondent weight calibrated to the adult

population and the estimates here do not use it, and Appendix E shows that neither weighting reproduces the model's population. It is an equality, so it conditions the schedule below.

In order, the steps between the inference and the aggregate are

$$\underbrace{\mathcal{C}_R}_{\text{inference}} \longrightarrow \underbrace{d_{\text{obs}}}_{\text{data}} \longrightarrow \underbrace{\mathcal{L}_C}_{\substack{\text{outcome and}\\ \text{dynamic bridge}}} , \underbrace{\mathcal{L}_Z}_{\text{signal bridge}} , \underbrace{(e_0, e_1)}_{\text{state bridge}} , \underbrace{\mathcal{W}}_{\substack{\text{representativeness}\\ \text{bridge}}}$$

$$\longrightarrow \underbrace{\mathcal{M}^{\text{data}}(\kappa)}_{\substack{\text{assignments the bound}\\ \text{leaves admissible}}} \longrightarrow \underbrace{Y\Big(\mathcal{M}^{\text{data}}(\kappa)\Big)}_{\text{the aggregate}} .$$

The outcome and signal bridges enter the calculations only through the single composite κ, which is represented as one scalar sensitivity, and the state bridge is exact given its two rates together with Assumption 2.

5.3 The Consumer Expectations Survey

The Consumer Expectations Survey is a monthly panel of households in eleven euro area countries, Austria, Belgium, Finland, France, Germany, Greece, Ireland, Italy, the Netherlands, Portugal and Spain, running from April 2020. It carries 1,315,684 household-month observations on 156,554 households over seventy five rounds, and the master file of consecutive-round revisions, from which each map takes the observations its own outcome and signals leave, is 1,072,891 observations on 87,893 households. Expectations and spending are recorded for the same household in the same month, which is what allows the two maps to be compared: the comparison below is estimated on the households that answer both. Estimated separately the two use their own samples and those differ, because the open-ended spending questions carry a non-response code often enough to cost about one revision in five and a half, which is roughly one round in ten, while the expectation questions carry none.

Three things define the exercise. The economic state is proxied by a household's own report of whether it has sufficient liquidity, a binary question about meeting an unexpected expense, taken in the previous round so that it is predetermined with respect to the signal. That is crossed with the household's disposable income quintile, computed within country and round, giving ten cells. The two are related without being the same thing: within the constrained group the cell weights fall monotonically across quintiles, from 8.5 percent of the sample in the lowest to 2.3 percent in the highest, and within the unconstrained group they rise from 9.5 to 19.9 percent. Liquidity is correlated with income and is not income under another name. The outcome is household spending, which the survey asks about in four ways: a retrospective and a forward-looking question, each in an open-ended form recording a percentage change over twelve months and in an ordinal form recording a category. The two open-ended measures are the ones used and the two ordinal ones corroborate. And three kinds of variation are used. The first is external, the monthly change in the country's annual HICP inflation rate and in its unemployment rate, taken from the ECB Data

Portal and dated so that the release had occurred before the interview. The second is external too, the intraday surprise around Governing Council announcements from the euro area monetary policy database of Altavilla et al. (2019). It is run over the same grid of twenty-four configurations and detects nothing either, before or after a multiplicity correction, so Table 3 reports the price and unemployment family. Appendix E returns to the surprises for what is specific to them, that the timing rule binds hardest there and that rotating them leaves the statistic unchanged. The internal one, which we call survey-common variation, is the round-level mean revision of an expectation across all households, so it is the common movement in the survey's own measured expectations, which no identification makes a shock.

What the survey measures is a response, estimated within respondent and between consecutive rounds,

$$\Delta C_{ict} \;=\; \alpha_i + \tau_{ct} + \sum_{g,q,k} \beta_{gqk}\, \mathbf{1}_{igq}\, Z^k_{ct} + \varepsilon_{ict}, \tag{47}$$

where $\mathbf{1}_{igq}$ picks out the cell with liquidity classification g and income quintile q, both dated $t-1$. The classification is the binary $\widehat{C}$ the bridges of Section 5.2 and Appendix E use, so the ten cells are its two values crossed with the five quintiles. The household state is predetermined at $t-1$, the signal is dated so that it had been published before the interview, and country-by-round effects absorb every movement common to a country in a month. What is identified is whether cells inside the same country and month respond differently, so the estimand is the coefficient matrix demeaned across cells. The primary spending outcomes are the two open-ended questions, which record a percentage change. The two ordinal questions corroborate and nothing rests on them, since a difference of an ordinal index is not a change in consumption under any scoring.

Table 3 reports what the two variations reveal. Under an external anchor nothing is detectable on either side. Under the survey's own common variation two expectation-map tests reject while the same variation reveals nothing in spending, across twelve configurations built from the two open-ended measures. Those estimates are diagnostic, because their signal is built from the outcome sample itself, so every structural bound below uses the external-signal specification and Appendix E reports what the survey-common one does under a construction that separates the signal from the estimation sample.

Comparing the two maps to one another does not change that. Separate tests against zero do not establish that the maps differ, and neither does a test of equality between them, since the outcomes are measured in different units and (43) leaves the bridge between them unidentified. What survives a change of units is whether one scalar bridge reconciles them, and that is not rejected in any of the *four* pairings, with the inverted sets for the scalar all containing zero. Table 9 reports them. The bound below uses the external-signal specification of the first row. It comes from inverting a test on the projected contrast instead of on the spending map itself, for the reason given after the table, and the third row reports what that inversion returns.

Table 3: The two heterogeneity maps and the observable bound they yield

	p-value for $H_0 : R = 0$		
source of variation	expectations	spending	interpretation
external macro signals	1.0000 / 0.9117	0.9512	nothing detectable
survey-common variation	0.0013 / 0.0064	0.0649	generated signal, a diagnostic and not evidence
	ninety-five percent upper bound		
observable contrast	—	$d_{\text{obs}} = 0.1778$	the input to what follows

Notes. Entries are p-values for $H_0 : R = 0$ on the cell-demeaned map, with cells given by liquidity crossed with income lagged one round. Under external signals they are the largest over the segment of timing mixtures with one share common to every country and round, which requires both candidate signals for the same observation and so uses 999,144 revisions for the inflation-expectation map, 999,161 for the income-expectation map, 813,014 for the expected-spending map and 809,828 for the retrospective-spending map, which is the one the bound below is attained on. The survey-common construction is contemporaneous and admits no timing choice. The spending column is the smallest p-value over the two open-ended measures, and under survey-common variation over the six signals crossed with them. The reported entries use the rule of Davezies et al. (2025) without modification, which sets the two-way arm aside when its own statistic is negative. A variant also sets it aside wherever the estimated two-way matrix fails to be positive definite. That variant is reported in Appendix E. Removing an arm from a minimum can only raise the statistic, so the variant makes rejection easier, which is the unfavourable direction for a finding that is a non-detection: it gives 0.5716, 0.9117 and 0.8412 in the same three cells, so non-detection does not turn on the choice. The bound is the largest value of the economic contrast the data cannot rule out at ninety-five percent, maximised over the timing segment with one share common to every country and round, and is an observable quantity throughout. Table 11 reports how it moves when the share may differ by country or by year, as a numerical sensitivity. Its units are percentage points of a twelve-month spending revision per standard deviation of signal, which is the scale of the non-detection: contrasts above 0.1778 in those units are the ones this realised confidence set rules out. On the sample supporting it, 0.2533 of households are classified as constrained. That share describes the estimation sample, and the bound reported here restricts an observable contrast. Appendix E gives the inference and the ordinal corroboration.

The dependence is two-way, since households recur across rounds and share them. We use the procedure of Davezies et al. (2025) because it is the only candidate with approximately correct size in both validation designs, and Appendix E reports the comparison. Since the interview day is not recorded, inference is required to hold under every admissible signal timing.

The survey records the round of interview but not the day, so respondents inside one country and round may have seen different releases and the true signal is a household-level mixture. Write d_{ct} for the difference between the two candidate signals and L_i for the indicator that respondent i had seen the later one, so the true regressor is $Z_{c,t-2} + d_{ct}L_i$ and its interaction with the cell is $\mathbf{1}_{igq}Z_{c,t-2} + \mathbf{1}_{igq}d_{ct}L_i$. If the chance of having seen the later release does not depend on the cell within a country and round, the second piece has conditional mean $\mathbf{1}_{igq}d_{ct}\pi_{ct}$ and the rest is orthogonal noise, so the mixture is a pure design with the convex signal

$$Z(\pi) \;=\; (1-\pi)\, Z_{c,t-2} \;+\; \pi\, Z_{c,t-1}. \tag{48}$$

The share is π_{ct} in the derivation and a single π in the signal, which restricts it to be the same in every country and round. The nuisance is then π itself, a single number, and the two designs

the lags name are its endpoints. Every configuration is evaluated along the whole segment and summarised by

$$p_{\text{robust}} \;=\; \max_{\pi \in [0,1]} p(\pi), \tag{49}$$

which rejects only if every admissible mixture does. Taking the endpoints alone would not deliver that, and the difference matters for the bound alone: the p-value is maximised at an endpoint, while the bound is maximised in the interior, at 0.45, where it is 0.1778 against 0.1345 at the endpoints. The rule binds: in the monetary-surprise family the liquidity-by-income cells on the ordinal spending measure reject at one month and come nowhere near rejecting at two.

A single share is a restriction, and the bound is sensitive to it. Table 11 in Appendix E relaxes it in two directions, letting the share take a separate value in each of the eleven countries or in each of the seven survey years. The two-way fixed effects enter linearly, so each block's interaction is residualised once and the bound can be recomputed for any vector of shares. It is maximised by coordinate ascent from six starting points and checked by a second search over every vertex of the share box and 500 uniform draws. On the retrospective measure the largest bound found is 0.1960 with shares by country and 0.1914 with shares by year, against 0.1778 with one share. On the expected measure the three figures are 0.1551, 0.1469 and 0.1260. These are numerical maxima. They bound the supremum over each family from below and are not certified, since no bound on how fast the objective varies between evaluated points is computed, and on the expected measure the two searches stop at different local maxima. Neither family covers shares that differ arbitrarily by country and round, which would need one share per country-round cell and is not searched here. Because a numerical maximum can fall short of the supremum, it cannot serve as a confidence bound for its family, and the largest figure found, 0.1960, 10 percent above the one-share value, is reported as a sensitivity. The confidence statements below use the one-share bound and are made for the single-share family only. That bound is itself located numerically, on a grid whose refinement in Appendix E leaves it unchanged at the reported precision, which is numerical stability and not a certified supremum over the segment. A valid bound would be at least 0.1960 over shares by country and at least 0.1914 over shares by year. The p-values of Table 3 keep the single share. Enlarging the family can only raise a maximum p-value, so the non-detections reported there survive either relaxation.

The bound that feeds the proposition comes from inverting the test instead of from an estimate. Inversion gives a confidence set for the map, and the observable analogue of the matching contrast is a linear functional of the estimated response map, namely the difference between the average response in the classified group and the average response overall. The equilibrium uses that functional, and the functional has one coordinate per signal, *two* of them here against *eighteen* for the map. The inference is therefore conducted on the projected contrast directly. Inverting a simultaneous test on the whole map and reading a low-dimensional functional off it is conservative for a reason that has nothing to do with the economics: it pays for the directions of the map that the aggregate never loads on. Appendix E gives the two constructions and reports what the

whole-map version delivered.

Over the two open-ended measures the timing-robust bound on the observable contrast is 0.1778 per standard deviation of signal, the largest value (62) that the confidence set for the projected contrast admits, maximised over the timing segment with one share common to every country and round and attained on the retrospective question, with 0.2533 of the sample classified as constrained on the sample that supports it. The reported construction is the componentwise one, whose validity conditions are the weaker of the two. With one share the single-ellipsoid majorant gives 0.1672 against 0.1778 and is reported as a check.

What that bound identifies is not the covariance of (46). The outcome is a spending response and the object the theory needs is a loading, so the two are connected by the model. Under Assumption 2, writing y for the measured response, $\mathrm{Cov}(y, \widehat{C}) = (1 - e_0 - e_1)\,\mathrm{Cov}(y, \mathbf{1}_C)$, and expanding both covariances in terms of group contrasts,

$$\Delta_{\text{true}} \;=\; \frac{\gamma\,\Delta_{\text{obs}}}{h\,(1 - e_0 - e_1)}, \qquad \gamma \;=\; (1 - e_1)\,h + e_0\,(1 - h), \tag{50}$$

where Δ_{obs} is the measured contrast between the classified group and the sample and Δ_{true} is the model's contrast between the constrained group and the population. The classified share is not free: the constrained share and the two rates imply it. Under perfect classification $\gamma = h$ and the correction is one.

Section 4 has already settled what Δ_{true} is in the two-type economy this section works in. It is the contrast (34) under the rule $\dot{z} = 0$, fixed before any survey is read. The object the spending contrast disciplines is not the covariance $\mathrm{Cov}(T^C, \mathbf{1}_C) = M_0 - M_S$ up to any scale. The covariance vanishes at the independent assignment, which is interior to the feasible set. In each economy used below the contrast vanishes nowhere on that set. A positive scale carries zeros to zeros, so it cannot carry one into the other. Both objects restrict the assignment, and they restrict it differently.

The sign of the contrast is not governed by the incidence alone. Differentiating $h = \bar{\lambda}/W$ with $W = \lambda_{\text{out}} + \bar{\lambda} + \theta Y(\xi - 1)$ gives $\dot{h} = h^2\theta(1 - \xi)\dot{Y}/\bar{\lambda}$, and substituting into (34) with $\dot{z} = 0$ factors it exactly,

$$\Delta_{\text{true}} \;=\; (1 - \xi)\,\dot{Y}\left\{\frac{h^2\theta}{\bar{\lambda}}\,(\xi Y + \sigma r - \mu\psi Y) - 1\right\}. \tag{51}$$

Three terms govern it. The reversal at $\xi = 1$ is an identity, since the first factor vanishes there and so does $\dot{h}$. Which side is positive is not: it depends on the sign of $\dot{Y}$ and on whether the brace crosses zero, and both are properties of the economy. In the economies swept below the brace stays in $[-1.004, -0.974]$ and $\dot{Y}$ is negative throughout, so the contrast is positive below proportional incidence and negative above, and Figure 6(a) shows that. An economy with a large enough aggregate response puts the brace above zero and reverses the pattern.

Proposition 5 rules the benchmark out as the economy to read a survey in, and it also accounts for a coincidence in Appendix B: the economy in which the frozen and equilibrium-consistent calculations agree is the economy in which the spending contrast is orthogonal to the assignment. A survey measuring that contrast without error would still learn nothing about the assignment.

Figure 6(a) shows the contrast at five assignments spanning the feasible set: every one crosses zero at $\xi = 1$ and reverses sign there, so the degeneracy is a property of the incidence itself.

The economy of Section 4 therefore serves here as an analytical benchmark. The quantitative exercise is run instead as a sensitivity in the income incidence, over the grid Table 6 already sweeps, along which the marginals-only width runs from 18.04 percent at $\xi = 0.4$ to 51.97 percent at $\xi = 1.8$.

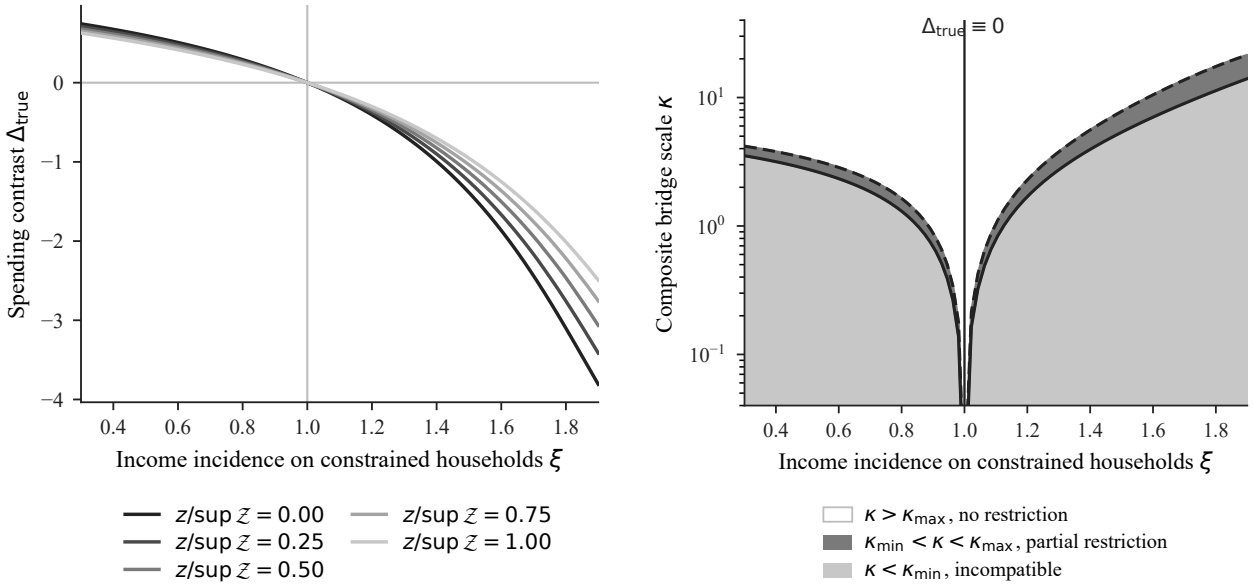
At each incidence the observable bound leaves admissible the assignments

$$\mathcal{M}^{\text{data}}(\kappa) \;=\; \left\{ m \in \mathcal{M} \;:\; |\Delta_{\text{true}}(m)| \;\leq\; \frac{\kappa\,\gamma\,d_{\text{obs}}}{h(m)\,(1 - e_0 - e_1)} \right\}, \tag{52}$$

which replaces the symmetric band the covariance would have delivered. Writing $\kappa_{\min}$ and $\kappa_{\max}$ for the smallest and largest value the ratio on the right must take to admit an assignment, three regimes follow. Below $\kappa_{\min}$ no assignment survives, so the model, the bridges and the data are jointly incompatible. The theoretical identified set is never empty. Between the two the bound excludes some assignments and leaves the rest. Above $\kappa_{\max}$ nothing is excluded and the marginals are all that bind. Figure 6(b) maps the three regions.

The level of κ carries a change of units as well as a bridge, so only the ratio of the two thresholds is interpretable, never their position. That ratio is the factor in the unidentified scale over which the survey discriminates at all, and it is narrow. Under perfect classification it is 1.20 at $\xi = 0.4$, 1.22 at $\xi = 0.6$, 1.42 at $\xi = 1.4$ and 1.54 at $\xi = 1.8$. The window is wider above proportional incidence than below it, and below the benchmark it narrows as incidence falls, so it does not order by distance from the benchmark. Misclassification moves both thresholds by nearly the same factor and leaves the ratio almost unchanged, at 1.41 and 1.41 for $\xi = 1.4$ where the two rates sum to one fifth and two fifths. The thresholds are proportional to $1/d_{\text{obs}}$. Under the one-share bound the window is $[3.16, 3.79]$ at $\xi = 0.4$ and $[11.31, 17.44]$ at $\xi = 1.8$. Evaluated at the block-wise sensitivity value both thresholds fall by the factor 0.907, to $[2.87, 3.44]$ and $[10.26, 15.82]$, and Table 11 reports the four incidences. The ratio is the same under any value of d_{obs}, which follows from the proportionality and is no separate confirmation. These shifted levels carry no coverage guarantee, for the reason given in Section 5.3. Appendix D plots how fast the width falls across that window, and Table 7 there reports which combinations of incidence, position in the window and classification error produce a substantial reduction.

One more distinction belongs here. The quantity d_{obs} is the upper end of a ninety-five percent confidence set for the observable contrast, with the sampling error that implies, and $\mathcal{M}^{\text{data}}(\kappa)$ is built from it. Proposition 4 is stated for a known bound, so the set the data leave admissible is not an identified set. It is a set that covers the identified set for the aggregate with asymptotic probability at least ninety-five percent, at the level inherited from d_{obs}, with no adjustment for the intersection and conditional on the bridges holding exactly (Imbens and Manski, 2004; Chernozhukov et al., 2007). The marginals-only widths are of the other kind: they are computed from assumed primitives, carry no sampling content, and are identified-set widths.



(a) Structural sensitivity of the spending contrast

(b) When survey information restricts assignment

Figure 6: Whether the spending contrast can discipline the assignment

Notes. Every parameter other than ξ is held at its value in Table 1. Panel (a) plots (34) with $\dot{z} = 0$ at five assignments, indexed by position in that economy's own feasible interval. Panel (b) plots the two thresholds in the composite scale κ that (52) implies under perfect classification, on a logarithmic axis. Both fall to zero as ξ approaches one and leave the plotted range there. At $\xi = 1$ the spending contrast is identically zero for every admissible assignment even though the assignment continues to move the aggregate. Away from that benchmark the contrast regains sensitivity to the assignment, and depending on the quality of the bridge the data either reject the maintained bridge and model combination, partially restrict the feasible assignments, or add no restriction. The regions in panel (b) are conditional on the representativeness bridge and on the classification of Assumption 2, and they are statements about a ninety-five percent confidence bound on the observable contrast.

5.4 What would identify the statistic

The survey does not identify the statistic, and precision is not what stands in the way. Theorem 4 and Theorem 5 are the two poles. A classification unrelated to the economic state leaves the identified set exactly where the marginals put it, at any sample size, and one whose groups lie inside the states delivers the statistic outright from group shares and group averages, with no distributional object and no panel. Between them is everything Theorem 3 covers, where a restriction tying the groups to the states shrinks the identified set without closing it. The classification the survey supplies is intermediate in that way, tied to the state without lying inside it. The poles are separated by a property of the classification, so the binding constraint is one of measurement. A register recording liquid balances and income at the household level can implement the split between constrained and unconstrained households directly instead of proxying it, and a repeated cross-section carrying the same information suffices, since the theorem needs group averages and never a within-household object.

The requirement has been met in practice, on data other than the survey above. Ekici et al. (2026) study how household expectations respond to financial-crisis events in the Consumer Fi-

nance Monthly survey, a monthly repeated cross-section of United States households. Their design supplies what Theorem 5 requires: households are classified as poor hand to mouth, wealthy hand to mouth or unconstrained by the Kaplan et al. (2014) conditions, which gives the group shares, and their estimating equation gives a group-specific coefficient on exposure to a dated shock. Because the theorem needs group averages instead of household-level responsiveness, a repeated cross-section suffices and no panel is required.

Three mappings stand between their estimates and the statistic. The first concerns the regression. Exposure is an indicator that a retained event fell on the interview day or one of the two before it, so the coefficients are responses if the timing of an interview relative to those events is unrelated to whatever else moves expectations, within the controls and the liquidity groups. The second mapping reads the classification as the economic state. The third is that their coefficient measures responsiveness of inflation expectations, while ρ is the loading through which expectations reach consumption, and the two coincide under a mechanism in which a household applies one parameter to every forecast. Appendix F states all three, together with the estimating equation, the exposure construction, the sign convention, the testing restriction and the six published specifications.

The exercise runs the chain end to end on real data, from an observable classification tied to the economic state through group shares and group averages to an assignment statistic. The primary specification does not reject the hypothesis that mean responsiveness is the same across liquidity states, while multiple imputation over the full sample rejects the same restriction. The non-rejection rests on configurations that give one liquidity group negative responsiveness. Requiring the group responses to be non-negative, the primary specification rejects at $p = 0.046$ under a critical value valid whichever constraint binds, close to the five percent level, and the sorting ratios acquire confidence sets just below their independence values. The restriction reaches the regression only through the mapping from forecast coefficients to responsiveness, so the rejection is conditional on that mapping. Appendix F reports the testing restriction, the six published specifications, the selection analysis and the estimates.

6 Conclusion

Two economies can agree on every distribution a survey reports and still respond differently to the same policy. That is the paper's answer to its opening question: knowing the distribution of expectations mechanisms is not enough to identify a macroeconomic counterfactual when their assignment across heterogeneous households is unknown. Under a scalar responsiveness channel and the model's resource condition, everything the assignment does to equilibrium runs through one statistic, the responsiveness located among unconstrained households, and rearrangement bounds that statistic sharply. In the reference economy the identified set for the aggregate spans a quarter of its own midpoint.

General equilibrium reshapes the problem. Once household composition responds to the assignment, the admissible set depends on the composition each candidate generates and has to be

solved as a fixed-point object. In the reference economy proportional income incidence switches that feedback off, which is what makes it a clean benchmark. The same benchmark has an empirical implication. There the observable spending contrast is zero under every admissible assignment even though the aggregate keeps moving with it, so survey heterogeneity is informative only when the observed classification is tied to the economic state and the observable response loads on the assignment dimension the aggregate depends on. Precision is not what stands in the way, measurement is.

The counterfactuals studied here hold the assignment fixed while composition moves, which the model supports when the rate at which households become constrained can vary with responsiveness. Under that condition the macroeconomic effect of expectations heterogeneity depends on a small number of assignment statistics and can still remain set identified. Measuring how prevalent each mechanism is will therefore not close the set. A classification tied to the economic state can, from nothing more than group shares and group averages in a repeated cross-section. What policy analysis needs is to know which households hold which mechanisms, and the paper says what kind of data would deliver that.

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A Derivations

The statements in the main text carry their proofs. This appendix collects the derivations that are compressed there, in the order they are used.

The partial derivatives of the equilibrium system come first. With G and H as in (25) and (26) and the hazard as in (24), write $\lambda_{\text{in}} = \bar{\lambda} - \theta a$ with $a = Y + \sigma r - \psi Y M_S/(1-h)$. Then

$$\mathcal{G}_Y = h\xi + \psi M_S, \qquad \mathcal{G}_h = \xi Y + \sigma r, \qquad \mathcal{G}_M = \psi Y, \tag{53}$$

$$\mathcal{H}_Y = -(1-h)\,\theta\left(1 - \frac{\psi M_S}{1-h}\right), \qquad \mathcal{H}_h = 1 - \lambda_{\text{in}} - \lambda_{\text{out}} + \frac{\theta\,\psi\,Y\,M_S}{1-h}, \qquad \mathcal{H}_M = \theta\,\psi\,Y. \tag{54}$$

The expression for $\mathcal{H}_h$ uses $\partial a/\partial h = -\psi Y M_S/(1-h)^2$, so that $(1-h)\,\partial\lambda_{\text{in}}/\partial h = \theta\psi Y M_S/(1-h)$.

Three of these deserve a word. $\mathcal{G}_M = \psi Y$ is the direct demand channel: responsiveness located among the unconstrained raises their consumption in proportion to the aggregate. $\mathcal{H}_M = \theta\psi Y$ is the composition channel: the same consumption is saving foregone, so the inflow hazard rises. Both are strictly signed only when $\theta > 0$, $\psi > 0$ and $Y \neq 0$, and all three conditions are carried explicitly wherever a strict sign is claimed.

The factorization in Corollary 3 follows from a cancellation between the two summands of the numerator. From (53) and (54),

$$\begin{aligned} N &= \mathcal{G}_M(1-\mathcal{H}_h) + \mathcal{G}_h\mathcal{H}_M = \psi Y\left(\lambda_{\text{in}} + \lambda_{\text{out}} - \frac{\theta\psi Y M_S}{1-h}\right) + (\xi Y + \sigma r)\,\theta\psi Y \\ &= \psi Y\left[\lambda_{\text{in}} + \lambda_{\text{out}} - \frac{\theta\psi Y M_S}{1-h} + \theta\,(\xi Y + \sigma r)\right]. \end{aligned}$$

Substituting $\lambda_{\text{in}} = \bar{\lambda} - \theta\,(Y + \sigma r - \psi Y M_S/(1-h))$, the two terms in $\theta\psi Y M_S/(1-h)$ cancel and so do the two in $\theta\sigma r$, leaving

$$N \;=\; \psi Y\left[\bar{\lambda} + \lambda_{\text{out}} - \theta Y + \theta\xi Y\right] \;=\; \psi Y\left[\lambda_{\text{out}} + \bar{\lambda} - \theta Y(1-\xi)\right] \;=\; \psi\,Y\,W.$$

Theorem 8 then replaces W by $\bar{\lambda}/h$. Note that the cancellation occurs before the feasibility identity is used, so the factorization $N = \psi Y W$ is an algebraic fact about the partial derivatives, while $W = \bar{\lambda}/h$ is the equilibrium content.

The system (12) is bilinear and cannot be solved by treating it as linear in (Y, h). The term $h\xi Y$ in G and the term $h\,\lambda_{\text{in}}$ in H are both products of the two unknowns, which is why the reduction in Proposition 6 produces a quadratic, not a linear equation. Eliminating the constrained share through the feasibility identity, without eliminating the aggregate, makes that reduction a quadratic in Y with coefficients explicit in the primitives. Among its at most two roots, interiority of the constrained share is generally not enough to select one, and the determinant requirement does. At $\xi = 1$ the quadratic term disappears and the constrained share is pinned at (39), so the question does not arise.

The two-type case reduces to a single free parameter. With F_ρ carrying mass q at $\rho = 1$ and

mass $1-q$ at $\rho=\mu$, an assignment is described by the four cell masses of a two-by-two table, and the marginals leave one degree of freedom. Taking $z=\mathbb{P}(s=C,\rho=1)$ as that degree of freedom, the four cells are z, $q-z$, $h-z$ and $1-h-q+z$, so

$$M_S \;=\; 1\cdot(q-z) \;+\; \mu\cdot(1-h-q+z), \tag{55}$$

which is (8). Non-negativity of the four cells is

$$\max\{0,\; h+q-1\} \;\le\; z \;\le\; \min\{h,\, q\}, \tag{56}$$

the Fréchet–Hoeffding interval for the table, and once h responds to the assignment the bounds are evaluated at $h(z)$, not at a fixed value. The feasible set for z is therefore the continuous image of $\mathcal{M}$ under the inverse of (55), which is how connectedness transfers from Theorem 6 to the two-type economy.

B Assignment feasibility

This appendix supports two claims made in the general-equilibrium analysis. The first is that a condition placed on the two summands of N separately is neither sufficient nor necessary for the endpoint characterization. The second is that the set of admissible assignments is an equilibrium object once the constrained share responds, so that a Fréchet interval computed at a frozen value of that share is the wrong object.

Take the first. Proposition 3 requires the sign of N to be constant on the feasible set. A more intuitive requirement would be that the direct assignment channel and the distributional channel reinforce one another at every assignment, that is, that $G_z(1-H_h)$ and G_hH_z agree in sign pointwise. That requirement is neither sufficient nor necessary, and both failures occur.

Agreement at each assignment does not prevent both summands from migrating together across assignments, so the pointwise requirement is not sufficient. If both are positive at one end of the feasible set and both are negative at the other, the pointwise requirement holds everywhere and N still changes sign. It fails to be necessary because N can hold its sign while the summands disagree throughout, which happens whenever one summand dominates the other uniformly.

Consider a class of economies in which the assignment enters the derivatives as well as the levels,

$$\begin{aligned}\mathcal{G}(Y,h,z) &= g_0+g_YY+c_1h+c_2h^2+(e_0+e_1Y+e_2h)\,z,\\ \mathcal{H}(Y,h,z) &= h_0+d_1Y+d_2Y^2+h_hh+(f_0+f_1Y+f_2h)\,z,\end{aligned} \tag{57}$$

so that $G_z=e_0+e_1Y+e_2h$ and $H_z=f_0+f_1Y+f_2h$ both vary with the equilibrium. A class in which G_z, H_z and H_h are constants cannot exhibit the failure at all, since $G_z(1-H_h)$ is then a constant whose sign cannot vary with the assignment.

Table 4 cross-tabulates the two candidate conditions against the claim over 32,782 economies

in this class with a unique stable equilibrium on the whole assignment path.

Table 4: The pointwise condition against constancy of N

Summands agree pointwise	Sign of N constant	Economies	Endpoint failures
yes	yes	10,785	0
yes	no	33	32 (97.0 percent)
no	yes	17,645	0
no	no	4,319	4,098 (94.9 percent)

Notes. Economies drawn from (57). An economy is counted only if a unique stable equilibrium exists at every assignment on the path. An endpoint failure is a path on which the aggregate leaves the interval spanned by its two endpoint values.

The second row is the case that matters. The pointwise requirement holds at every assignment on the path, and the extremal assignments nonetheless fail to bound the identified set in 97.0 percent of those economies. In the most extreme such economy the numerator moves from -0.905 at one end of the feasible set to $+0.802$ at the other, and the two endpoint values of the aggregate miss the identified set by 0.122 on an interior range of 0.154, so the reported set would understate the true one by most of its width. The third row shows the other side: 17,645 economies violate the pointwise requirement, keep the sign of N, and bound correctly.

Now the second. With h fixed, the admissible assignments are the interval (56). Once h responds, that interval cannot be formed, and the correct object is $\mathcal{Z}$ as defined in (38). The two differ in both directions.

Locating the endpoints of $\mathcal{Z}$ requires solving the two restrictions jointly with the equilibrium, since each is a comparison between the assignment and a constrained share that depends on it. Over 1,548 economies drawn widely, $\mathcal{Z}$ extends below the lower endpoint of the frozen construction in 21.3 percent of cases and above its upper endpoint in 7.5 percent, so the two objects differ in 28.7 percent of economies. In the worst case encountered, the frozen construction overstates the width of the identified set by 6,106 percent by admitting assignments that no equilibrium supports. The reverse error, excluding assignments that are feasible under their own $h(z)$, is the more common one.

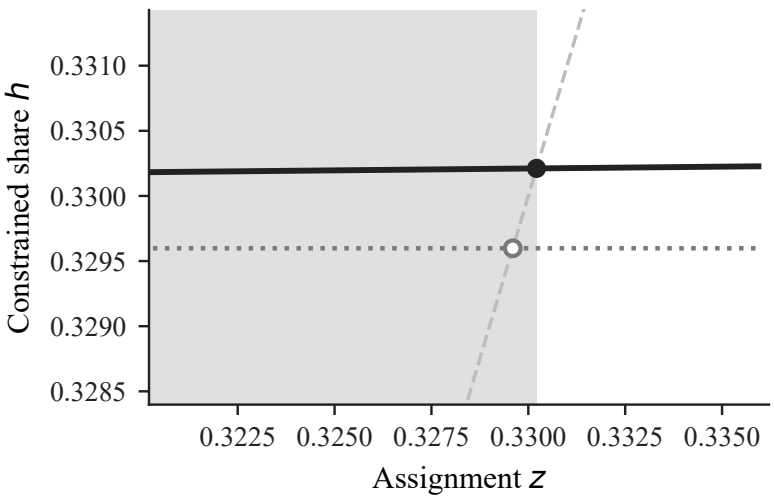


Figure 7: An economy in which composition responds to the assignment

Notes. Every reference parameter of Table 1 is held fixed and only the income incidence is moved to $\xi = 1.4$, which allows the constrained share to respond. The reference economy itself has $\xi = 1$ and cannot display this mechanism. The solid line is the constrained share along the equilibrium branch, the dotted line is the value that a frozen-composition construction would use, and the markers are the two upper endpoints.

The illustrative economy of Figure 7 shows the second error. Every reference parameter is held fixed and only ξ is moved to 1.4, which is enough for the constrained share to vary over $[0.3289, 0.3302]$ across the feasible set. The true upper endpoint is 0.3302 and the frozen construction returns 0.3296, so assignments that an equilibrium supports are discarded, and they are discarded at the extreme of the set, which is exactly where the sharp bounds are attained. Moving ξ to the other side of one reverses the direction: at $\xi = 0.4$ the true endpoint is 0.3359 against a frozen construction of 0.3361, so the naive object is too large instead of too small.

Sharpness over the corrected object holds. Taking the two extreme elements of $\mathcal{Z}$ and evaluating the aggregate there bounds the aggregate over the whole of $\mathcal{Z}$ in 1,685 of 1,685 economies, with 0 failures.

Theorem 6 reduces the bounds to two global extremes. Writing $\varphi(z) = h(z) - z$, the restriction $z \geq h(z) + q - 1$ is $\varphi(z) \leq 1 - q$, so

$$\mathcal{Z} \;=\; \varphi^{-1}([\,0,\,1-q\,]) \cap [\,0,\,q\,], \tag{58}$$

and $\mathcal{Z}$ is an interval whenever φ is monotone. Since $\varphi'(z) = dh/dz - 1$, a sufficient condition is that the constrained share respond less than one for one to the assignment.

That route does not close. Over 161,570 admissible points, $dh/dz \geq 1$ at 53 of them, or 0.033 percent, with a maximum of 3.81. Checking φ directly, it is monotone in 2,459 of 2,469 economies, or 99.6 percent. In the worst non-monotone economy φ ranges over $[0.244, 0.753]$ while $1-q = 0.339$, so the binding restriction sits where φ turns, which is the configuration that could disconnect the set. It did not disconnect: $\mathcal{Z}$ was an interval in 2,255 of the 2,255 economies whose feasible set

meets the grid in at least three points, which is the least that distinguishes an interval from a gap. In the rest of the 2,469 the feasible set is too small on the grid for the question to arise, and every non-monotone economy is among those tested.

What freezing composition costs

Definition 4 is a fixed point because the rearrangement bounds are evaluated at the composition the assignment itself induces. Section 3 states that freezing composition is neither an inner nor an outer rule. This proves it for the two-type economy.

Proposition 8 (What a feasible frozen composition delivers)**.** *Let F_ρ place mass q at $\rho = 1$ and mass $1-q$ at $\rho = \mu$ with $\mu \in [0,1)$. Suppose that at every composition arising on $\mathcal{M}$ one has $h + q < 1$ and $h < q$, that $\mathcal{M} = [m_-, m_+]$ is a compact interval, and that both endpoints are set by the rearrangement bounds, not by the boundary of I. Write $A = q + \mu(1-q)$ and $\mathcal{M}^F(\bar{h}) = [\underline{M}(\bar{h}), \overline{M}(\bar{h})]$ for the interval a frozen composition $\bar{h}$ delivers. Then*

(i) $\underline{M}(h) = A - h$ and $\overline{M}(h) = A - \mu h$, so $m_- = A - h(m_-)$, $m_+ = A - \mu h(m_+)$ and $\mathcal{M}^F(\bar{h}) = [A - \bar{h},\ A - \mu\bar{h}]$;

(ii) for $\mu \in (0,1)$, $\mathcal{M} \subseteq \mathcal{M}^F(\bar{h})$ if and only if $h(m_-) \le \bar{h} \le h(m_+)$, and $\mathcal{M}^F(\bar{h}) \subseteq \mathcal{M}$ if and only if $h(m_+) \le \bar{h} \le h(m_-)$. At $\mu = 0$ the upper bound is A whatever the composition, so $m_+ = A$ and the two conditions lose their second half: the first inclusion holds if and only if $\bar{h} \ge h(m_-)$ and the second if and only if $\bar{h} \le h(m_-)$;

(iii) consequently, for every feasible *anchor $\bar{h} = h(m)$ with $m \in \mathcal{M}$,*

$$
\begin{aligned}
h_M > 0 &\implies \mathcal{M} \subseteq \mathcal{M}^F(\bar{h}),\\
h_M < 0 &\implies \mathcal{M}^F(\bar{h}) \subseteq \mathcal{M},\\
h_M = 0 &\implies \mathcal{M}^F(\bar{h}) = \mathcal{M}.
\end{aligned}
$$

Proof. In the stated regime the rearrangement constraint is $z \in [0, h]$, and $M(z;h) = (q - z) + \mu(1 - h - q + z) = A - \mu h - (1-\mu)z$ is strictly decreasing in z because $\mu < 1$. Hence $\overline{M}(h) = M(0;h) = A - \mu h$ and $\underline{M}(h) = M(h;h) = A - h$, which gives $\mathcal{M}^F(\bar{h})$. Both endpoints of $\mathcal{M}$ being set by those bounds, they satisfy $m_- = \underline{M}(h(m_-))$ and $m_+ = \overline{M}(h(m_+))$, which completes (i). For (ii), $\mathcal{M} \subseteq \mathcal{M}^F(\bar{h})$ holds exactly when $A - \bar{h} \le A - h(m_-)$ and $A - \mu\bar{h} \ge A - \mu h(m_+)$. The first is $\bar{h} \ge h(m_-)$. The second is $\bar{h} \le h(m_+)$ when $\mu > 0$ and is $A \ge A$ when $\mu = 0$, which restricts nothing. The reverse inclusion reverses both. This is where μ may not be divided out. For (iii), h_M keeps its sign along the branch by Theorem 9, so h is monotone on $\mathcal{M}$ and a feasible anchor satisfies $h(m_-) \le h(m) \le h(m_+)$ when $h_M > 0$ and the reverse when $h_M < 0$, which (ii) asks for in each case. At $\mu = 0$ only the condition on $h(m_-)$ remains and a feasible anchor satisfies it in the same way, so (iii) holds on the whole of $\mu \in [0,1)$. □

The proposition covers anchors inside the range of compositions the feasible set generates. Table 6 in Appendix D freezes at the composition prevailing at one assignment. It is not a statement about widths. Since $|\mathcal{M}^F(\bar{h})| = \bar{h}(1-\mu)$ and $|\mathcal{M}| = h(m_-) - \mu\, h(m_+)$,

$$|\mathcal{M}| - |\mathcal{M}^F(\bar{h})| \;=\; \left[h(m_-) - \bar{h}\right] \;+\; \mu\left[\bar{h} - h(m_+)\right], \tag{59}$$

whose sign depends on where $\bar{h}$ sits, not on the sign of h_M: freezing at the composition of either endpoint reverses the comparison against freezing at the no-feedback value, at the same h_M. Which interval is wider is therefore not a question about the response, and inclusion is.

Both statements carry to the identified set for the aggregate. Inclusion is preserved by any map applied to both sides, and the map applied is the same one. Monotonicity, which (32) supplies, is what makes the endpoints of the image the images of the endpoints.

Appendix D already varies ξ with every other primitive fixed and reports the frozen endpoint next to the correct one, so the sizes are there. They are small: the two endpoints nowhere differ by more than 0.0026, and the widths of the identified set they deliver by more than 0.32 percentage points, both gaps widest in the row with the strongest incidence. The aggregate moves the constrained share by very little at these transition rates.

What is not in Appendix D is how far the response has to go before the difference stops being small. Raising it while holding ξ fixed, the frozen calculation overstates the width by five percent of it once $|h_M|$ reaches somewhere between 0.035 and 0.069, and by twenty-five percent once it reaches between 0.069 and 0.087. These are ranges because $|h_M|$ orders the effect at a given ξ and is not a sufficient statistic for it across ξ.

Across the 86 economies visited in this exercise the fixed-point feasible set was connected in every one, 0 exceptions. The hypothesis of Theorem 6 is undemanding in this family.

C Numerical verification

The numbers in this appendix are neither model magnitudes nor estimates. They are agreement tolerances between an analytical expression and its numerical counterpart, and counts of drawn economies. Every statement in the paper is proved analytically, and the checks exist because several proofs pass through algebra that is easy to get wrong, and because a claim that a condition is not vacuous is a claim about a class of economies. One economy does not make it. A check that failed to exercise its claim is labelled.

Table 5 lists each check against the statement it tests.

Table 5: Verification of the analytical statements

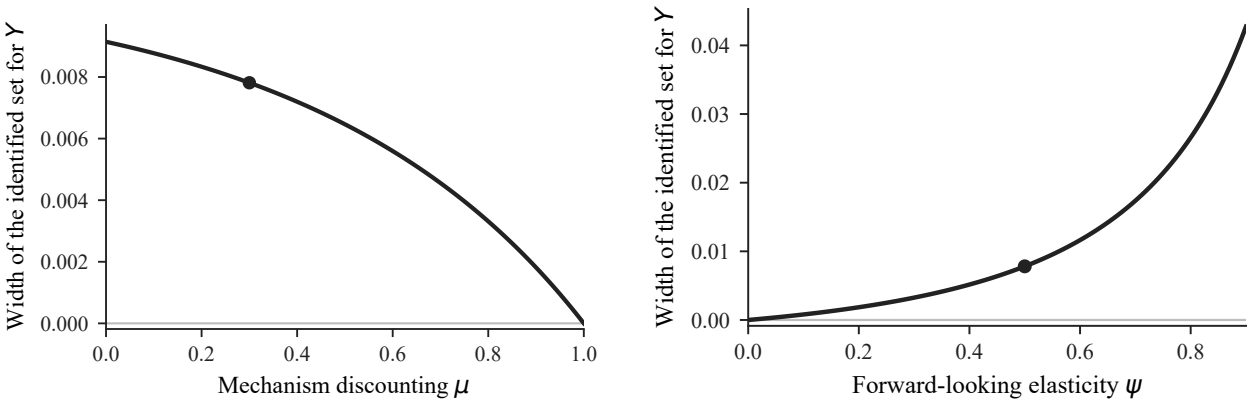
Statement	Check	Result
The six partials of (53) and (54)	each against a numerical derivative of its block	agreement to 3.17×10^{-10}
The factorization of the numerator in (30)	both sides evaluated independently	agreement to 2.22×10^{-16}
The feasibility identity $h\,W = \bar{\lambda}$ in (29)	evaluated at 155,783 equilibria, interiority of h not imposed	maximum relative departure 3.88×10^{-7}
Cramer's rule for dY/dM_S in (13)	against a numerical derivative of the solved equilibrium path	agreement to 1.66×10^{-7}
The differential equation (37)	$-c(z)Y$ against a numerical derivative at 30,758 equilibria	maximum relative departure 4.51×10^{-4}
Positivity of W at every interior equilibrium	203,591 admissible equilibria drawn widely	0 with $W < 0$
Constancy of the sign of N on the feasible set	106,855 complete assignment paths	0 sign changes of YW
Monotonicity and the sign of Y from (37)	53,767 complete paths	0 sign changes of Y, 0 endpoint failures
Strict decrease of $\lvert Y \rvert$ in the assignment	largest step ratio $\lvert Y(z_{k+1})\rvert/\lvert Y(z_k)\rvert$ over all paths	0.383, below one as required

Notes. Economies are drawn over wide ranges of the primitives and retained when a unique equilibrium exists with an interior constrained share and a positive determinant. Tolerances for analytical identities are absolute. Tolerances for comparisons against numerical derivatives are relative and reflect the differencing step.

At $\xi = 1$ the identity $h\,W = \bar{\lambda}$ with $W = \lambda_{\text{out}} + \bar{\lambda}$ fixes the constrained share at $\bar{\lambda}/(\lambda_{\text{out}} + \bar{\lambda})$ at every assignment, so the binding restriction is $z \leq h(z)$ with h constant and the supremum is available in closed form. The reported feasible set $[0.0000, 0.3333]$ and identified set $[0.02623, 0.03404]$ agree with it. The sign relation between W and h is not a numerical finding either: (29) forces $\operatorname{sign}(W) = \operatorname{sign}(h)$ whenever $\bar{\lambda} > 0$.

D Additional comparative statics

The magnitudes here are computed from assumed primitives, like those in the main text and unlike the estimates in the application. Two parameters are jointly necessary for the assignment to matter at all, by Theorem 9 and Corollary 5, and Figure 8 shows how each moves the width of the identified set in the reference economy. The width vanishes as the alternative mechanism approaches the responsive one, since there is then nothing for the assignment to sort, and it rises with the strength of the forward-looking channel, since that channel is what the mechanism acts on. At $\mu = 0$ the width is 0.00914, and at the upper end of the plotted range for ψ it is 0.04274.



(a) Discounting by the alternative

(b) Strength of the forward-looking channel

Figure 8: Width of the identified set across two parameters

Notes. Each panel moves one parameter of the reference economy of Table 1 and holds the rest fixed. The vertical axis is the width of the identified set for the aggregate.

The two panels move parameters that enter the necessity condition of Proposition 1, so what they trace is how wide the set is across economies in which the assignment matters at all. Each panel runs to the boundary of that condition, and the width goes to zero there: at $\mu = 1$ the two mechanisms coincide and at $\psi = 0$ demand stops being forward looking, and in either case who holds which mechanism does not reach the aggregate. What neither panel touches is whether the set is the right object to compute in the first place.

A third parameter does. The incidence of aggregate income on constrained households does not enter that necessity condition, so it cannot switch the assignment problem off, but it is the parameter that decides whether composition responds to the assignment. Where it does, the rearrangement bounds are evaluated at a composition the assignment itself moves and the feasible set is the fixed point of Definition 4, not an interval computed once. The table below therefore reports two things the panels cannot: how the width varies with that incidence, and how far the endpoint a frozen composition returns sits from the one the equilibrium admits.

Table 6: The identified set and the feasible set as income incidence varies

ξ	$\sup \mathcal{Z}$	Frozen-h endpoint	Identified set for Y	Width, percent of midpoint
0.4	0.3359	0.3361	[0.01875, 0.02246]	18.04
0.6	0.3352	0.3354	[0.02073, 0.02536]	20.10
1.0	0.3333	0.3333	[0.02623, 0.03404]	25.93
1.4	0.3302	0.3296	[0.03546, 0.05086]	35.69
1.8	0.3241	0.3215	[0.05329, 0.09071]	51.97

Notes. Every parameter other than ξ is held at its value in Table 1, and the row $\xi = 1.0$ is the reference economy. The frozen-h endpoint applies (56) at the constrained share the equilibrium induces at the coupling $z = 0.15$, which is one assignment, not a composition read off outside the model. The lower endpoint of $\mathcal{Z}$ is zero in every row. Endpoints are located by bisection on the restrictions in (56).

The width of the identified set rises steeply in income incidence, from 18.04 percent of the midpoint at $\xi = 0.4$ to 51.97 percent at $\xi = 1.8$ (Table 6). The economics is that ξ enters

the aggregate-demand channel G_h, so a stronger incidence makes the aggregate more sensitive to household composition, and composition is what the assignment moves. This is a distinct margin from the two in the main text. There, the assignment mattered or did not depending on whether mechanisms differ and demand is forward looking. Here the assignment matters for a given degree of mechanism heterogeneity, and how much depends on how strongly the constrained group transmits aggregate income.

The error made by freezing the constrained share reverses direction at $\xi = 1$. Below one, the frozen construction returns an endpoint above the true one and therefore admits assignments that no equilibrium supports. Above one, it returns an endpoint below the true one and discards assignments that are feasible. At $\xi = 1$ the two coincide exactly, because the identity $h\,W = \bar{\lambda}$ with $W = \lambda_{\text{out}} + \bar{\lambda}$ then fixes the constrained share at $\bar{\lambda}/(\lambda_{\text{out}} + \bar{\lambda})$ independently of the assignment, and a constant marginal makes the Fréchet interval and the feasible set the same object. The reference economy sits at that point, which is why Figure 7 uses an illustrative economy off it, and why the agreement of the two constructions in the reference economy is an identity.

How fast the informative window closes

Section 5 reports two thresholds in the composite scale at each income incidence, and the ratio between them instead of their level, because the level carries a change of units. Between the two the observable bound excludes some assignments, not others, and how quickly it stops excluding anything is a separate question from how wide the window is. Figure 9 answers it, with each economy's scale divided by its own lower threshold so that four economies sit on one axis.

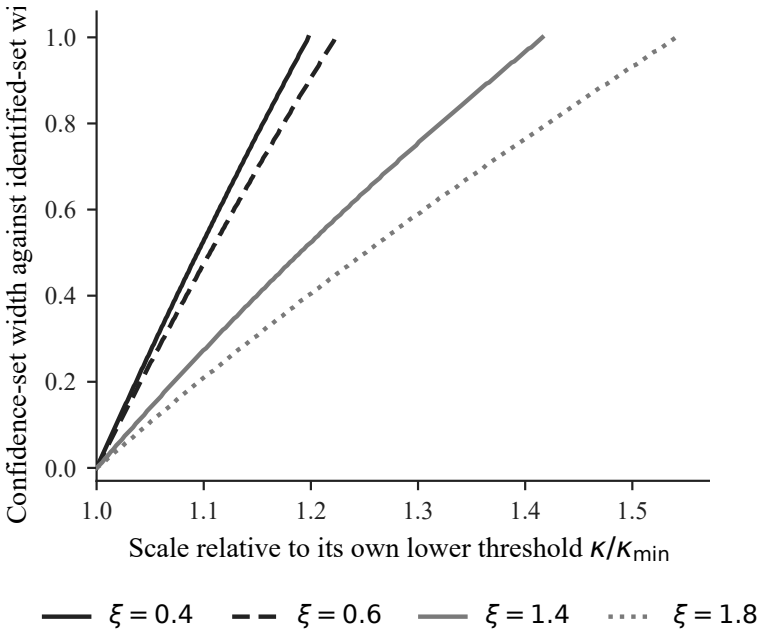


Figure 9: The width of the data-admissible set across the informative window

Notes. Every parameter other than ξ is held at its value in Table 1, and the classification is perfect. The vertical axis divides two widths of different type. The numerator is the width after intersecting with the observable bound, which is a confidence set for the identified set at the level inherited from d_{obs} and conditional on the bridges. The denominator is the width the two marginals deliver on their own, which is computed from assumed primitives and carries no sampling content, so it is an identified-set width. One means the survey adds nothing. Each curve begins where a single assignment survives and ends where none is excluded, and its horizontal extent is the ratio reported in Section 5. The $\xi = 1$ benchmark is excluded because its lower threshold is zero, so the normalisation is undefined. There the spending contrast is uninformative for every positive scale.

Table 7 reads the same object at three positions inside the window and adds the classification error. Two regularities organise it. The reduction depends on where the scale falls inside the window and hardly at all on the incidence or the error rate. A scale a quarter of the way in removes about three quarters of the marginals-only width, and one three quarters of the way in removes about a fifth. What incidence and misclassification change is the location of the window. Errors summing to two fifths move both thresholds down by roughly half, so a worse classification makes the same observable bound informative at a smaller composite scale. A substantial reduction therefore requires the composite scale to lie in the lower part of a window whose two ends differ by a factor between 1.20 and 1.54, and the data say nothing about where in the scale the economy sits.

Table 7: Reduction of the identified set across incidence, scale and classification error

		Window $[\kappa_{\min}, \kappa_{\max}]$, by $e_0 + e_1$			Reduction at position in window		
ξ	Marginals only	0	0.2	0.4	25%	50%	75%
0.4	18.04%	[3.16, 3.79]	[2.31, 2.76]	[1.59, 1.90]	73%	48%	23%
0.6	20.10%	[2.33, 2.85]	[1.70, 2.08]	[1.17, 1.43]	73%	47%	23%
1.4	35.69%	[3.95, 5.59]	[2.86, 4.05]	[1.96, 2.77]	72%	46%	22%
1.8	51.97%	[11.31, 17.44]	[8.16, 12.52]	[5.58, 8.51]	72%	46%	22%

Notes. Every parameter other than ξ is held at its value in Table 1, and the assignment is held fixed along the experiment, $\dot{z} = 0$. The second column is the width of the identified set for the aggregate from the marginals alone, in percent of its midpoint. The window is the range of the composite scale over which the observable bound excludes some assignments and not others, under symmetric error rates summing to the column heading. The last three columns give the reduction in that width, relative to the second column, when the scale lies a quarter, half and three quarters of the way from $\kappa_{\min}$ to $\kappa_{\max}$. They are computed under perfect classification and differ by at most 0.4 percentage points under the other two error rates. The second column repeats Table 6. The rest is computed on a grid of 2001 couplings, which returns marginals-only widths at most 0.02 percentage points below that table and converges to it as the grid is refined, and a grid sixteen times finer moves no reduction by more than 0.1 percentage points. The window is built from the ninety-five percent bound d_{obs}, so the reductions describe a confidence set conditional on the bridges.

E Inference for the survey estimand

The bound in Section 5.3 inverts a test on the projected contrast, so it inherits the properties of that test. This appendix sets out the estimand, the inference the reported bound uses, the size of that inference under nulls that hold the design at its data values, and the sensitivity of the results to weighting and to rank. Where an alternative procedure was available its size is reported alongside.

The estimand

The reported object is the coefficient matrix of (47) demeaned across cells, since the response common to all cells is absorbed by the country-by-round effects. It is stored with cells $\times$ signals entries but lies in a subspace of dimension $k = (\text{cells} - 1) \times \text{signals}$, so every variance estimate of it is singular by construction: ten cells and two signals give twenty coefficients and $k = 18$. Inverting the singular object with a pseudo-inverse at a default tolerance is not a fix, because whichever numerically small eigenvalues survive an inclusion-exclusion subtraction are then inverted and carry enormous weight. The problem is written in the identifiable subspace instead. With Q an orthonormal contrast basis of shape $(\text{cells}, \text{cells} - 1)$ satisfying $Q'Q = I$ and $Q'\mathbf{1} = 0$, the reduced estimand and variance are $(I \otimes Q')R$ and $(I \otimes Q')V(I \otimes Q)$, the latter exactly k by k and inverted by an ordinary solve. Rank, extreme eigenvalues and condition number are computed on every row, and no p-value is returned when the rank falls short of k or the condition number exceeds 10^{10}.

The comparison between the two maps is made on the same cells, the same signals and the same inference, with the outcome swapped, precisely so that an asymmetry cannot be attributed to using a survey-built regressor on one side and an external one on the other. Where the two maps are stacked, as in the two-measure exercise below, the covariance of the combined coefficient vector is estimated jointly across equations, not block by block: the equations share households

and rounds, and their influence functions correlate at 0.462 in the case reported, so treating them as separate problems would be a different exercise.

The two ordinal spending questions are not coded in economic order. One is an increase a lot, two a decrease a lot, three an increase a little, four a decrease a little and five no change, so the order is $2 < 4 < 5 < 3 < 1$. The categories are therefore reordered before any index is built, and because attaching numbers to ordered categories is a choice, the index is constructed under three monotone scorings: $(-2, -1, 0, 1, 2)$, a version that pulls the extremes in at $(-1.25, -1, 0, 1, 1.25)$ and one that pushes them out at $(-4, -1, 0, 1, 4)$. The last two are not affine transformations of the first. Across both ordinal outcomes, both signal families and the three scorings the smallest timing-robust p-value is 0.3599, and most exceed 0.6.

These measures are used only as corroboration. The difference of an ordinal index is not a change in consumption under any scoring, so they cannot feed the structural bridge, and the bound in Section 5.3 comes from the open-ended questions, which record a percentage change.

The fixed effects are re-estimated inside every replicate, not held at their full-sample values, and levels that a resampling draw leaves without support are dropped, not absorbed by a pseudo-inverse.

The construction of Davezies et al. (2025) delivers asymptotically valid inference under conditions on the two-way array: the scores are separately exchangeable in the household and round indices up to the included effects, both dimensions grow, and the limiting one-way variance components are non-degenerate in the directions being tested. Two further conditions matter. The first concerns the multivariate test. Davezies et al. (2025) prove the minimum-of-three F-test under an additional condition on the limiting variance components, whose leading sufficient case is that the sum of the two one-way components be invertible, and they show by example that it cannot be dropped. Over the whole timing segment the estimated sum has smallest eigenvalue at least $6.7e-04$ with condition number at most 54, so the sample satisfies the sufficient case. Their same theorem shows that the componentwise construction dispenses with the condition entirely, which is a further reason to prefer it. The second concerns the regression estimator. The ratio of the extreme eigenvalues of the variance of the summed moment vector must stay bounded, so that different linear combinations of the moments converge at compatible rates and the influence-function linearisation holds. That condition lives in the space the estimator lives in, not in the projection, and they give linear-regression examples in which it fails while the two-way structure is intact. Their regression result is stated for a parameter of fixed dimension, while (47) absorbs household and country-by-round effects whose number grows with both clustering dimensions. What is assumed here is that residualising those effects leaves the cell-by-signal coefficients with the asymptotically linear two-way score their result requires. The part of these conditions that leaves a trace in the sample is measured. The sample for the retrospective-spending map, which is the one the reported bound is attained on, carries $809,828$ revisions on $74,374$ households over 69 rounds, and neither dimension rests on a few heavy units: the inverse Herfindahl of the per-household contributions is $42,312$ and of the per-round contributions 65.4, with the largest single household contributing 0.00006 of the observations and the largest round 0.019. Write $\mathcal{P}$ for the matrix that carries an

estimated response map into the difference between the classified group's average response and the average, one row per signal, so that $\widehat{\Delta} = \mathcal{P}\widehat{R}$ is the projected contrast the bound is a supremum over. In those directions both one-way components are well away from degeneracy over the whole timing segment: the smallest eigenvalue of $\mathcal{P}V_1\mathcal{P}'$ is $4.98e-04$ and of $\mathcal{P}V_2\mathcal{P}'$ is $4.77e-04$, with condition numbers of 2 and 2, so the arm the minimum-of-three statistic would lose to a degenerate component is nowhere near being lost. The bounded-ratio condition is a statement about the full *eighteen*-dimensional object, and there the ratio of the extreme eigenvalues over the segment reaches 46 for V_1, 182 for V_2 and 43 for their positive semidefinite majorant. The experiments establish that the procedure attains its nominal rate under two concrete null laws, one holding the country-round signals at their observed values and one redrawing the design.

Size experiments

Design A holds the survey design at its real value, the households, their observation pattern, the predetermined states and the country-by-round signals, and draws only the errors, which is the conditional target a score-based sandwich approximates. Design B draws the design as well, which is closer to what a resampling scheme reproduces. In both the map is exactly zero, and the errors carry household and round dependence through a household loading on a common round factor, calibrated in A to the observed decomposition of the residual variance.

Table 8: Rejection rates against a true null, nominal five percent

inference law	survey design held fixed	design also drawn
inclusion-exclusion, Cameron et al. (2011)	0.395	0.375
minimum of three, Davezies et al. (2025)	0.035	0.025
pigeonhole, Owen (2007)	0.000	0.000
product of Rademacher weights	0.100	0.192

Notes. 200 simulations with bootstraps of 399 in the first column and 120 with bootstraps of 199 in the second. Monte Carlo standard errors near five percent are 0.015 and 0.020. The heterogeneity map is exactly zero in both. The designs, the distribution of the p-value under the null and the reason the usual estimator fails are given below.

The rejection rate at five percent is one point of a distribution (Table 8). Under design A the quantiles of the p-value under the null are 0.000, 0.001, 0.002, 0.013 and 0.073 for the usual estimator against a uniform reference of 0.01, 0.05, 0.10, 0.25 and 0.50, so its entire distribution is displaced. For the minimum of three they are 0.024, 0.077, 0.137, 0.256 and 0.471, mildly conservative. For pigeonhole resampling the first quantile is 0.846: ninety-nine percent of its p-values under the null exceed that, which is why it is not used. Because the simulations share draws across laws the comparisons are paired, and the usual estimator rejects 0.360 more often than the minimum of three with a paired standard error of 0.034.

The mechanism also explains a diagnostic in the data. The usual estimator is a difference, $V_1 + V_2 - V_{12}$, so it need not be positive semidefinite and the subtraction can leave very little

variance along particular directions. A Wald statistic depends on the matrix in the direction of the estimand, not on the average of its diagonal. In the data the ratio of resampled to usual standard errors along the eigen-directions of the usual matrix rises monotonically from 1.44 in the widest to 3.38 in the narrowest, while the round-clustered matrix is flat at 1.00 throughout. Across the grid of specifications the minimum of three almost always selects the household-only matrix, not the usual one.

Two nuisances remain. The interview date is handled by (49) in the main text, and the rule binds: in the monetary-surprise family the liquidity-by-income cells on the ordinal spending measure give 0.0023 at one month and 0.7667 at two. The other is the sample restriction. Requiring every candidate state to be observed in the previous round binds only through employment status, whose coverage is zero in every round to July 2022 and one in two months of every three thereafter, so that restriction selects rounds, not households. It is therefore reported as a period split, run on all observations in each block, not through the variable that encodes it, and the split does not reproduce the rejection that the round-selected version produces at nearly identical sample size.

Two exercises are not robustness checks. Rotating the monetary surprises, from the short end, the long end or by principal components, returns an identical statistic to four decimals, because the Wald statistic on the full interaction block is invariant to an invertible linear transformation of the signals. And the expected financial situation is not admitted as a structural state, since conditioning on it puts an expectation on the classification side.

Inference on the projected contrast

The bound is a supremum of $|\Delta_{\text{obs}}(u)| = |u' B \widehat{R}|$ over unit combinations, so what it needs is the projected contrast $\widehat{\Delta} = \mathcal{P}\widehat{R}$ of the size experiments above. It has one coordinate per signal: *two* of them here against *eighteen* for the map. Inverting a simultaneous test on the map and reading the functional off the result pays for directions the aggregate never loads on, and it produced a numerical problem. The two-way matrix is a difference, $V_1 + V_2 - V_{12}$, and in eighteen dimensions it stops being positive semidefinite at interior mixtures. An indefinite quadratic form has an acceptance region that is not an ellipsoid: moving out along a positive eigendirection and a negative one together keeps the form inside the band while the vector grows without limit, so the region is unbounded and the support function of a union containing it is infinite. Taking the largest of three ellipsoid support functions returns a finite number from a set that is not bounded.

After projection that does not arise here. The smallest eigenvalue of $\mathcal{P} V_u \mathcal{P}'$ is positive over the whole admissible segment, at least 4.53×10^{-4}, minimised over π by continuous optimisation, not read off a grid, with the minimum at $\pi = 0.515$. The confidence regions are bounded and the indefiniteness of the eighteen-dimensional covariance is irrelevant for this functional.

Two constructions are used, both of them Davezies et al. (2025)'s. The reported one is componentwise. For each coordinate the standard error is the largest of the two one-way ones and the positive part of the two-way one, the normal quantile is taken at $1 - \alpha/(2K)$ with K the number of signals, and the supremum of the Euclidean norm over the resulting box is attained at a corner.

It is reported because its validity conditions are the weaker of the two. That alternative is their positive semidefinite majorant, $\max(V_u, \max(V_1, V_2))$ in the matrix sense, which is one ellipsoid by construction and yields 0.1672 against 0.1778, both with one timing share.

Both were sized. Under $\widehat{\Delta} = 0$, with disturbances from a two-way wild bootstrap on the residuals of the restricted fit so that the dependence in the data is preserved, the componentwise rule rejects 0.060 of the time and the majorant 0.045, against a nominal 0.05 with a Monte Carlo standard error of 0.0154 over 200 draws at $\pi = 0.45$.

The construction the manuscript first used gives 0.2859 over the segment and 0.2209 at the two lags. It is the radius of a confidence region for the whole map, so it bounds every direction of the map at once and not the one combination the aggregate loads on. And it is not the inversion of the published test, which sets the two-way arm aside only when that arm's own quadratic form is negative: where the two-way covariance is indefinite its acceptance region is unbounded, since the quadratic form stays below the critical value along directions of arbitrary length, so the published rule has no finite support bound there at all. The finite figure comes from the modified procedure of the whole-map subsection below, which sets the arm aside on the matrix.

The whole-map tests and the sign check

Table 3 reports tests of $R = 0$, which are hypotheses about the map itself and not about the projected contrast, so the indefiniteness above is live for them. Davezies et al. (2025) replace the two-way arm by infinity when its quadratic form comes out negative. The check is one-sided: where the same indefiniteness leaves the form small and positive it survives, and a statistic that is a minimum is then set by it. At $\pi = 0.150$ the three arms of the inflation expectation map are 2.41 against 21.01 and 46.00, with the smallest eigenvalue of the two-way matrix negative, and the p-value at that one mixture is 1.0000.

Table 3 reports that rule as published. A variant applies the same check to the matrix, not to one of its forms, setting the two-way arm aside wherever $V_1 + V_2 - V_{12}$ fails to be positive definite. Along a continuous path of mixtures the smallest eigenvalue reaches zero before it turns negative, and at that boundary the matrix is singular rather than indefinite, so its inverse does not exist and (61) would be undefined under the narrower wording. Dropping an arm from a minimum raises the statistic, so the variant makes rejection easier, which is the unfavourable direction for a finding that is a non-detection. In the three cells where it is active it gives 0.5716, 0.9117 and 0.8412 against 1.0000, 0.9117 and 0.9512 under the published rule, so nothing detectable is found either way. Its size was measured the same way as above, at *three* interior mixtures with 200 draws each: the published rule, the modified one and the rule that never uses the two-way arm all reject 0.060 of the time, and the two-way matrix was indefinite in at most 0.5 percent of draws. Under the null the modification is almost never active and does not change the rate. It is active in the sample.

Comparing the two maps

The scalar-bridge null of Section 5.3 is tested over every direction of the half circle and the reported p-value is the supremum, so it rejects only where no direction supports a common scalar. Inverting the same statistic returns a set for the scalar.

Table 9: Whether one scalar bridge reconciles the expectation and spending maps

signal	spending measure	p for proportionality	inverted set for the scalar
inflation	expected, open ended	0.3829	$[-0.75,\ 0.86]$
inflation	retrospective, open ended	0.7922	$[-1.03,\ 1.00]$
own income growth	expected, open ended	0.9011	$[-1.46,\ 0.47]$
own income growth	retrospective, open ended	0.4634	$[-1.59,\ 0.88]$

Notes. Each row pairs one expectation map with one spending map. The null is that a single scalar a reconciles them, the spending map equalling a times the expectation map, which is the statement that survives a change of units. The reported p-value is the supremum over directions of the half circle, so it rejects only if every direction does, and the inverted set is the set of a the data do not reject at ninety-five percent. Every set contains zero, which is why the comparison is reported as uninformative about whether the two maps differ rather than as evidence that they agree.

A signal built from the outcome

The size experiments above hold the country-round signals at their real values. For the external series that is the right target, since a published price index is not a function of the sample. It is the wrong target for the survey-common signal, which is the round mean of the same revision that is the dependent variable. Holding it fixed removes exactly the feature that makes it a generated regressor, so the rejection rate it delivers says nothing about the specification that uses it.

The channel is short. Let two cells have mean disturbances a and b in a round. The signal is $(a+b)/2$ and the demeaned cell contrast is $(a-b)/2$, so their covariance is $[\mathrm{Var}(a) - \mathrm{Var}(b)]/4$. It vanishes only if the cells have the same disturbance scale, which is not true here: the cell scales run from 4.69 to 9.08 for inflation and from 4.78 to 10.55 for own income growth. Country-round effects attenuate the term, because the signal pools eleven countries while the contrast is taken inside one, but they do not remove it.

Design C measures what is left. The null has no heterogeneous response,

$$y_{it} \;=\; \mu_{c,t} \;+\; \sigma_{g(i)}\, c_t\, e_{it}, \tag{60}$$

with the country-round means, the cell scales and the round scales held at their data values and e resampled from the real standardised residuals, so the shape and the tails are the data's own. Each of the three pieces is calibrated. The country-round means fix how much genuine common movement the signal carries relative to sampling noise, which decides whether the mechanism operates. The cell scales are what switch it on. The round scales are what the two-way law has to get right. Disturbances are drawn independently across households, since the household and cell pieces are the same size in the real data, standard errors of 0.00540 against 0.00542. The round-level dependence in the score does not come from the disturbances in any case: every household in

a round faces the same signal, so the score sums to that signal times a cell aggregate.

Over 200 draws, with a Monte Carlo standard error of 0.0154 at the nominal five percent:

Table 10: Rejection rates when the signal is built from the outcome

	signal held fixed	signal rebuilt	leaving the cell out
inflation	0.060	0.055	0.050
own income growth	0.035	0.100	0.130

For inflation the two constructions agree. For own income growth the rebuilt signal puts the rejection rate at twice the nominal five percent, four Monte Carlo standard errors above the fixed-signal arm, so the reported p-value for that specification does not have the validity the fixed-signal experiment claims for it.

The third column shows that leaving a cell out of its own signal does not remove the channel. It removes that cell's disturbances from the regressor multiplying its own dummy, but the estimand is demeaned across cells, so the same disturbances enter through the regressors of every other cell. The measured rate is no better and is worse.

A split by household removes part of it: one half builds the signal, the other half is estimated on, and the regressor is then a function of data that contains none of the respondents whose residuals it multiplies. That disposes of the mechanical channel. Under a disturbance with a household loading on a common round factor, which is the dependence this appendix maintains throughout and simulates in its own size experiments, a signal built in one half still inherits the round factor that the other half's residuals carry, and the country-by-round effects absorb only its common level, not its cell-specific part. The split-specific p-values therefore require the builder half and the estimation half to be conditionally orthogonal within a round after the included effects, and household disjointness alone does not deliver that. The aggregation below is reported under that condition. The split is by household, not by observation, since a household recurs across rounds. At half the sample a single split is imprecise, and the two orientations of one split can give p-values an order of magnitude apart, so the split is repeated 100 times and aggregated by the rule of Meinshausen et al. (2009): if each split gives a valid p-value then the ς-quantile of $\{p_b/\varsigma\}$ is one too, which at $\varsigma = 1/2$ is twice the median. The level is written ς because γ is the classified share throughout this paper. For inflation the median is 0.0559 and twice it is 0.1118, with 0.46 of splits rejecting. For own income growth the median is 0.1977 and twice it is 0.3955, with 0.17 rejecting. Searching over ς with the penalty that result requires gives 0.1113 and 0.6183. Neither survives.

The two survey-common rejections in Table 3 are therefore reported as a diagnostic and nothing is built on them. The external-signal results are untouched, since those signals are published series, and so is d_{obs}, which is built from the external-signal spending map.

Inverting the test

This subsection describes the whole-map construction, whose radius is the figure quoted above for comparison. It is not the construction behind the reported bound: that one is componentwise, its region is the rectangle whose sides are the coordinatewise intervals, and its support bound is the corner of that rectangle. For the whole map the acceptance region is a union of ellipsoids instead, since the statistic does not reject when at least one of its three quadratic forms is small, so the confidence set is

$$\mathcal{C}_\tau \;=\; E_u \cup E_1 \cup E_2, \qquad E_j = \left\{ v : (\widehat{R} - v)' V_j^{-1} (\widehat{R} - v) \le q_{1-\alpha} \right\}, \tag{61}$$

where v is a candidate value of the map and $q_{1-\alpha}$ the critical value of the reference distribution. No simulation is needed. For a positive definite V_j each region is an ellipsoid and $\sup_{v \in E_j} \|v\|$ is a trust-region problem with an exact solution through a scalar root-find, so the norm bound is the largest radius of the three. Inverting along one direction instead would bound only that direction. The two one-way matrices are positive definite everywhere. The two-way matrix is positive definite at the two lags and indefinite at interior mixtures, as the subsection on inference for the projected contrast reports, and where it is indefinite its region is not an ellipsoid: moving out along a positive eigendirection and a negative one together keeps the form inside the band while both coordinates grow, so the region is unbounded and the support function over the union would be infinite. That arm is therefore dropped from the union wherever the matrix fails to be positive definite, which covers the boundary at which its smallest eigenvalue reaches zero and the inverse ceases to exist as well as the interior of the indefinite region, which is the same guard the test uses and is applied at the same mixtures.

Dropping it changes nothing here. The bound is a maximum over the arms, and at the mixture where it is attained the two-way arm gives the smallest of the three quadratic forms along the functional's direction, 1.364 against 1.505 for the household arm and 1.665 for the round arm, in units of 10^{-3}. The largest sets the bound and that is the round arm. With the two-way arm in the union the bound is 0.2859 and without it 0.2859. Had the degenerate arm been the largest instead, the bound would have been infinite.

Coverage does not need its own simulation. The statistic depends on $\widehat{R} - v$. With a fixed design and a linear model the distribution of $\widehat{R} - R$ does not depend on R, and neither do the residuals from which the three matrices are built. The size measured against a zero map is therefore also the size of a test of the true map. The timing-robust bound takes the maximum over the timing shares searched, so it covers at least as often as a test at the true mixture when the true shares lie in a family that was searched and the search reached that family's supremum, and when timing is unrelated to the cell within a round. With one share common to every country and round the family is the convex segment of Section 5.3, a single dimension, and the reported bound is its maximum on a grid of step 0.025. A grid of step 0.005 followed by a continuous refinement around the best point raises it by at most 0.00003 on either open-ended measure, to 0.17776 at $\pi = 0.447$ on the

retrospective one. That establishes numerical stability of the one-share maximum, not a certified global supremum over the continuous segment, so the bound is a nominal ninety-five percent bound computed numerically and its uniform coverage over the segment holds to the extent the search reaches the supremum. This is the family for which the confidence statements of the main text are made. With shares by country or by year the search is numerical and its maximum is not certified, so no coverage statement is made for those families, as the subsection on timing shares below records. The size experiments above hold the candidate country-round signals fixed and so validate the inference at a given point of that segment.

Collapsing the set to its radius discards geometry, and the equilibrium needs a linear functional. Write R_{gq} for the block of the response map belonging to the cell with classification g and income quintile q, π_{gq} for that cell's sample share and $\gamma = \sum_q \pi_{1q}$ for the share classified as constrained. For a unit combination u of the signals, the observable contrast between the classified group and the population is

$$\Delta_{\mathrm{obs}}(u) \;=\; \sum_{q} \frac{\pi_{1q}}{\gamma}\, u' R_{1q} \;-\; \sum_{g,q} \pi_{gq}\, u' R_{gq}, \tag{62}$$

which is linear in the map, so it is $b'v$ for a b fixed once u and the cell shares are. Both sums run over the five quintiles inside each classification, so the functional uses the whole ten-cell map and is not a two-cell contrast. For a linear functional the support function of an ellipsoid is closed form, $\sup_{v \in E_j} |b'v| = |b'\widehat{R}| + \sqrt{q_{1-\alpha}\, b' V_j b}$, and d_{obs} is the largest value of $|\Delta_{\mathrm{obs}}(u)|$ in (62) over the three matrices, the admissible lags and the unit combinations u, the combination entering equilibrium being unknown and the larger of the two coordinate bounds not bounding a combination. Routing through the functional instead of the radius turns a norm bound of 1.0013 into 0.1345 on the open-ended measure at the endpoint the manuscript first used, and its supremum over the one-share timing segment is 0.1778. Applied to the modified whole-map region the same routing gives 0.1778 over the segment, where the radius of that region is 0.2859. The difference between the two measures the region against the functional. The next subsection reports how the bound moves when the share may differ by country or by year.

Timing shares by country and by year

The one-share segment restricts the share of respondents who had seen the later release to be the same in every country and round. Table 11 relaxes that restriction in two directions. In the first the share takes one value per country, π_c, and in the second one value per survey year. The signal of (48) becomes $Z_{c,t-2} + \pi_b\, d_{ct}$ for the block b containing the observation, and its interaction with the cell splits into a fixed part and one column per block, each scaled by its share. The two-way fixed effects are removed by a linear projection, so every column is residualised once and the projected contrast and its three covariance matrices are rebuilt for any vector of shares without re-estimating the fixed effects. On the retrospective measure, setting every share to 0 and to 0.45 reproduces the one-share bounds 0.1345 and 0.1778 to the fourth decimal, which checks the decomposition.

The bound is maximised over the share box by coordinate ascent, on a step of 0.1 refined to

0.025, from six starting points. They are the one-share maximiser, the two corners at which all shares are zero or one, and three random points. A second, independent search evaluates every vertex of the box, 2^{11} for countries and 2^7 for years, and 500 uniform draws, and restarts the ascent from the best point found. On the retrospective measure both searches reach 0.1960 with shares by country and 0.1914 with shares by year. On the expected measure the first finds 0.1551 and the second 0.1526 by country, and both find 0.1469 by year, so the objective has more than one local maximum and the larger value is the one reported. Neither search bounds how fast the objective varies between evaluated points, so the figures are numerical maxima and bound each family's supremum from below. Neither family covers shares that vary arbitrarily by country and round.

A numerical maximum of an upper confidence bound can fall short of the supremum over the family, and a confidence statement for the family needs the supremum, so these figures are not used as confidence bounds. They are a sensitivity. They show that the single-share restriction has quantitative consequences, and that a valid bound would be at least 0.1960 over shares by country and at least 0.1914 over shares by year, ratios of 1.102 and 1.076 to the one-share bound. A certified bound would need a global optimisation with verifiable error bounds over the share box, which is not attempted here. The main text keeps the one-share bound and states its coverage for that family. Panel B of Table 11 evaluates the thresholds of (52) at the largest figure found. They scale with the inverse of the bound, so their levels fall by the factor 0.907 and their ratio does not move, which is a property of the proportionality and not a separate check.

Table 11: Sensitivity of the observable bound to country-specific and year-specific timing shares

Panel A. Ninety-five percent bound d_{obs}, by timing family					
	Common share	Share by country		Share by year	
Spending measure	bound (π^*)	bound	ratio	bound	ratio
Retrospective, open-ended	0.1778 (0.45)	0.1960	1.102	0.1914	1.076
Expected, open-ended	0.1260 (0.65)	0.1551	1.231	0.1469	1.166
Independent search, retrospective		0.1960		0.1914	
Independent search, expected		0.1526		0.1469	

Panel B. Thresholds in the composite scale, perfect classification					
	$d_{\text{obs}} = 0.1778$		$d_{\text{obs}} = 0.1960$		
ξ	$\kappa_{\min}$	$\kappa_{\max}$	$\kappa_{\min}$	$\kappa_{\max}$	$\kappa_{\max}/\kappa_{\min}$
0.4	3.16	3.79	2.87	3.44	1.20
0.6	2.33	2.85	2.12	2.59	1.22
1.4	3.95	5.59	3.58	5.07	1.42
1.8	11.31	17.44	10.26	15.82	1.54

Notes. Panel A reports the Bonferroni bound on the projected contrast of Section 5.3. The first column restricts the share of respondents who had seen the later release to one value in every country and round and reports its supremum over a grid of step 0.025, attained at the share in parentheses. The next columns let the share differ across the eleven countries or across the seven survey years, and report the largest bound found by coordinate ascent from six starting points, with ratios to the first column. The two rows below repeat the search from the best of all 2048 or 128 vertices of the share box and 500 uniform draws. The block-wise entries are numerical maxima. They bound the supremum over each family from below and are not certified, and neither family covers shares that differ arbitrarily by country and round. Panel B gives the thresholds of (52) at four incidences under the common-share bound, which is the one the confidence statements use, and under the largest bound in Panel A, a numerical sensitivity without a coverage guarantee for its family. Both thresholds scale with $1/d_{\text{obs}}$, by the factor 0.907, so their ratio, the last column, is the same under either bound.

Weighting and the target population

The survey supplies a respondent weight and the estimates in the main text do not use it. The weight is calibrated to each country's adult population, by age, sex and region, so a respondent's weight is the number of adults it represents. Summed within a country and round it reproduces that country's adult population, and pooled it targets the adult population of the eleven countries the survey covers. The theory has a unit mass of households and the outcome is household spending, so neither weighting delivers the model's population: unweighted estimates average over the survey's respondent households, weighted estimates average over the adults those households represent. The representativeness bridge of Section 5.3 is the gap between either of these and the population the covariance is defined on, and no weighting choice closes it.

Which weighting supports inference is a separate question, and it is settled by size. Both designs of the size experiments were rerun with the survey weight carried through the estimator, so that the coefficient solves $(X'WX)\beta = X'Wy$, the bread is $(X'WX)^{-1}$ and the score is $w_i x_i u_i$. The weight is part of the design and is never resampled, so a replicate weight is $w_{it} c_i d_t$. Table 12 reports the rejection rate at a nominal five percent.

Table 12: Rejection rates against a true null under the two weightings, nominal five percent

design	weighting	inclusion-exclusion	minimum of three
design held fixed	unweighted	0.395	0.035
design held fixed	adult-weighted	0.529	0.050
design redrawn	unweighted	0.354	0.050
design redrawn	adult-weighted	0.544	0.100

Notes. The first two rows hold the real survey design fixed and draw only the errors, as in Table 8. The last two redraw the design, resampling whole households from the panel so that a household arrives with its country, its pattern of rounds, its predetermined cell and its entire weight path, and generating only the errors. The two laws reported are analytic, so the redrawn rows use a thousand simulations and no bootstrap, with a Monte Carlo standard error of 0.007 near five percent. Holding the design fixed, the weighted minimum of three was also run with the loading of the null pinned at four values spanning the plausible range, giving 0.033 to 0.055, the upper end less than one Monte Carlo standard error above nominal.

The two designs answer different questions and give different answers. Holding the design fixed, the weighted procedure is calibrated: it rejects at the nominal rate and stays there across the loadings of the null. Redrawing the design, it rejects twice as often as it should, while the unweighted procedure stays at nominal in both. Its coverage relies on the particular design the survey happens to have, and does not survive this block-resampled redraw of that design.

The main text therefore uses the unweighted procedure for inference, since it is the one that holds size under both designs. The weighted estimates enter as a sensitivity to the target population, and only through objects that do not pass through the confidence set: the coefficient map, and the classified share, which moves from 0.2526 to 0.2463 under weighting. That share is computed on the sample this run uses, which conditions on one lag of the signals. The timing-robust construction of Section 5 requires both lags to be present and gives 0.2533 on the smaller sample that survives. The weighted p-values are not used, and neither is a weighted d_{obs}, because that bound is the support function of the confidence set over the functional and inherits its coverage. The weighted bound is larger, at 0.3874 against 0.1345 at the same endpoint.

Rank

A rank test needs its own validation and does not survive it. The natural candidate is the minimum distance to the set of matrices of rank at most j, computed under each of the three matrices and minimised, but that construction is not derived by Davezies et al. (2025), and the null $\text{rank}(R) \leq j$ is not regular where the true rank is strictly below j, so size must be measured across the whole null, not at its boundary alone. On the full nine by six object of six expectation measures it fails, rejecting a true rank of one at $H_0 : \text{rank} \leq 1$ every time while rejecting essentially never in the interior. Against a known covariance the same minimisation is correctly sized at the boundary, so the failure is in estimating the two-way covariance, not in solving the minimum-distance problem.

On a nine by two object the test is usable but not pivotal, its boundary size depending on the unknown true rank-one matrix, with a ninety-fifth percentile from 15.74 to 21.11 across a grid of scales against a chi-square reference of 15.51. The stacked map of two expectation measures is

nonzero, and rank one is not rejected: the statistic is 9.29, below both the size-calibrated value and the naively liberal chi-square one, so the conclusion does not depend on the calibration.

F The dated-news exercise

The testing restriction, the six published specifications, the selection analysis and the estimates behind the dated-news exercise are collected here.

The survey is a monthly repeated cross-section of United States households running from 2005:2 to 2013:6, with more than 25,000 observations and a few hundred respondents per month, recording expectations about inflation, unemployment and interest rates alongside the respondent's income, assets, liabilities and credit limits. Households are sorted into poor hand to mouth, wealthy hand to mouth and unconstrained by the Kaplan et al. (2014) conditions, applied to liquid wealth, illiquid wealth, income and unsecured credit limits, and about 35 percent of the sample is hand to mouth. By Theorem 5 the statistic needs group averages, which are estimable from repeated cross-sections.

Exposure comes from the timing of financial-crisis events relative to the interview. A published timeline lists 305 events, of which the retained ones are those where the stock market falls, or its implied volatility rises, by more than one standard deviation in a two-day window, leaving 58 selected on the market index and 42 on volatility. Since the survey records the interview day, a respondent is exposed when a retained event fell on that day, the day before, or two days before, so exposure varies within a calendar month according to when the household happened to be interviewed.

The estimating equation is a pooled regression of inflation expectations on hand-to-mouth status, the shock and their interaction,

$$y_i \;=\; \alpha \;+\; \delta_p\, p_i + \delta_w\, w_i \;+\; \beta_0\, \text{shock}_i \;+\; \beta_p\, p_i\, \text{shock}_i + \beta_w\, w_i\, \text{shock}_i \;+\; x_i'\eta \;+\; \varepsilon_i, \qquad (63)$$

with p_i and w_i indicating the two hand-to-mouth categories and unconstrained households omitted. The five lagged macroeconomic indicators and the state indicators enter every column. The column labels describe what is added to them: nothing in the first of each block, then age category, college education, sex and race, then log household income as well. The full control set of the third column is therefore age category, college education, sex, race and log household income, five lagged macroeconomic indicators, and state indicators, with the heteroskedasticity-robust standard errors of the source paper, which the estimates reported here replace as explained below. The unconstrained response is β_0 and the two hand-to-mouth responses are $\beta_0 + \beta_p$ and $\beta_0 + \beta_w$, which Theorem 5 uses together with the three shares.

Three mappings connect these estimates to the statistic. The first, exposure exogeneity, concerns the regression itself and is taken up at the end of this appendix. The two that follow concern its interpretation. The first reads the hand-to-mouth classification as the economic state, treating both hand-to-mouth categories as constrained. Evidence that wealthy hand-to-mouth households carry high marginal propensities to consume supports it (Kaplan et al., 2014). The second reads

relative cross-group responsiveness of inflation forecasts as relative ρ, the loading through which expectations reach consumption in (23). Under cognitive discounting in the sense of Gabaix (2020) a household applies one parameter to every forecast, so the two coincide and the mapping is an identity. Under mechanisms in which attention is allocated across variables the two come apart and the mapping is a restriction. The constant of proportionality is absorbed into ψ and does not affect Λ_S or R_S.

Responsiveness is non-negative, so the sign of a coefficient is not its sign. The shock measures the magnitude of an adverse financial event, which lowers realized inflation and lowers inflation expectations, so the signal points toward lower future inflation, responding in its direction means revising down, and responsiveness is proportional to minus the coefficient. One sign flip is applied to every group in every specification, with no absolute values and no truncation.

Writing β_p and β_w for the two interactions of the shock with hand-to-mouth status, the hypothesis $R_S = 1$, that responsiveness is not sorted in the mean across states, is equivalent to the linear restriction

$$\Delta_{\text{assignment}} \;=\; \pi_p\beta_p + \pi_w\beta_w \;=\; 0, \tag{64}$$

because $\mathbb{E}[\rho \mid S] - \mathbb{E}[\rho \mid g = p]$ and $\mathbb{E}[\rho \mid S] - \mathbb{E}[\rho \mid g = w]$ are exactly those two coefficients. The imprecisely estimated main effect drops out. (64) is therefore the object we test, and Λ_S and R_S are reported afterwards as descriptive ratios.

The null is weaker than independence. Independence of responsiveness and state implies $R_S = 1$, but not conversely: a law placing half its unconstrained mass at zero and half at twice the mean, with every constrained household at the mean, has $R_S = 1$ while the two conditional distributions share no mass. What makes the weaker restriction the right target here is Theorem 1. The aggregate depends on the assignment only through M_S, so a first moment is all the counterfactual can see, and testing more would be testing something the model does not use.

From the published replication package we reproduce all six columns of their Table 1 to the precision it reports, every coefficient and every sample size. In their baseline specification, which uses 8,579 observations with shares $\pi_p = 0.102$, $\pi_w = 0.221$ and $\pi_n = 0.677$, the result is reported three ways.

Exposure is assigned at the interview date, not at the respondent, so households seen on the same day share the regressor and a disturbance common to that day enters all their scores together. An observation-level heteroskedasticity-robust variance omits those cross-products and is valid only if no such component exists, which nothing in the design supplies. Standard errors are therefore clustered on the interview date, of which the baseline sample has $1,790$. The group shares are estimated on the same sample as the coefficients, so the restriction is a weighted sum whose weights carry sampling error of their own. Treating them as fixed reports a standard error conditional on the realised composition. Adding their influence terms and the covariance with the coefficients raises the standard error by at most a factor of 1.00 across the six specifications, so the distinction changes no reported conclusion and the conditional version is what the table shows. The correction is not what carries the result. Across the six specifications the clustered standard errors average

0.88 times the observation-level ones, and 4 of 6 reject at five percent under either law. The table carries the clustered law alone and the comparison comes from the same fits.

Without the support restriction, the data do not reject the hypothesis that mean responsiveness is the same across liquidity states. In the primary specification $\Delta_{\text{assignment}} = -0.0123$ with a clustered standard error of 0.0091, against 0.0097 under the observation-level law. Neither rejects at five percent.

The point estimate places less responsiveness among unconstrained households than independence implies, $R_S = 0.248$ against one and $\Lambda_S = 0.168$ against 0.677. These are descriptive ratios. The baseline coefficient is the numerator of R_S, and it also enters the mean responsiveness both ratios divide by, since that mean is the baseline together with the combination (64) isolates. It is what that equation removes for being imprecisely estimated: 0.0041 against a standard error of 0.0112 on the same scale, so 0.36 standard errors from zero, on both sides of a ratio. A point cannot summarise a ratio in that configuration, and the confidence sets reported below replace the two points. The object with an inference attached is $\Delta_{\text{assignment}}$, whose sign is the same in all six published specifications and which is distinguishable from independence in four of the five alternatives. Figure 10 shows all six with their intervals and Table 13 gives the levels behind them.

Those levels also answer a question the framework asks and the exercise had left open. Responsiveness lives in $[0, \bar{\rho}]$, so the implied group values must be non-negative under the single sign flip. In 2 of the six they are not. Both are specifications that add household income, and one of them is the baseline, where the implied value for the poor hand-to-mouth group is -0.0023 against 0.0041 for the unconstrained and 0.0628 for the wealthy hand-to-mouth. Nothing is truncated, since truncating would conceal exactly what the restriction asks about. The negative value is small next to its own standard error.

The lower end of the support can be imposed in the inference instead of checked at the point estimate. Write $m = (m_n, m_p, m_w)$ for the three group responses under the sign flip, a linear map of the three coefficients with date-clustered covariance Ω. For a scalar functional f of m, the values v for which the smallest Wald distance $(m - \hat{m})'\Omega^{-1}(m - \hat{m})$ over $m \geq 0$ with $f(m) = v$ stays below a critical value form a confidence set that uses the restriction. Only non-negativity is imposed. The upper end $\bar{\rho}$ is not, because the coefficients identify responsiveness only up to the scale absorbed into ψ, so no finite bound on m follows from it. Each minimisation is a convex quadratic programme in three variables and is solved exactly. The reference critical value is the χ^2_3 quantile, which projects the joint confidence set for m and is valid whichever non-negativity constraints bind at the true m. The χ^2_1 quantile profiles a single restriction and is valid only where none of them binds there, which cannot be assumed here, so its results are reported for comparison and nothing below rests on them.

The ratios are undefined at $m = 0$, so the first question is whether the data admit that point. They do not. The distance from $\hat{m}$ to the origin is 11.52 in the primary specification, $p = 0.009$ against χ^2_3, and the origin is outside the set in all six. Mean responsiveness is then bounded away from zero over the set, and the ratios have informative sets. In the primary specification,

under the projection, Λ_S lies in $[0.000, 0.672]$, below the independence value 0.677, and R_S lies in $[0.000, 0.992]$, below one. Both upper ends are close to the independence values, so the sets exclude independence by a small margin. For the tested object the restriction matters. Without it the set for $\Delta_{\text{assignment}}$ is $[-0.0300, 0.0055]$ and contains zero. With it, under the projection, the set is $[-0.0377, -0.0002]$. The constrained distance at $\Delta_{\text{assignment}} = 0$ is 7.98, so $p = 0.046$ against χ^2_3, close to the five percent level. Across the six specifications the set excludes zero in 4 without the restriction and in 5 with it under the projection. The profiled critical value gives narrower sets, which exclude zero in 6 specifications, and is not used for the conclusion.

The reason is visible in the primary specification. The configuration with $\Delta_{\text{assignment}} = 0$ closest to the estimates gives the poor hand-to-mouth group a response of -0.091, which its standard error of 0.075 makes easy to reach and non-negativity rules out. Once the group responses are required to be non-negative, equal mean responsiveness across liquidity states is rejected in the primary specification, at the margin. That conclusion is conditional on the second mapping. Non-negativity is a property of ρ, and it reaches the inflation-forecast coefficients only if those coefficients, after the sign flip, measure the same responsiveness that enters consumption. The rejection is therefore a statement about the coefficients read through that mapping. It is not independent evidence on how households with different responsiveness are distributed across liquidity states.

The difference between the baseline and the alternatives is driven by sample composition, not by controlling for income. Restricting the demographic specification to respondents with non-missing income removes the result before income is added as a control. A pre-specified selection analysis is inconclusive. The weighting fits a logit for whether income is reported on the demographic sample, on the same covariates as the specification, clips the fitted probabilities to $[0.02, 0.98]$ and uses stabilised weights, the reporting rate over the fitted probability, in weighted least squares. The imputation draws income from a Bayesian normal linear model on those covariates together with the outcome, the shock and the interactions, with proper parameter draws, and combines $fifty$ completed samples by Rubin's rules. Every variance is clustered on the interview date, and the bootstrap resamples dates instead of respondents. Weighting gives $p = 0.066$, and $p = 0.093$ when the propensity score is re-estimated in each replication instead of held fixed, while imputation over the full sample of 12,295 observations gives $p = 0.002$ and so rejects where the baseline does not. The point estimates from the full sample, from weighting and from imputation agree closely, so the methods differ about precision. There is little observable imbalance for a selection-on-observables method to correct.[6]

[6] Weighting is mild, with an effective sample size of 7,736, and the standardized differences between income reporters and non-reporters are below 0.08 on every regressor that enters the restriction being tested.

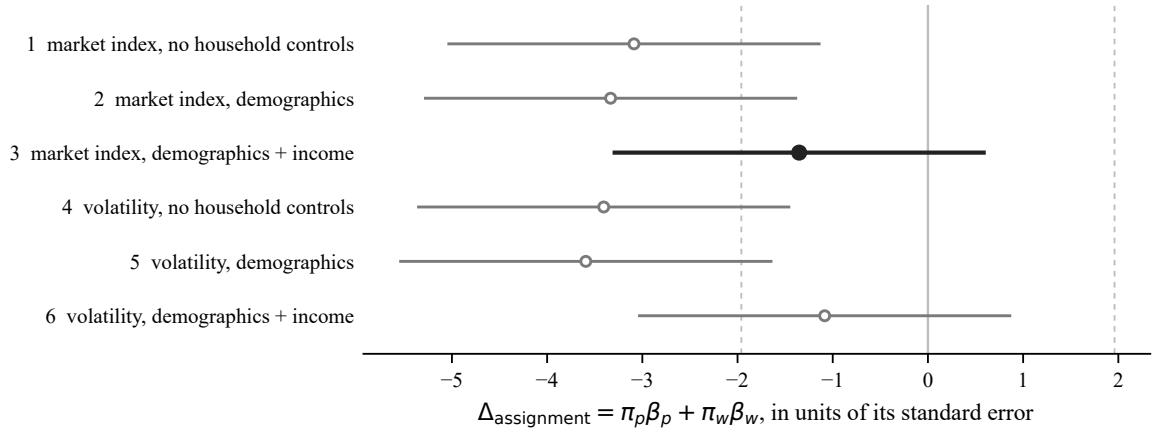


Figure 10: Departures from equal mean responsiveness across liquidity states

Notes. Each row reports $\Delta_{\text{assignment}} = \pi_p\beta_p + \pi_w\beta_w$ in units of its own standard error, so that specifications using different shock measures are comparable on one axis, with the horizontal bar covering plus and minus 1.96 standard errors. The solid vertical rule marks zero, which is no sorting in the mean, and the dashed rules mark plus and minus 1.96, so a bar crossing the solid rule is a specification that does not reject. The third row, which adds income to the demographic controls, is the primary specification. It is drawn in the darker shade. Estimates and shares are recomputed within each specification's own estimation sample.

The exercise establishes that the chain runs end to end on real data, from an observed classification tied to the economic state through group shares and group averages to an assignment statistic.

Is exposure as good as random

Section 5.4 names exposure exogeneity as the first of three mappings. The condition concerns unobservables. Its observable implication is: if the timing of an interview relative to the retained events were unrelated to who the respondent is, the observed characteristics would not predict exposure. That regression is run here, of the exposure indicator on the liquidity classification, the demographics and the state indicators, testing the liquidity and demographic block jointly. Its standard errors are clustered on the interview date, as the estimates are, since a diagnostic for a date-assigned regressor cannot treat respondents seen on the same day as independent draws. The test is a Wald statistic on the seven-coefficient block against an F reference distribution with the number of dates as the second degrees of freedom. The macroeconomic controls are excluded from the tested block because they are functions of the date and are mechanically related to which dates carry events, so including them would test the calendar instead of the respondents.

The two shock families do not answer alike. In the equity-price family, on which the baseline is built, the block does not predict exposure: $F(7, \cdot) = 1.18$ with $p = 0.314$ on an exposure rate of 0.071, and the largest single t statistic in the block is 2.17. In the volatility family it does: $F(7, \cdot) = 2.25$ with $p = 0.028$ on an exposure rate of 0.060, and the largest single t statistic, 2.50, falls on the poor hand-to-mouth indicator, which is one of the two indicators whose interaction with exposure the exercise reads.

In the volatility family exposure is not unconditionally as good as random, so the identifying content rests on the controls, and the dependence runs through the classification the mapping to the assignment statistic uses. The volatility columns are therefore read as corroboration of a chain.

Table 13: The six published specifications

	Equity-price shock			Volatility shock		
	(1) none	(2) demog.	(3) + income	(4) none	(5) demog.	(6) + income
Shock, β_0	-0.0036	-0.0060	-0.0041	-0.0091	-0.0241	-0.0126
	(0.0104)	(0.0098)	(0.0112)	(0.0751)	(0.0723)	(0.0762)
Poor hand to mouth, β_p	-0.0495	-0.0472	0.0064	-0.3556	-0.3797	0.1887
	(0.0366)	(0.0377)	(0.0740)	(0.2345)	(0.2433)	(0.4053)
Wealthy hand to mouth, β_w	-0.0541	-0.0576	-0.0587	-0.3791	-0.3866	-0.3516
	(0.0157)	(0.0145)	(0.0184)	(0.0928)	(0.0922)	(0.1164)
Implied $m_n \propto -\beta_0$	0.0036	0.0060	0.0041	0.0091	0.0241	0.0126
Implied $m_p \propto -(\beta_0 + \beta_p)$	0.0532	0.0532	-0.0023	0.3646	0.4038	-0.1761
Implied $m_w \propto -(\beta_0 + \beta_w)$	0.0577	0.0635	0.0628	0.3882	0.4107	0.3642
$\Delta_{\text{assignment}}$	-0.0173	-0.0178	-0.0123	-0.1214	-0.1253	-0.0588
	(0.0056)	(0.0053)	(0.0091)	(0.0356)	(0.0349)	(0.0542)
Share poor hand to mouth	0.102	0.102	0.102	0.100	0.100	0.101
Share wealthy hand to mouth	0.226	0.226	0.221	0.226	0.226	0.221
Observations	12314	12295	8579	12177	12159	8470
Interview dates	1891	1891	1790	1873	1873	1772

Notes. Columns reproduce Ekici et al. (2026) Table 1. Columns one to three use the equity-price shock and four to six the volatility shock, with no household controls in the first column of each block, demographics in the second and demographics with log household income in the third. Standard errors are clustered on the interview date, which is the level at which exposure is assigned, and appear in parentheses. The implied group rows apply the single sign flip of Section 5.4 and are not truncated, so that the reader can see whether the values lie in the support $[0, \bar{\rho}]$ that Section 2 imposes on responsiveness. $\Delta_{\text{assignment}} = \pi_p \beta_p + \pi_w \beta_w$ is the object the test in (64) restricts.

G Data, code and declarations

Data and code

The inventory below names every display in the paper. Three groups use no data. The theory illustrations, Figures 3 and 2 in Section 2 and Figure 4 in Section 4, are drawn with responsiveness distributions stated in their own notes, which are not the reference economy. The reference economy of Table 1 is an assumed parameter vector, never an estimate, and Figure 1, Table 2, Figure 5, Table 6, Figure 8 and Figure 7 are computed from it. Figure 1 changes nothing but the assignment, Table 2 with Figure 5 holds its other ingredients fixed while changing the responsiveness marginal, and Table 6, Figure 8 and Figure 7 move one primitive at a time. Table 4 reports economies drawn from the class of (57), and Table 5 records the numerical verification of Appendix C. The code for all three groups will be deposited with the article and needs no data to run.

The remaining displays use survey data. Table 3, Table 10, Table 8, Table 12, Table 9 and Panel A of Table 11 are computed on the estimation samples or on resamplings of the survey's own

design, Figure 9, Table 7, Panel B of Table 11 and the second panel of Figure 6 condition on the observable bound those samples deliver, while the first panel of that figure varies income incidence in the reference economy and uses no data, and Table 13 and Figure 10 come from the third-party package described below.

The euro area exercise of Section 5.3 uses the European Central Bank Consumer Expectations Survey. The microdata are available to researchers on application to the European Central Bank and cannot be redistributed, so the deposit carries the estimation code, the variable definitions and the sample-selection steps in place of the data. The macroeconomic series used to build the signals are public releases from the same institution and are retrieved by the code.

The exercise of Section 5.4 re-estimates the published replication package of Ekici et al. (2026), which is available from the publisher. The deposit carries the code that reads that package and produces Table 13 and Figure 10. The package itself is third-party material and is not redistributed.

Declarations

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors. The author declares no competing interests. The views expressed are the author's own and do not represent those of the European Central Bank or of any institution with which the author is affiliated.